EXCEL INTERMEDIAT TRAINING

Second Edition

Said Fawzy

Aug 2025

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Preface to the Second Edition

This Second Edition of *Intermediate Excel Training* builds on the strengths of the first edition, with enhanced explanations, updated visuals, and brand-new exercises to make your learning experience even more effective.

What’s New in This Edition

- **Expanded Chapter 7 – Charts in Excel**
  More detailed coverage of chart formatting options, including advanced formatting tools, customization tips, and professional design techniques. New hands-on exercises have been added to help you create clear, professional visuals.
- **New Chapter 10 – Conditional Formatting**
  A complete, step-by-step guide for using conditional formatting to highlight trends, patterns, and outliers in your data. Includes practical examples, color scales, icon sets, and data bars.
- Updated examples, screenshots, and instructions to reflect the latest Excel interface and features.
- Additional tips and shortcuts throughout the book to improve productivity.

This edition continues the book’s practical, “learn-by-doing” approach while making sure the content reflects the most current capabilities of Excel.

Introduction

This book is my second training guide on Microsoft Excel, designed for **intermediate-level learners** — people who already know their way around Excel but want to **master it more effectively** and work with it more powerfully.

The approach is **100% practical**. Every topic is explained through step-by-step exercises you can try yourself, so you get hands-on experience ("getting your hands dirty") rather than just reading theory.

All the required training materials, including exercise files, solution files, and PowerPoint slides, are available on my GitHub profile:
🔗 **https://github.com/saidfawzy**

You can also watch the complete video training for this course on my YouTube channel, organized in playlists for easy navigation:
📺 **www.youtube.com/saidfawzy**

How to Use This Book

- Work through each chapter in order, completing the exercises as you go.
- At the end of each chapter, you'll find a **self-test exam** — try it without looking at the answers to check your progress.
- If you want feedback, feel free to send me your completed exam file and I'll be happy to review it and share my comments.

I recommend following along with the videos on YouTube while using this book and its accompanying presentation and practice files. **You won't truly learn Excel until you try things yourself.**

Stay Connected

If you'd like to connect or ask questions, you can reach me through my LinkedIn profile:
🔗 **www.linkedin.com/in/saidfawzy**

You can also join my Power BI communities:

- Facebook Group
- WhatsApp Group
- LinkedIn Group
- Telegram Group

This book and its materials are **completely free**. You’re welcome to share them with others, and you should never hesitate to contact me if you need help.

Said Fawzy
Manager of Information Center
Arab Contractors
30 April 2025

Chapter 1: Named Ranges

What are Named Ranges

Instead of specifying a cell or range of cells individually every time you want to reference the data they contain; you can name the cell or cells—in other words, create a *named range.*

Exercise 01A: Calculation Using Ranges

1. Calculate the Values in the right tables using the suitable Function.

Exercise 01 Named Ranges

Order #	Customer	Quantity	Price Per Unit	Total
3211	Staples	27,095	£ 0.20	£ 5,419.00
2955	WH Smith	48,696	£ 0.15	£ 7,304.40
3159	The Art Supply Store	29,970	£ 0.25	£ 7,492.50
3004	Brush Strokes	15,667	£ 0.20	£ 3,133.40
4534	Crafty Business	46,321	£ 0.20	£ 9,264.20
2220	Pens and Things	47,003	£ 0.20	£ 9,400.60
1796	Color Me Silly	41,595	£ 0.20	£ 8,319.00
3558	Sketch & Co	41,552	£ 0.20	£ 8,310.40
1437	The Desinger Centre	23,299	£ 0.30	£ 6,989.70
1336	Waterstones	34,014	£ 0.15	£ 5,102.10
2044	Painterly	43,247	£ 0.15	£ 6,487.05
3320	J Howard and Sons	28,305	£ 0.15	£ 4,245.75
1534	Art Attack	39,218	£ 0.20	£ 7,843.60
3989	Doodles	30,468	£ 0.20	£ 6,093.60

Average Quantity	
No. of Orders	
Minimum Sales	
Maximum Sales	
Total Sales	

2. The result must be like that:

Order #	Customer	Quantity	Price Per Unit	Total
3211	Staples	27,095	$ 0.20	£ 5,419.00
2955	WH Smith	48,696	$ 0.15	£ 7,304.40
3159	The Art Supply Store	29,970	$ 0.25	£ 7,492.50
3004	Brush Strokes	15,667	$ 0.20	£ 3,133.40
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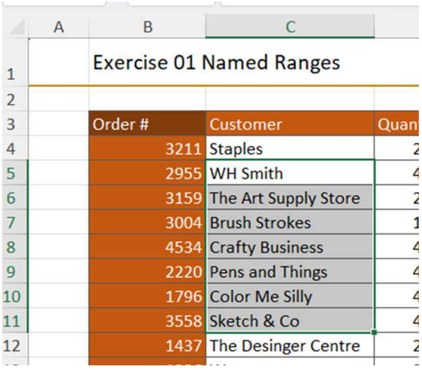
1534	Art Attack	39,218	$ 0.20	£ 7,843.60
3989	Doodles	30,468	$ 0.20	£ 6,093.60

Average Quantity	35,461
No. of Orders	14
Minimum Sales	$3,133
Maximum Sales	$9,401
Total Sales	$95,405

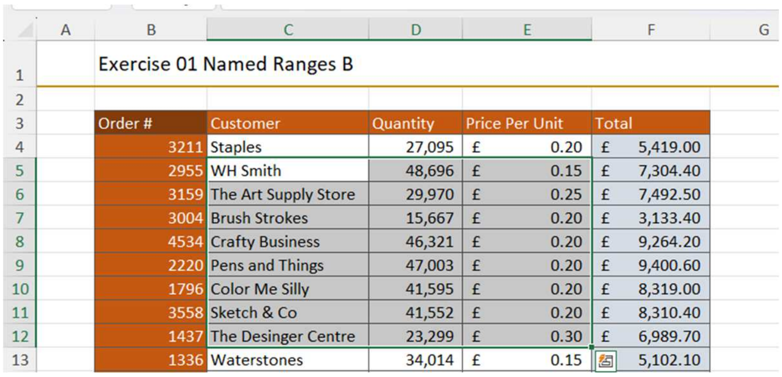
Named Ranges

- When you select a group of cells they are range and can have names
- For example, in the following figure we selected range (C5:C11) and (C5:F12)



Exercise 01 Named Ranges

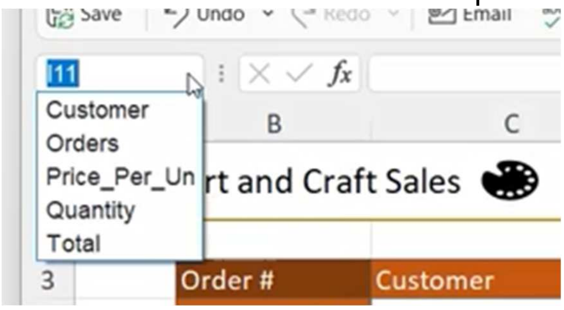
Order #	Customer	Quan
3211	Staples	2
2955	WH Smith	4
3159	The Art Supply Store	2
3004	Brush Strokes	1
4534	Crafty Business	4
2220	Pens and Things	4
1796	Color Me Silly	4
3558	Sketch & Co	4
1437	The Desinger Centre	2



Exercise 01 Named Ranges B

Order #	Customer	Quantity	Price Per Unit	Total
3211	Staples	27,095	£ 0.20	£ 5,419.00
2955	WH Smith	48,696	£ 0.15	£ 7,304.40
3159	The Art Supply Store	29,970	£ 0.25	£ 7,492.50
3004	Brush Strokes	15,667	£ 0.20	£ 3,133.40
4534	Crafty Business	46,321	£ 0.20	£ 9,264.20
2220	Pens and Things	47,003	£ 0.20	£ 9,400.60
1796	Color Me Silly	41,595	£ 0.20	£ 8,319.00
3558	Sketch & Co	41,552	£ 0.20	£ 8,310.40
1437	The Desinger Centre	23,299	£ 0.30	£ 6,989.70
1336	Waterstones	34,014	£ 0.15	5,102.10

- You can give the range of cells a name for later reference.
- You can find them later in the drop box on the left od the worksheet



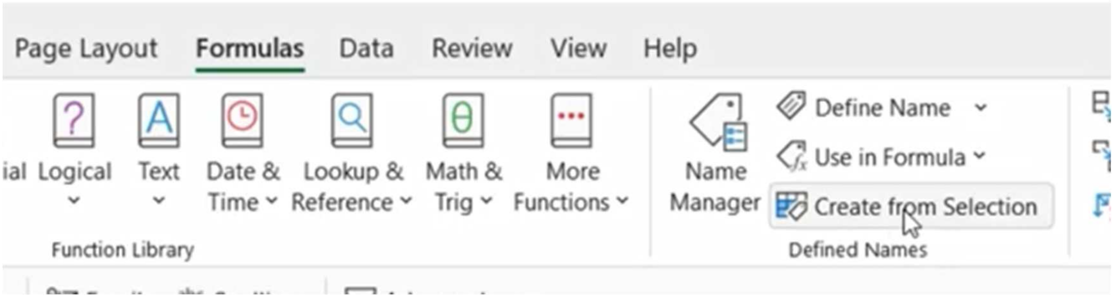
Exercise 01 B: Create Named Ranges

1. You can duplicate your sheet if you want to.
2. Select Range (B3:B17) and in the Name Box Name it **Orders**.

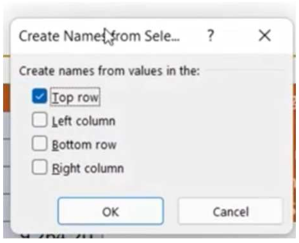
Art and Craft Sales

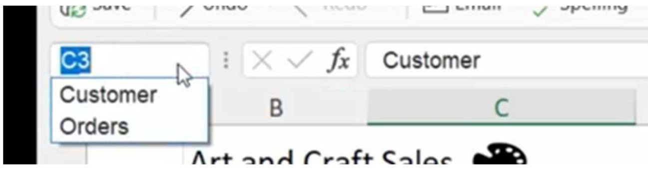
Order #	Customer
3211	Staples
2955	WH Smith
3159	The Art Supply Sto
3004	Brush Strokes
4534	Crafty Business
2220	Pens and Things
1796	Color Me Silly
3558	Sketch & Co
1437	The Desinger Cent
1336	Waterstones
2044	Painterly
3320	J Howard and Son:
1534	Art Attack
3989	Doodles

3. Make sure that Name has no space, you can only use _.
4. Select **Customer** column (including the heading).
5. Formulas ➔Defined Names➔Create From Selection.

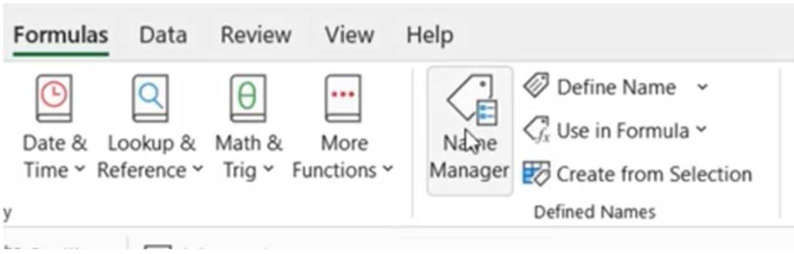


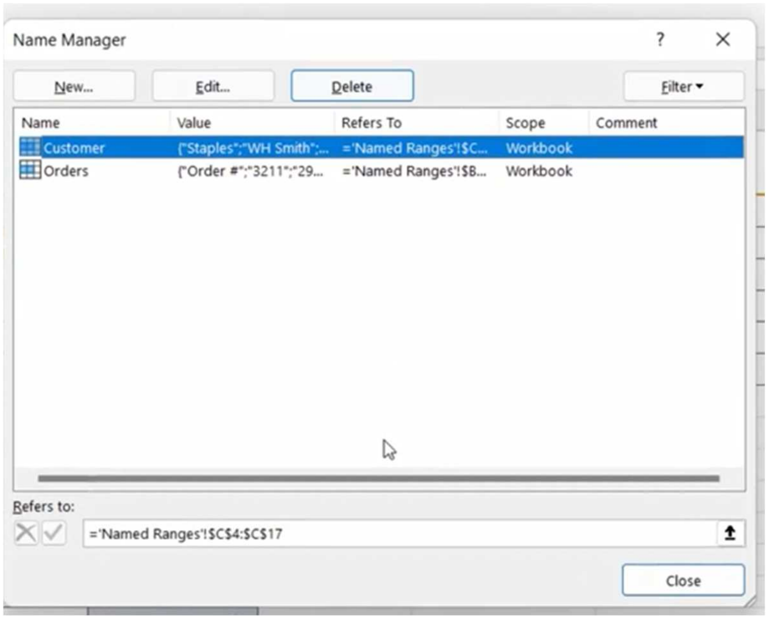
6. Use Top Row as name



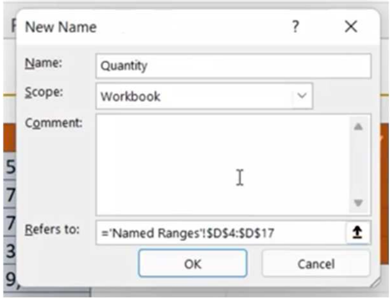


7. Click on Name Manager (Ctrl+F3)

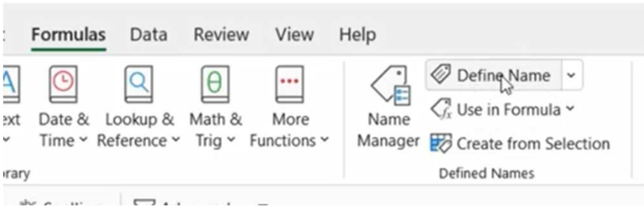




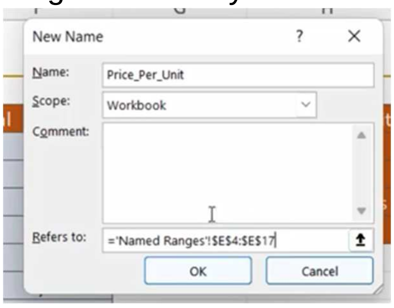
8. Click on New Button
9. Name it: **Quantity** and select the range and make it available through the workbook and press OK.



10. Use Define a name to create a new **Price_Per_Unit** Range Name. from Ribbon Data➔Defined Names➔Define Name.



11. Notice that the Refer Ranges are always Absolute Ranges.



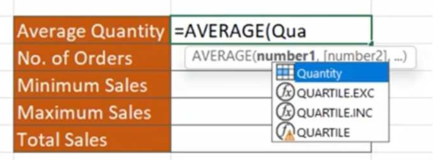
12. Use Create from Selection to create the **Total** Name range.

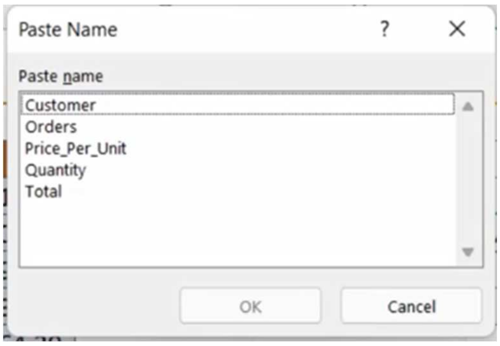
Using Named Ranges for Calculation

13. Now Fill the list in the right with calculation using the Named Ranges instead of Ranges as in Exercise 01 A.
14. You can use the first letter of the Named range to select from the list or press **F3** to open the list to choose from.
15. For example, write **Q** and when the Quantity appears in the list click the **tab key** to select the Quantity will be highlighted in blue and also the range will be selected in the list in blue.
16. If you have many names, you can press F3 Key to select from the Names.

Using Named Ranges for Navigation

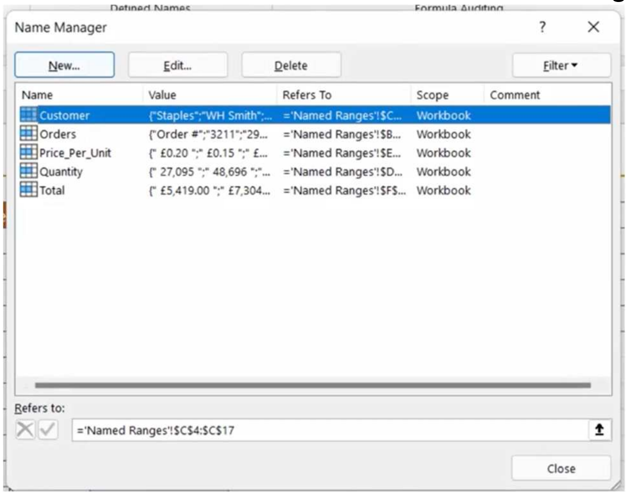
17. Create a new Sheet and while inside and try to Navigate to any range using the Name Box.



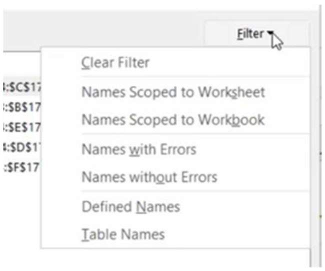


Exercise 01 C: Mange Named Ranges

1. Click on Name Manager.
2. Formulas ➔Defined Names➔Name manager



3. It is where you can create, Edit and Delete any Name.
4. It shows the **name**, the **Values** in, what Range it **refer** to and the **scope** (Workbook or Worksheet).
5. Notice also you have a Filter button. Try (Edit)



6. Try to rename **Quantity** to **Quantity1**.
7. Notice that Excel has corrected the formula for you in the list.
8. Delete **Total** Name.
9. You will get the name Error.

Average Quantity	35,461
No. of Orders	14
Minimum Sales	#NAME?
Maximum Sales	#NAME?
Total Sales	#NAME?

10. Create it again with the **same name** the error is corrected.
11. Get back to 1st sheet that you have used ranges in calculation.
12. Use Define Name ➔ Apply Names, to use names instead of ranges in all formulas.

Exam 01 Named Ranges

Manage Ranges

1. Name the following data ranges:

    A2:A52, "Customer_Name"
    B2:B52, "Quantity"
    C2:C52, "Price_Per_Item"
    D2:D52, "Total"
    E2:E52, "Sales_Tax"
    F2:F52, "Total_Inc_Tax"

2. Name H2:I2, "Tax_Rate"

NOTE: Try and use all of the different methods for creating named ranges.

Using Named Ranges in Formulas

1. Replace the reference to cell I2 in the **'Sales Tax'** formula in column E with the named range.

2. Re-calculate the formulas in cell range I4:I8, using the named ranges instead of cell references.

Try to use Apply Names Button.

Chapter 02: Lists and Tables

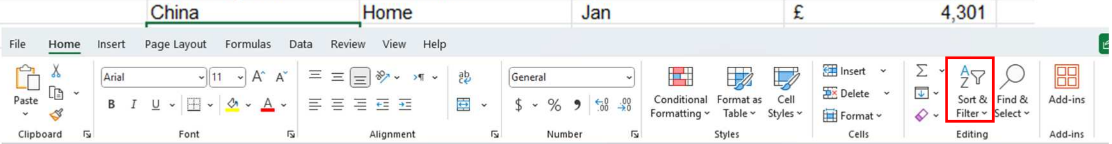
Exercise 2 A: List Single Level Sorting

1. Single level sorting means we sort with one column only.
2. In the List I want to sort data by **Region** Column.

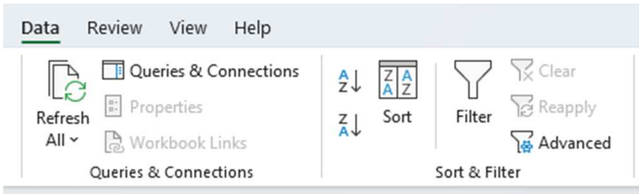
SORTING A LIST - Single Level Sorting

Region	Country	Department	Sales Period	Sales Amount
Europe	Germany	Clothing	Jan	£ 7,738
Europe	Germany	Home	Feb	£ 5,641
Europe	Germany	Accessories	Mar	£ 3,023
Europe	Germany	Electrical	Apr	£ 6,439
Europe	Germany	Clothing	May	£ 5,328
Europe	Germany	Home	Jun	£ 6,608
Europe	France	Accessories	Jan	£ 4,243
Europe	France	Electrical	Feb	£ 8,171
Europe	France	Clothing	Mar	£ 7,408
Europe	France	Home	Apr	£ 5,556
Europe	France	Accessories	May	£ 6,714
Europe	United Kingdom	Electrical	Jan	£ 8,065
Europe	United Kingdom	Clothing	Feb	£ 4,207
Europe	United Kingdom	Home	Mar	£ 6,937
Europe	United Kingdom	Accessories	Apr	£ 7,074
Asia	China	Home	Jan	£ 4,301
Asia				
Asia				

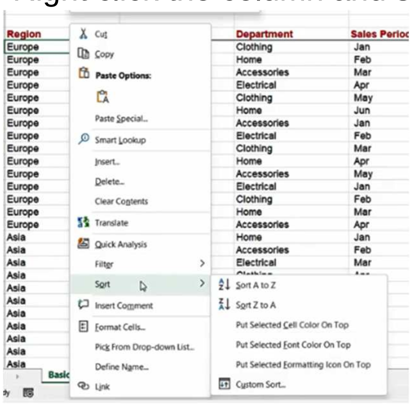
3



b. Data Ribbon➔Sort and Filter➔Sort

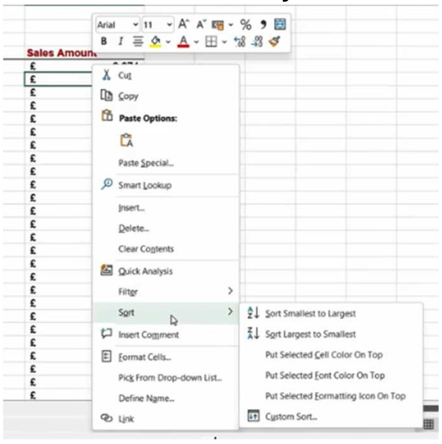


c. Right click the column and select Sort



4. Click Sort A➔Z.
5. You will have Asia then Europe then North America.
6. Now Sort A➔Z in the column of **Country**.
7. Remember the single sort can only sort by one column.
8. Notice that two sorts we have done were **Text Fields**.

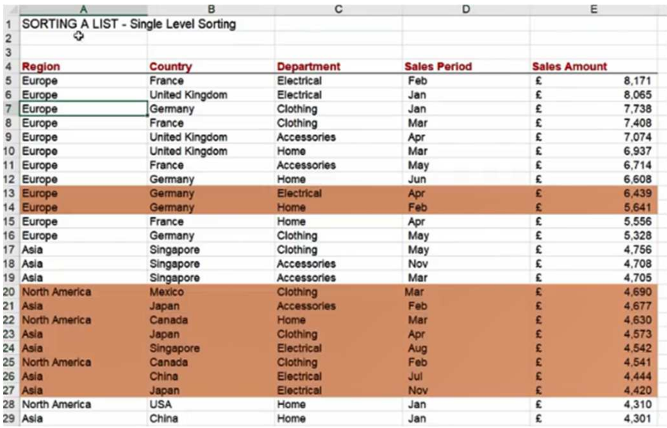
9. I want now to sort by Sales Amount (Numerical Value).



10. Notice here we have (Smallest to Largest) or (Largest to Smaller).
11. Sort Heights Sales First.

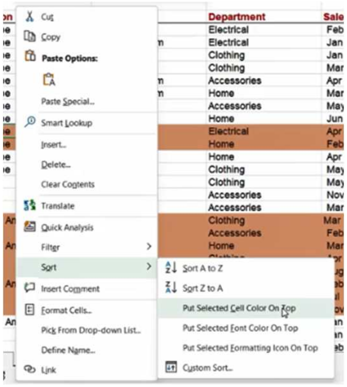
Sort Using Color

12. Go and shade some rows with a same color



	A	B	C	D	E	
1	SORTING A LIST - Single Level Sorting					
2						
3						
4	Region	Country	Department	Sales Period	Sales Amount	
5	Europe	France	Electrical	Feb	£	8,171
6	Europe	United Kingdom	Electrical	Jan	£	8,065
7	Europe	Germany	Clothing	Jan	£	7,738
8	Europe	France	Clothing	Mar	£	7,408
9	Europe	United Kingdom	Accessories	Apr	£	7,074
10	Europe	United Kingdom	Home	Mar	£	6,937
11	Europe	France	Accessories	May	£	6,714
12	Europe	Germany	Home	Jun	£	6,608
13	Europe	Germany	Electrical	Apr	£	6,439
14	Europe	Germany	Home	Feb	£	5,641
15	Europe	France	Home	Apr	£	5,556
16	Europe	Germany	Clothing	May	£	5,328
17	Asia	Singapore	Clothing	May	£	4,756
18	Asia	Singapore	Accessories	Nov	£	4,708
19	Asia	Singapore	Accessories	Mar	£	4,705
20	North America	Mexico	Clothing	Mar	£	4,690
21	Asia	Japan	Accessories	Feb	£	4,677
22	North America	Canada	Home	Mar	£	4,630
23	Asia	Japan	Clothing	Apr	£	4,573
24	Asia	Singapore	Electrical	Aug	£	4,542
25	North America	Canada	Clothing	Feb	£	4,541
26	Asia	China	Electrical	Jul	£	4,444
27	Asia	Japan	Electrical	Nov	£	4,420
28	North America	USA	Home	Jan	£	4,310
29	Asia	China	Home	Jan	£	4,301

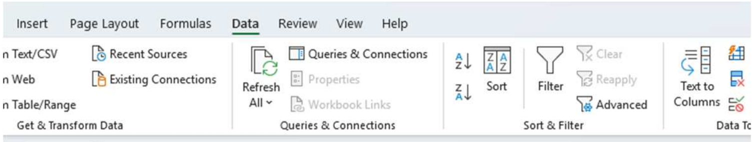
13. Click on one cell that has a color.
14. Right click and select➔ Put selected color on top.



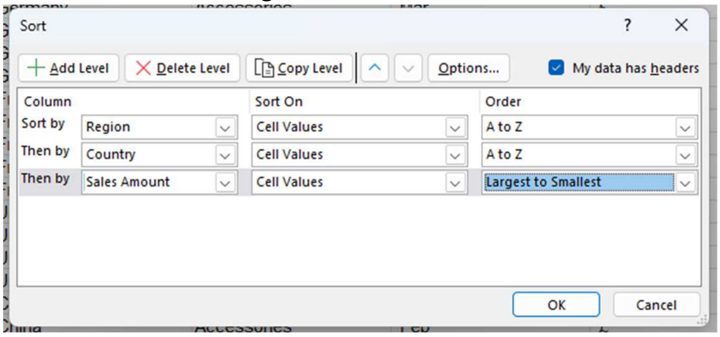
15. Put the color shade all to **No Color**.
16. Try now the same steps but with Front Color.
17. Right click and select **Put Selected Font on Top**.

Exercise 2 B: Multi-Level Sorting

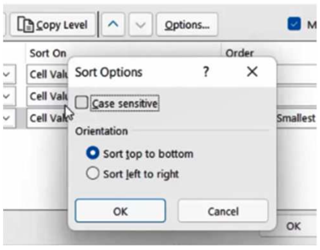
1. Suppose I want to sort the list in 3 levels: Region, Country and Sales.
2. Click anywhere in your list.
3. Data ➔Sort and Filter ➔Sort.



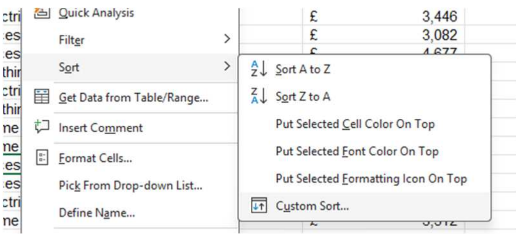
4. Make sure that check box **My Data has headers** is selected.
5. Add 3 levels of sorting:
   a. Region A➔Z.
   b. Country A➔Z.
   c. Sales Amount Largest to Smallest.



6. Check the options you have in that box:
   a. Delete level.
   b. Copy level.
   c. Up and Down Arrows.
   d. Options.



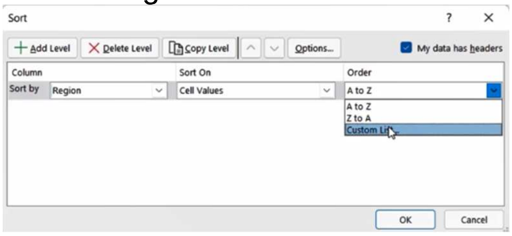
7. Click OK and see the sort result.
8. You can also find the dialogue box if you right click any cell in the list and choose Sort➔Custom Sort



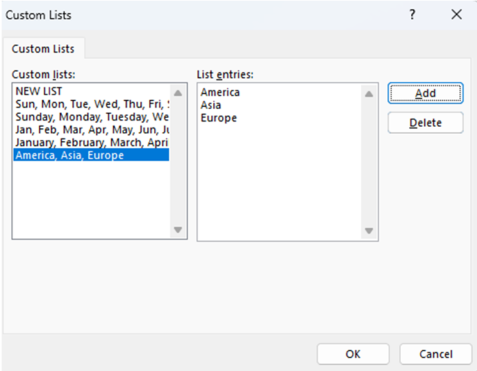
Exercise 02 C: Sort Using Custom List

1. Suppose I want to sort in the order: **North America, Asia, Europe**.

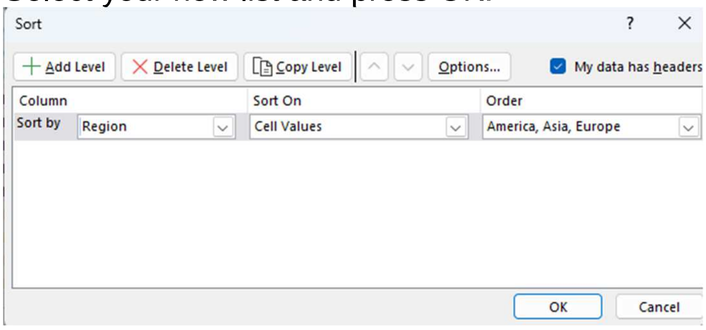
2. You can use your own list in this time.
3. Click anywhere in your data.
4. Open Sort Dialogue box.
5. Select Region and in order select **Custom List…**



6. The Custom List dialogue box appears.
7. Notice that we have many custom lists recorded in our Excel.
8. In Custom Lists select **NEW LIST**.
9. In List Entries type: North **America, Asia, Europe**.
10. Make sure that every item is separated by “, “(Comma then Space).



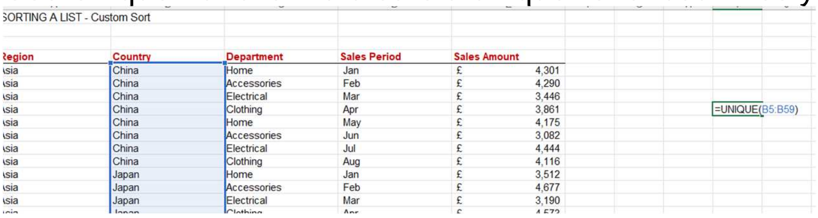
11. Click Add button to add the list.
12. Select your new list and press OK.



13. Click OK on the Sort Dialogue Box.
14. Now your data is sorted using your custom list.

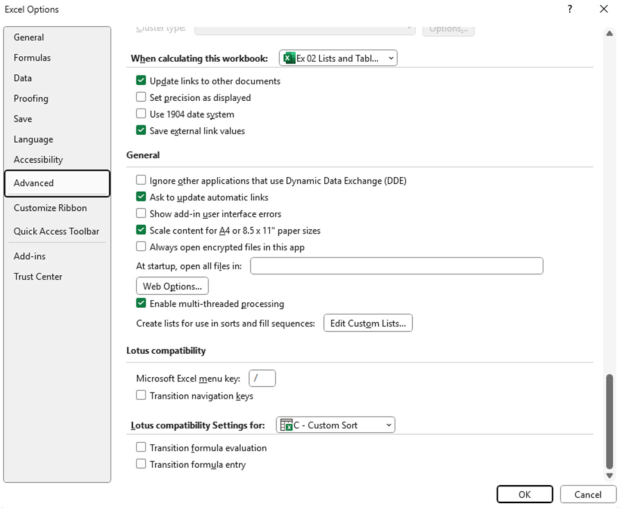
Create a Unique Custom List

15. If you want have Custom List for sorting Countries.
16. Use Unique Function to create a unique list of countries you have.

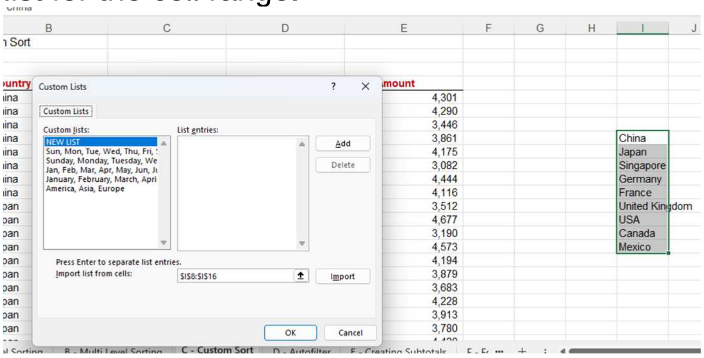


17. Select the created list **Copy** and the **Past Values** over the list with Dynamic Formula.

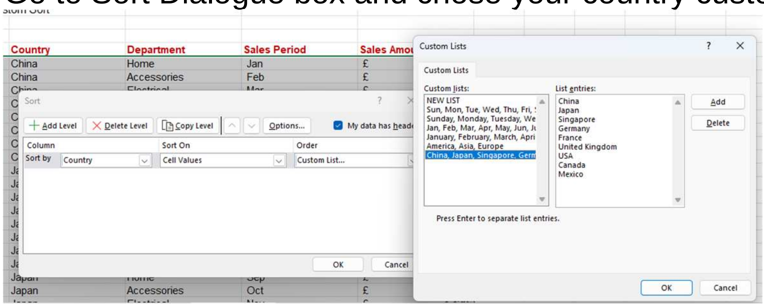
18. To add the list to your Excel.
19. File➔Options➔Advanced➔General➔Edit Custom List.



20. The dialogue box this time has import option you can use to get your list for the cell range.



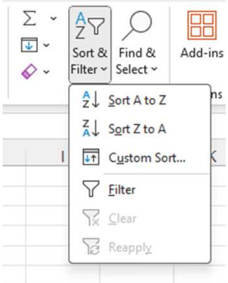
21. Select your country list.
22. Click Import then OK then OK again in the advanced Option window.
23. You can now delete your country cell range if you want to.
24. Go to Sort Dialogue box and chose your country custom list.



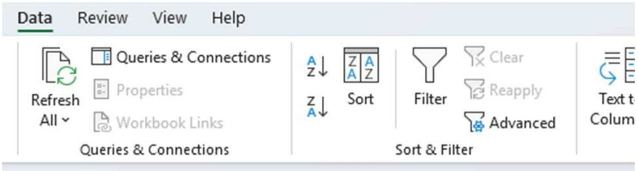
Exercise 02 D: Auto Filter

1. Select any cell in your data list.
2. You can reach **auto filter** in many ways:

a. Home ribbon ➔Sort and Filter➔Filter



b. Data ribbon ➔Sort & Filter➔Filter.

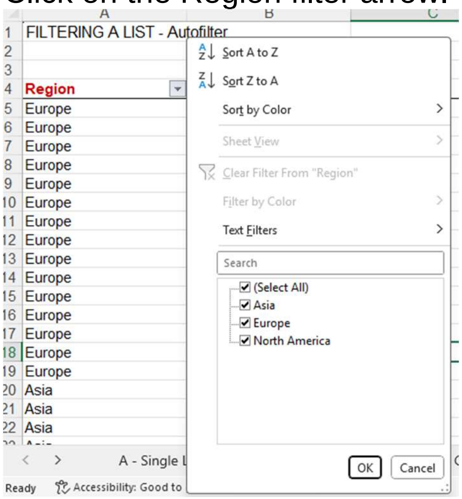


c. Use shortcut **Ctrl + Shift + L**

3. Notice you have drop down arrows in the top or your list.

	Region	Country	Department	Sales Period	Sales Amount
4	Region	Country	Department	Sales Period	Sales Amount
5	Europe	Germany	Clothing	Jan	£ 7,738
6	Europe	Germany	Home	Feb	£ 5,641
7	Europe	Germany	Accessories	Mar	£ 3,023
8	Europe	Germany	Electrical	Apr	£ 6,439
9	Europe	Germany	Clothing	May	£ 5,328

4. You can toggle auto filter on and off clicking on the Funnel icon.
5. Now you can filter using one column or many columns.
6. Click on the Region filter arrow.

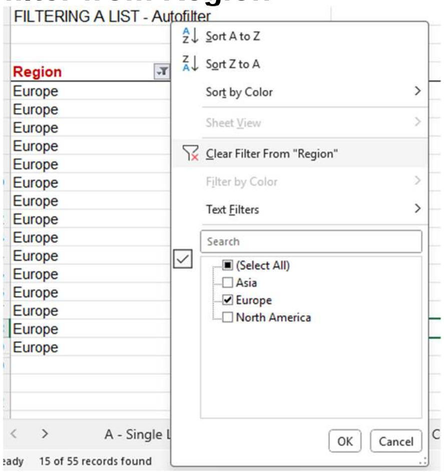


7. Notice you have **sort** option here.
8. Deselect anything but Europe from the list so you filter for only Europe.

	Region	Country	Department	Sales Period	Sales Amount
5	Europe	Germany	Clothing	Jan	£ 7,738
6	Europe	Germany	Home	Feb	£ 5,641
7	Europe	Germany	Accessories	Mar	£ 3,023
8	Europe	Germany	Electrical	Apr	£ 6,439
9	Europe	Germany	Clothing	May	£ 5,328
10	Europe	Germany	Home	Jun	£ 6,608
11	Europe	France	Accessories	Jan	£ 4,243
12	Europe	France	Electrical	Feb	£ 8,171
13	Europe	France	Clothing	Mar	£ 7,408
14	Europe	France	Home	Apr	£ 5,556
15	Europe	France	Accessories	May	£ 6,714
16	Europe	United Kingdom	Electrical	Jan	£ 8,065
17	Europe	United Kingdom	Clothing	Feb	£ 4,207
18	Europe	United Kingdom	Home	Mar	£ 6,937
19	Europe	United Kingdom	Accessories	Apr	£ 7,074

9. Notice that we have only **Europe** and a **funnel** instead of down arrow in the region column.
10. Also notice that the **Row Numbers has turned into blue**.

11. All these means I have filter on this column.
12. If you want to clear the filter, click on the small funnel then select **Clear filter from Region**.



13. You can also select “**Select All**” to clear the filter then press **OK**.

Filter Multiple columns

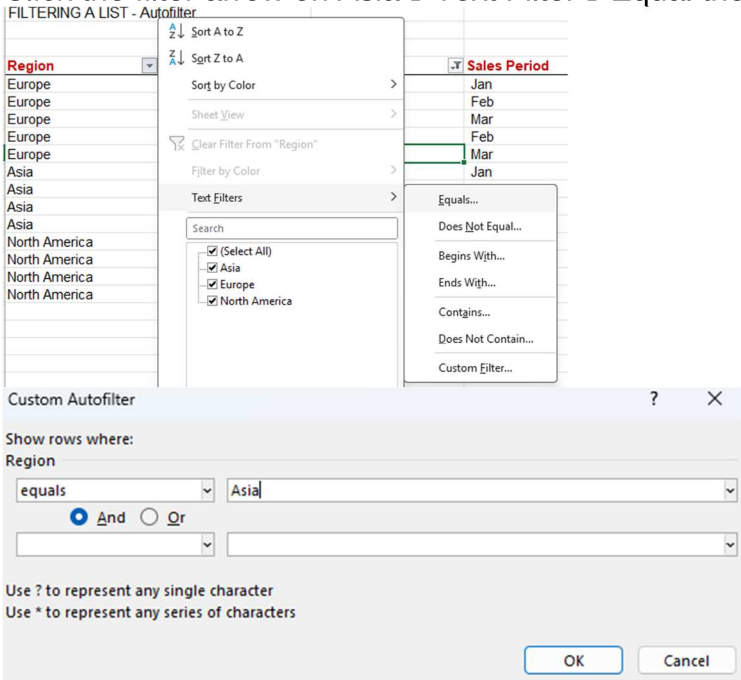
14. You can filter on many column try this:
    a. Filter department on: **Clothing** and **Home**.
    b. Filter Sales Period on: **Jan**, **Feb**, **Mar**.
15. You have now two filters applied to your data.

FILTERING A LIST - Autofilter

Region	Country	Department	Sales Period	Sales Amount	
Europe	Germany	Clothing	Jan	£	7,738
Europe	Germany	Home	Feb	£	5,641
Europe	France	Clothing	Mar	£	7,408
Europe	United Kingdom	Clothing	Feb	£	4,207
Europe	United Kingdom	Home	Mar	£	6,937
Asia	China	Home	Jan	£	4,301
Asia	Japan	Home	Jan	£	3,512
Asia	Singapore	Clothing	Jan	£	3,951
Asia	Singapore	Home	Feb	£	3,087
North America	USA	Home	Jan	£	4,310
North America	Canada	Clothing	Feb	£	4,541
North America	Canada	Home	Mar	£	4,630
North America	Mexico	Clothing	Mar	£	4,690

Text Filter

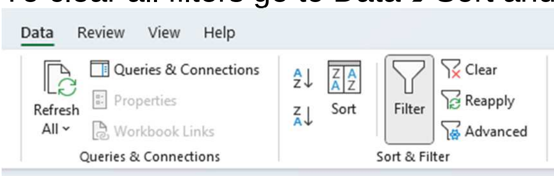
16. Click the filter arrow on Asia➔Text Filter➔Equal the write Asia.



17. Clear the filter from the region.

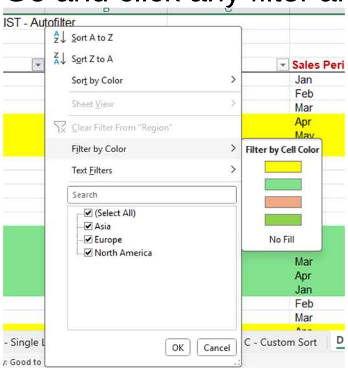
Number Filter

18. The same way filter on the Sales Amount Column to values > 7000 (Use Number Filter➔Greater Than…..)
19. To clear all filters go to Data➔Sort and Filter➔ Clear Button.



Filter in Colors

20. Change the back ground color of some rows to 3 different colors.
21. Go and click any filter arrow and chose Filter by Color

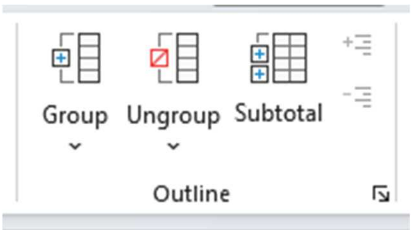


22. Choose the color you want to filter on.

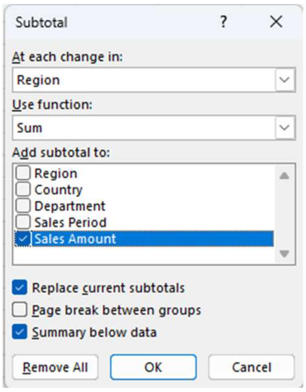
Region	Country	Department	Sales Period	Sales Amount	
Europe	United Kingdom	Electrical	Jan	£	8,065
Europe	United Kingdom	Clothing	Feb	£	4,207
Europe	United Kingdom	Home	Mar	£	6,937
Europe	United Kingdom	Accessories	Apr	£	7,074
Asia	China	Home	Jan	£	4,301
Asia	China	Accessories	Jun	£	3,082
Asia	China	Electrical	Jul	£	4,444
Asia	China	Clothing	Aug	£	4,116
Asia	Japan	Home	Jan	£	3,512
Asia	Japan	Accessories	Feb	£	4,677

Exercise 02 E: Subtotals in a List

1. I want to see subtotal after every change in the Range.
2. The easiest way is to use subtotal button in the Data ribbon.
3. Data➔Outline➔Subtotal



4. Select:
    a. In each change in: **Region**.
    b. Use Function: **Sum**.
    c. Add subtotal to: **Sales Amount**.
    d. Unselect **Page break between groups**.
5. Click OK.



C18

	A	B	C	D	E	
1	CREATING SUBTOTALS					
2						
3						
4	Region	Country	Department	Sales Period	Sales Amount	
5	Europe	Germany	Clothing	Jan	£	7,738
6	Europe	Germany	Home	Feb	£	5,641
7	Europe	Germany	Accessories	Mar	£	3,023
8	Europe	Germany	Electrical	Apr	£	6,439
9	Europe	Germany	Clothing	May	£	5,328
10	Europe	Germany	Home	Jun	£	6,608
11	Europe	France	Accessories	Jan	£	4,243
12	Europe	France	Electrical	Feb	£	8,171
13	Europe	France	Clothing	Mar	£	7,408
14	Europe	France	Home	Apr	£	5,556
15	Europe	France	Accessories	May	£	6,714
16	Europe	United Kingdom	Electrical	Jan	£	8,065
17	Europe	United Kingdom	Clothing	Feb	£	4,207
18	Europe	United Kingdom	Home	Mar	£	6,937
19	Europe	United Kingdom	Accessories	Apr	£	7,074
20	**Europe Total**				£	93,152
21	Asia	China	Home	Jan	£	4,301
22	Asia	China	Accessories	Feb	£	4,290
23	Asia	China	Electrical	Mar	£	3,446
24	Asia	China	Clothing	Apr	£	3,861
25	Asia	China	Home	May	£	4,175
26	Asia	China	Accessories	Jun	£	3,082

6. You will see subtotal rows for each region.
7. Also 3 vertical bars 1 – 2 – 3 on the left.
8. Click 1 ➔ Grand Total

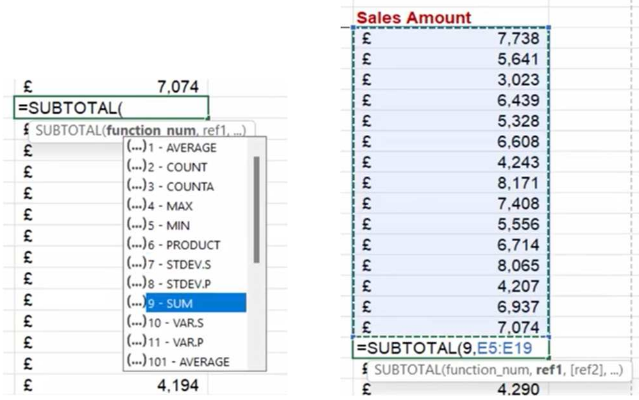
9. Click 2➔Subtotal for each region.
10. Click 3➔Expand all.
11. You can do it also clicking (+, -) signs.

Remove Subtotal

12. Click on the Subtotal button again and select **Remove All**.

Create Subtotals Using Formula

13. Insert a row below each region.
14. Write **Europe Subtotal** in 1st Cell then (CTRL+B) to make it bold.
15. In the Sales Amount cell enter the formula for Subtotal Function
16. Select 9.Sum for function Number to use

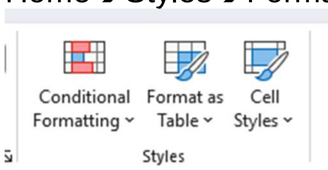


17. Repeat that for each region and for Grand Total
18. Make the line shaded color different.

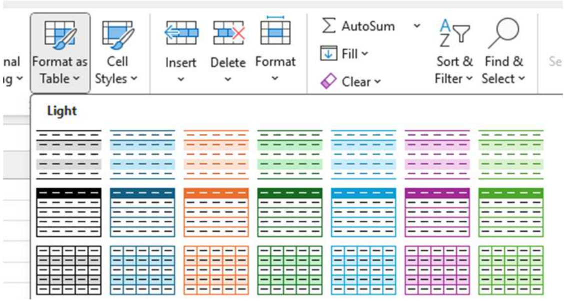
	A	B	C	D	E	
1	CREATING SUBTOTALS					
2						
3						
4	Region	Country	Department	Sales Period	Sales Amount	
5	Europe	Germany	Clothing	Jan	£	7,738
6	Europe	Germany	Home	Feb	£	5,641
7	Europe	Germany	Accessories	Mar	£	3,023
8	Europe	Germany	Electrical	Apr	£	6,439
9	Europe	Germany	Clothing	May	£	5,328
10	Europe	Germany	Home	Jun	£	6,608
11	Europe	France	Accessories	Jan	£	4,243
12	Europe	France	Electrical	Feb	£	8,171
13	Europe	France	Clothing	Mar	£	7,408
14	Europe	France	Home	Apr	£	5,556
15	Europe	France	Accessories	May	£	6,714
16	Europe	United Kingdom	Electrical	Jan	£	8,065
17	Europe	United Kingdom	Clothing	Feb	£	4,207
18	Europe	United Kingdom	Home	Mar	£	6,937
19	Europe	United Kingdom	Accessories	Apr	£	7,074
20	**Europ Subtotal**				**£**	**93,152**
21	Asia	China	Home	Jan	£	4,301

Exercise 02 F: Format as a Table

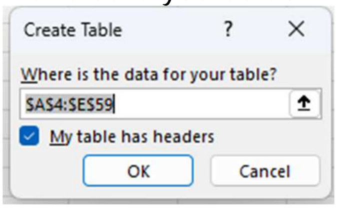
1. Let's start with the ribbon button.
2. Select any cell in the list.
3. Home➔Styles➔Format as a table.



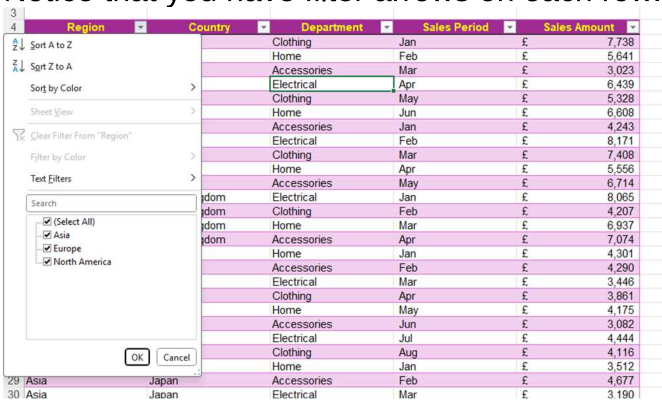
4. Select the style you want.



5. Make sure you select my table has headers.

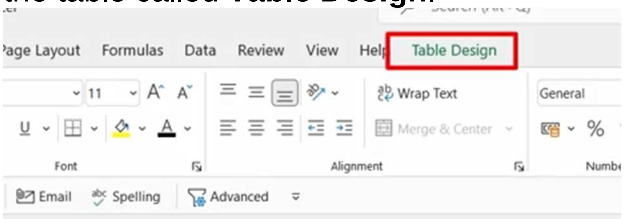


6. Click OK.
7. Notice that you have filter arrows on each row.



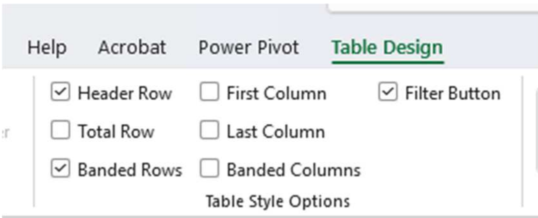
	Region	Country	Department	Sales Period	Sales Amount
5			Clothing	Jan	£ 7,738
6			Home	Feb	£ 5,641
7			Accessories	Mar	£ 3,023
8			Electrical	Apr	£ 6,439
9			Clothing	May	£ 5,328
10			Home	Jun	£ 6,608
11			Accessories	Jan	£ 4,243
12			Electrical	Feb	£ 8,171
13			Clothing	Mar	£ 7,408
14			Home	Apr	£ 5,556
15			Accessories	May	£ 6,714
16		gdom	Electrical	Jan	£ 8,065
17		gdom	Clothing	Feb	£ 4,207
18		gdom	Home	Mar	£ 6,937
19		gdom	Accessories	Apr	£ 7,074
20			Home	Jan	£ 4,301
21			Accessories	Feb	£ 4,290
22			Electrical	Mar	£ 3,446
23			Clothing	Apr	£ 3,861
24			Home	May	£ 4,175
25			Accessories	Jun	£ 3,082
26			Electrical	Jul	£ 4,444
27			Clothing	Aug	£ 4,116
28			Home	Jan	£ 3,512
29	Asia	Japan	Accessories	Feb	£ 4,677
30	Asia	Japan	Electrical	Mar	£ 3,190

8. Notice that you have a new *contextual ribbon* tab when you are inside the table called **Table Design**.



9. Click outside the table the tab disappears.
10. Undo the table using the Undo button (**CTRL+Z**).
11. Create the table again using the shortcut key (**CTRL+T**).
12. Notice that when you scroll down the table the column header stayed on top.

Table Style Options



   a. **Header Row**: Remove and add Header.
   b. **Total Row**: Add Total Row at the bottom of the table.

c. **Banded Rows**: make the even rows with different colors.
d. **Banded Columns**: Make even columns in different colors.
e. **Last Column**: Display different format for last column.
f. **First Column**: Display different format for first column.
g. **Filter Button**: show or hide filter button.

13. Create a meaningful name for your table.
14. Table Design➔Properties➔Table Name.
15. Name it : **Sales_Data** (There must be no space).

Expand Table

16. You can Expand the table adding new data in rows or add new columns.
17. Copy and paste the data from the Sheet: **Extra Data**.
18. Paste the data at the bottom of the table.
19. Use (**CTRL+A**), (**CTRL+C**),(**CTRL+** ↓), (**CTRL+V**).
20. Delete the copied header row.
21. The table is extended with the new data.
22. Create a new column Sales with Tax.
23. In the 1st cell of the column create a formula: Sales Amount * 1.2

Clipboard Font Alignment Number Styles

F5 =[@[Sales Amount]]*1.2

	A	B	C	D	E	F	G
	FORMAT AS A TABLE						
	Region	Country	Department	Sales Period	Sales Amount	Sales with Tax'	
	Europe	Germany	Clothing	Jan	£ 7,738	=[@[Sales Amount]]*1.2	
	Europe	Germany	Home	Feb	£ 5,641		

24. The whole table is propagated with the new formula.

FORMAT AS A TABLE

Region	Country	Department	Sales Period	Sales Amount	Sales with Tax'
Europe	Germany	Clothing	Jan	£ 7,738	£ 9,285.6
Europe	Germany	Home	Feb	£ 5,641	£ 6,769.2
Europe	Germany	Accessories	Mar	£ 3,023	£ 3,627.6
Europe	Germany	Electrical	Apr	£ 6,439	£ 7,726.8
Europe	Germany	Clothing	May	£ 5,328	£ 6,393.6
Europe	Germany	Home	Jun	£ 6,608	£ 7,929.6
Europe	France	Accessories	Jan	£ 4,243	£ 5,091.6
Europe	France	Electrical	Feb	£ 8,171	£ 9,805.2
Europe	France	Clothing	Mar	£ 7,408	£ 8,889.6
Europe	France	Home	Apr	£ 5,556	£ 6,667.2
Europe	France	Accessories	May	£ 6,714	£ 8,056.8
Europe	United Kingdom	Electrical	Jan	£ 8,065	£ 9,678.0
Europe	United Kingdom	Clothing	Feb	£ 4,207	£ 5,048.4
Europe	United Kingdom	Home	Mar	£ 6,937	£ 8,324.4
Europe	United Kingdom	Accessories	Apr	£ 7,074	£ 8,488.8
Asia	China	Home	Jan	£ 4,301	£ 5,161.2
Asia	China	Accessories	Feb	£ 4,290	£ 5,148.0

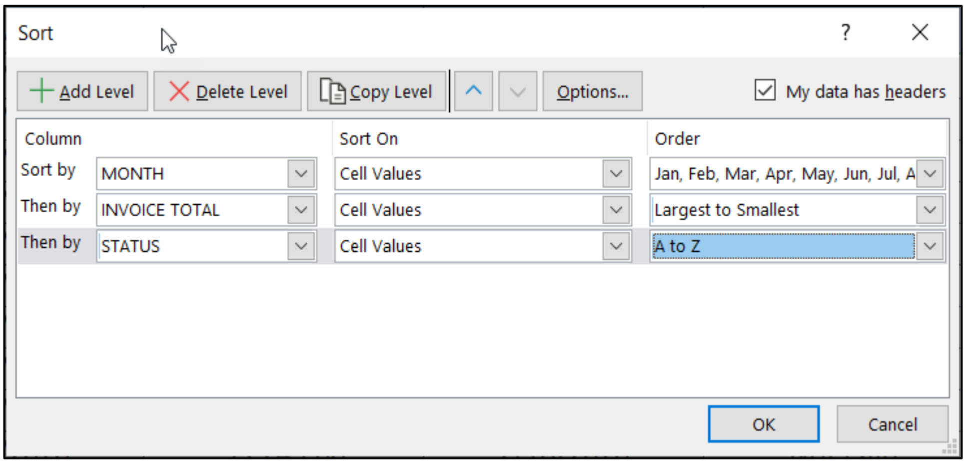
Exam 02: Lists and Tables

FORMAT AS A TABLE

1. Select the data on the worksheet 'Customer_Invoices'.
2. Format the data as a table (any table style).
3. Name the table, **'Invoices_2020'**

SORTING DATA

1. Sort the 'Invoices_2020' table data by **'Month'**, **'Invoice Total'** and **'Status'**.
2. Use a Custom List for the **'Month'** Order.



FILTERING DATA

1. Filter the data to show all invoices over **'$3000.00'** that have a status of **'Past Due'**.

MONTH	INVOICE NO.	DATE	INVOICE TOTAL	DUE DATE	STATUS
Feb	INV-00016	2/25/2020	$3,938.00	2/24/2020	Past Due
Apr	INV-00009	4/18/2020	$3,175.00	2/17/2020	Past Due
Sep	INV-00028	9/12/2020	$3,035.00	10/12/2020	Past Due

SUBTOTALS

1. Apply **Subtotals** to the data on the 'Customer_Invoices_Subtotal' worksheet.

2. At each change in **'Month'**, add the subtotal to the **'Invoice Total'**.

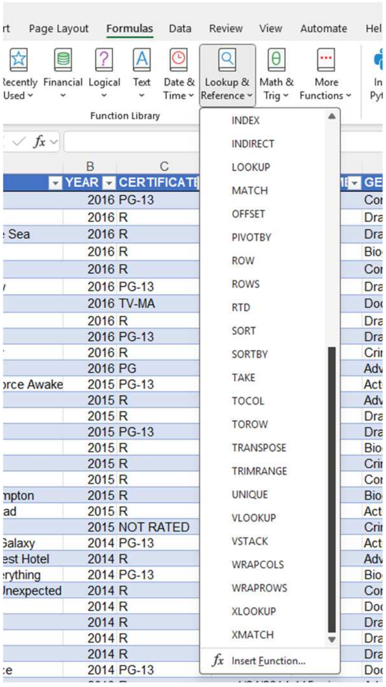
3. Collapse up the subtotals to show only the totals for each month.

MONTH	INVOICE NO.	DATE	INVOICE TOTAL	DUE DATE	STATUS
Jan Total			$16,320.00		
Feb Total			$11,568.00		
Mar Total			$6,780.00		
Apr Total			$9,133.00		
May Total			$9,564.00		
Jun Total			$4,536.00		
Jul Total			$9,384.00		
Aug Total			$5,607.00		
Sep Total			$10,085.00		
Oct Total			$11,616.00		
Nov Total			$7,350.00		
Dec Total			$10,656.00		
Grand Total			$112,599.00		

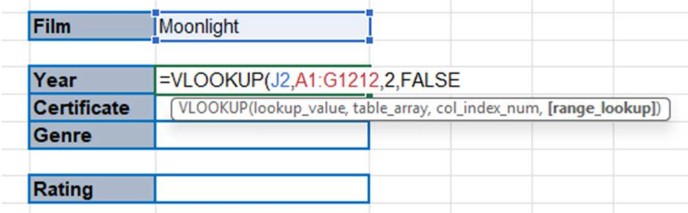
Chapter 03: Looking Up Information

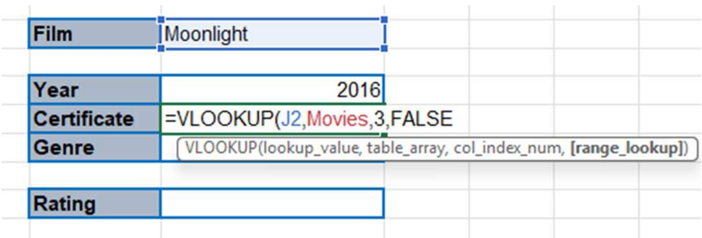
Exercise 03 A: VLOOKUP Function Using Exact matching

1. You can find VLOOKUP in the Formulas tab ribbon under Formula and references.

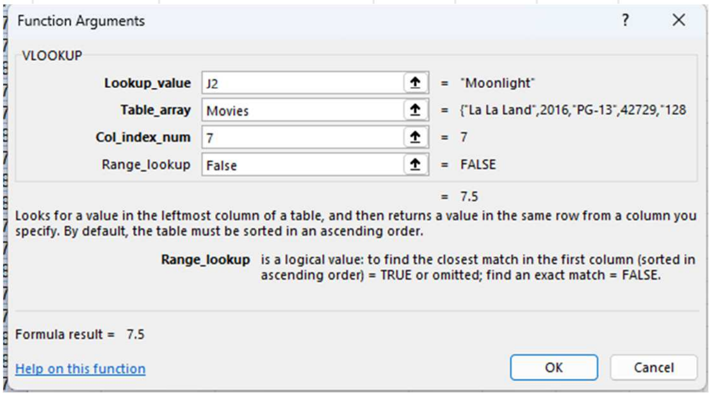


2. Now go and create **VLOOKUP** Function in the right table to get the information of the Film you typed in Film Cell.
3. Try to get Year using Range of Data.
4. Convert the List into table and name it movie to use in the Formula to get Certificate and Drama.
5. Try to Use the fx (Insert Function) to write your function.





Film	Moonlight
Year	2016
Certificate	R
Genre	=VLOOKUP(J2,Movies,6,False
	VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])
Rating	



6. Now try to write other movies to get its information

Film	Moonlight
Year	2016
Certificate	R
Genre	Drama
Rating	7.5

Film	La La Land
Year	2016
Certificate	PG-13
Genre	Drama
Rating	8.1

Exercise 03 B: VLOOKUP Approx. Match

1. I want to get the app for each user I have but that near the age in the table.

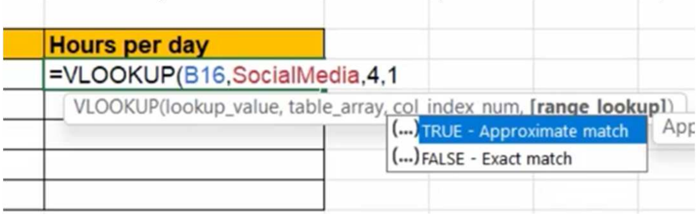
User	Age	App Preference	Hours per day
David Clark	56	=VLOOKUP(B16,$A$4:$D$12,3,TRUE)	
Jane March	22		
Lucy Bell	15		
Andrew Smith	37		
Brian Baker	78		

Low End	High End	App	Hours per day (Average)
0	10	GROM	1.5
11	20	TikTok	2
21	30	Instagram	3.5
31	40	Twitter	5
41	50	Twitter	4.75
51	60	Facebook	6
61	70	Facebook	1
71	Above	Facebook	0.3

User	Age	App Preference	Hours per day
David Clark	56	Facebook	
Jane March	22		
Lucy Bell	15		
Andrew Smith	37		
Brian Baker	78		

2. Do not forget to fix the table range using **F4**.
3. Fill down the formula for the rest of users.

4. Now let us use a **Named Range** for that List we have.
5. Name it **SocialMedia**.
6. Use it now to get the Hours per day column.
7. Press **F3** to get the Name of the Range you created while you create the Formula.
8. You can replace **TRUE** with **1** (o is for FALSE).

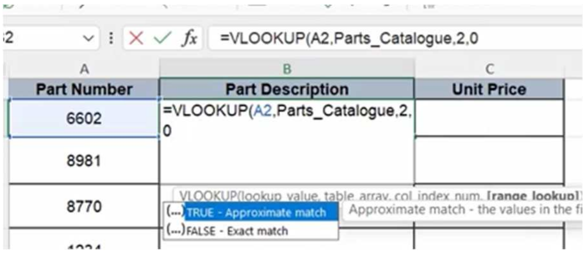


9. You have now full information.

User	Age	App Preference	Hours per day
David Clark	56	Facebook	6
Jane March	22	Instagram	3.5
Lucy Bell	15	TikTok	2
Andrew Smith	37	Twitter	5
Brian Baker	78	Facebook	0.3

Exercise 03 C: Error Handling

1. You have a table with **Part Number** and you want to get the description and unit price from the list in the sheet **Parts_Catalogue**.
2. We have Created **Named Range** for you **Parts_Catalogue** you can press **F3** to choose it while you are creating your formula.
3. Use **0** for the exact match.



4. Fill down the column you will get NA Error.

Part Number	Part Description	Unit Price
6602	Door Handles : Door Packs Modern : Urfic Door Pack Constance Satin Nickel	
8981	Door Handles : Lever On Backplate Modern : Bordeaux Lever Latch Satin Nickel	
8770	Door Handles : Mortice Sets Traditional : Jedo Fluted Door Knob Black Nickel 56mm	
1234	#N/A	
	Radiator Valves : Thermostatic Radiator Valves	

5. That means the part number is not found.

User IFNR Function

6. IFNR only deals with NA Error.
7. Handle the error and wrap the IFNR for VLOOKUP.

3	8981	Door Handles : Lever On Backplate Modern : Bordeaux Lever Latch Satin Nickel
4	8770	Door Handles : Mortice Sets Traditional : Jedo Fluted Door Knob Black Nickel 56mm
5	1234	Part Not Found
6	3500	Radiator Valves : Thermostatic Radiator Valves (TRVs) : Pegler Terrier II White & Chrome TRV
7	7506	Switches & Sockets : Crabtree Polished Brass LP : Crabtree 13A Sw FCU Polished Brass Flat Plate
8	6511	Part Not Found
	4275	Floor Paints : Standard Floor Paints : No

Use IFERROR Function

8. Create VLOOKUP for Unit Price column.

Unit Price
=VLOOKUP(A2,Parts_Catalogue,3,FALSE

VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])
(...) TRUE - Approximate match
(...) FALSE - Exact match

9. You also have NA errors.
10. Handle the error the same way But Use the General Function **IFERROR**

```
=IFERROR(VLOOKUP(A2,Parts_Catalogue,3,FALSE),"Price Not Found")
```

EXAM 03: Looking Up Information

LOOKING UP INFORMATION

1. On the '**Test_Scores**' worksheet, format the data as a table.
2. Name the table '**Test_Scores**'.
3. Create a **VLOOKUP** formula to lookup the Physics, Biology and Chemistry test scores for the student selected in cell M4.
4. Test that the formula is working by selecting a different student from the data validation drop-down list in cell M4.

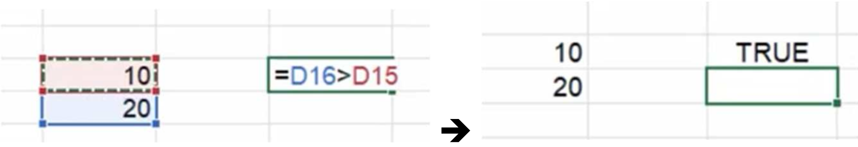
STUDENT	Mandy Chapman
PHYSICS	27
BIOLOGY	22
CHEMISTRY	67

1. On the '**Student_Grades' worksheet**', use **VLOOKUP** to find the Grade for each students average score for Science.
2. Copy the formula down for all students.
3. Notice any errors that appear in the cells.
4. Modify the formula so that instead of an error, the formula returns the text "Fail".

STUDENT	PHYSICS	BIOLOGY	CHEMISTRY	AVERAGE		GRADE
Alice Pierce	23	84	71	59		D
Dolores Parker	73	98	83	85		B
Mandy Chapman	27	22	67	39		Fail
Jeff James	45	48	69	54		E
Jerry Sparks	38	56	49	48		E
Alfonso Wagner	61	41	66	56		D
Clinton Greer	82	94	23	66		C

Chapter 04: Logical Functions

Logical Test



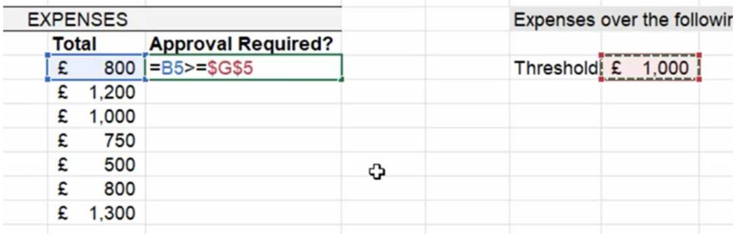
Condition	Operator
Equal to	=
Greater than	>
Less than	<
Greater than or equal to	>=
Less than or equal to	<=
Not equal to	<>

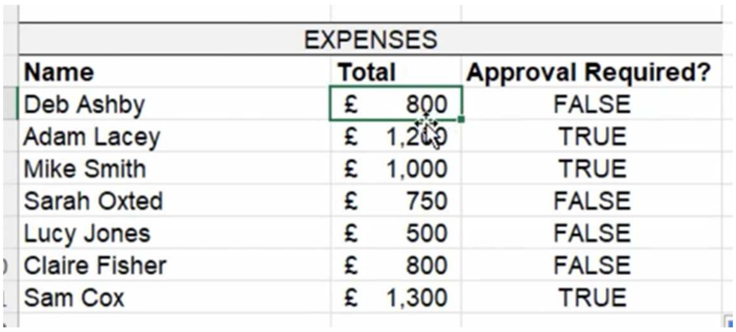
Logical Operators

Try those to get False or True.

Exercise 04 A: Basic Logical

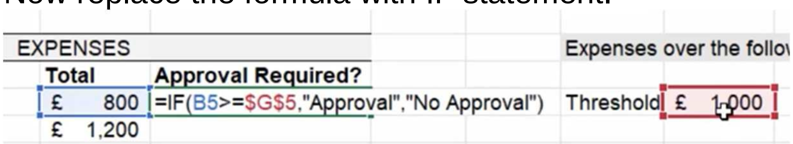
1. We want to approve only the ones > 1000.
2. Use **F4** to lock the value in Cell **G5**.





EXPENSES		
Name	Total	Approval Required?
Deb Ashby	£ 800	FALSE
Adam Lacey	£ 1,200	TRUE
Mike Smith	£ 1,000	TRUE
Sarah Oxted	£ 750	FALSE
Lucy Jones	£ 500	FALSE
Claire Fisher	£ 800	FALSE
Sam Cox	£ 1,300	TRUE

3. Now replace the formula with IF statement.



EXPENSES		
Name	**Total**	**Approval Required?**
Deb Ashby	£ 800	No Approval
Adam Lacey	£ 1,200	Approval
Mike Smith	£ 1,000	Approval
Sarah Oxted	£ 750	No Approval
Lucy Jones	£ 500	No Approval
Claire Fisher	£ 800	No Approval
Sam Cox	£ 1,300	Approval

Exercise 04 B: Examples on IF Statement

1. Try example 1 to test if the Student Pass or Not.

IF EXAMPLE 1

Name	Score	Result		Pass Mark	85
Deborah	93	=IF(B4>=$F$3,"Pass","Fail"			
Adam	65	IF(logical_test, [value_if_true], [value_if_false])			
Brooke	85				
James	79				

IF EXAMPLE 1

Name	Score	Result		Pass Mark	85
Deborah	93	Pass			
Adam	65	Fail			
Brooke	85	Pass			
James	79	Fail			
Rob	90	Pass			
Kata	60	Fail			

2. In Example 2 Use IF to make calculation

IF EXAMPLE 2

Product	Weight (kgs)	Price	Shipping Fee			
Office Chair	25	£ 500.00	=IF(B14>=$H$13,C14*$H$14,0		Weight	30
Oak Desk	50	£ 450.00	IF(logical_test, [value_if_true], [value_if_false])		Shipping Fee	20%
Storage Unit	30	£ 250.00				
Filing Cabinet	20	£ 175.00				
PC	5	£ 600.00				

IF EXAMPLE 2

Product	Weight (kgs)	Price	Shipping Fee			
Office Chair	25	£ 500.00	£ -		Weight	30
Oak Desk	50	£ 450.00	£ 90.00		Shipping Fee	20%
Storage Unit	30	£ 250.00	£ 50.00			
Filing Cabinet	20	£ 175.00	£ -			
PC	5	£ 600.00	£ -			

3. Try to Know How AND Function works in Example 3

AND

Name	Score 1	Score2	Result			
Claire	93	80	=AND(B23>=$H$22,C23>=$H$23)		Test 1	75
Julie	65	91			Test 2	65
Max	50	72				
Ben	78	93				
Courtney	38	30				

AND

Name	Score 1	Score2	Result			
Claire	93	80	TRUE		Test 1	75
Julie	65	91	FALSE		Test 2	65
Max	50	72	FALSE			
Ben	78	93	TRUE			
Courtney	38	30	FALSE			

4. Now try to make the result meaningful using IF with AND

```
=IF(AND(B23>=$H$22,C23>=$H$23),"Pass","Fail")
```

AND

Name	Score 1	Score2	Result
Claire	93	80	Pass
Julie	65	91	Fail
Max	50	72	Fail
Ben	78	93	Pass
Courtney	38	30	Fail

Test 1	75
Test 2	65

5. In Example 4 Find out How OR function works and combine with IF to get the result.

OR

Name	Score 1	Score2	Result
Claire	93	80	=IF(OR(B32>=$H$31,C32>=$H$32),"Pass","Fail")
Julie	65	91	
Max	50	72	
Ben	78	93	
Courtney	38	30	

Test 1	75
Test 2	65

OR

Name	Score 1	Score2	Result
Claire	93	80	Pass
Julie	65	91	Pass
Max	50	72	Pass
Ben	78	93	Pass
Courtney	38	30	Fail

Test 1	75
Test 2	65

Exercise 04 C: IF Statement

Use IF to perform logical test to calculate Shipping for orders over 1500.

Total Cost	Shipping	Total
£ 1,449.75	=IF(F4>$K$3,F4*$K$4,0	
£ 659.90	IF(logical_test, [value_if_true], **[value_if_false]**)	
£ 454.65		

For orders over:	1500
Shipping Charge (% of cost)	2%

Total Cost	Shipping	Total
£ 1,449.75	£ -	=SUM(F4+G4
£ 659.90	£ -	SUM(**number1**, [number2], ...)

Shipping	Total
£ -	£ 1,449.75
£ -	£ 659.90
£ -	£ 454.65
£ -	£ 419.85
£ -	£ 779.40
£ 52.00	£ 2,651.96
£ -	£ 649.50
£ -	£ 1,299.98
£ 39.59	£ 2,019.29

EXAM 04: Logical Functions

LOGICAL DECISIONS

1. On the **'Christmas_Party'** worksheet, use a formula to calculate which venue meets all required conditions.

2. In column G, calculate which venues can accommodate 400 or more attendees. Output the text "**Yes**" for a true result and "**No**" for a false result.

3. In column H, calculate which venues can accommodate 400 or more attendees **AND** are $4000 or less. Output the text "**Yes**" for a true result and "**No**" for a false result.

4. The last logical test will produce one venue that matches both conditions. Input the name of the venue to cell G22 by linking the cell.

VENUE	CAPACITY	AVAILABILITY	PRICE
The White Rooms	100	Available	$2,000
Fitzrovia House	500	Available	$10,000
The Belgrade	150	Available	$1,000
Waldorf Function Rooms	250	Available	$1,000
One Hotel	400	Available	$8,000
The Madison Lounge	220	Not Available	$3,000
Anderson Centre	650	Available	$9,000
The Marquee	700	Available	$7,000
Summertime House	400	Available	$3,000
Cranleigh Gardens	200	Available	$1,000
White Leaf Function Rooms	120	Available	$1,500
The Bell and Hound	50	Not Available	$750
Trident Place	75	Not Available	$800
Oak View Gardens	300	Available	$1,200
The Sky Garden	200	Available	$2,000
One Viewpoint Tower	150	Available	$1,500

Chapter 5: Date Functions

Exercise 05 A: Date Functions

Enter a dynamic Updated Date and Time

1. In the up left table, we want to enter today date.
2. Use **TODAY()** Function (It uses no arguments).

	A	B
	Date:	=TODAY()
	Date and Time:	

3. The Date is Dynamic and will always be updated when you open your worksheet.
4. If we want to enter date and time use **Now()** Function.

	A	B
1	**Date:**	4/28/2025
2	**Date and Time:**	4/28/2025 12:09

Enter Hard Coded Date and Time

5. If you want to put the date with no Update, use the shortcut (**CTRL+;**).
6. This will enter the date of today and never update.
7. To enter the date and time : (**CTRL+;**), Space, (**CTRL+SHIFT+;**).

WORKDAY Function

8. If you have table of your tasks and its start date and the working days of each and you want to know when the Finish date is excluding the weekends.

Task Name	Start Date	Working Days	Finish Date
Research	01/01/2020	5	
Book Venue	07/01/2020	1	
Order Wedding Dress	09/01/2020	3	
Order Bridesmaids Dress	11/01/2020	3	
Book Caterer	13/01/2020	2	

9. You can use the WORKDAY function
10. It takes two argumenta, the start date and the working days

Start Date	Working Days	Finish Date
01/01/2020	5	=WORKDAY(B6,C6,
07/01/2020	1	WORKDAY(start_date, days, [holidays])
09/01/2020	3	

11. Optionally you can add range of Holidays you want to exclude.

B	C	D
10/12/2021		
10/12/2021 15:19		
Start Date	**Working Days**	**Finish Date**
01/01/2020	5	=WORKDAY(B6,C6,$G$5:$G$14
07/01/2020	1	WORKDAY(start_date, days, [holidays])
09/01/2020	3	
11/01/2020	3	
13/01/2020	2	
17/01/2020	2	
20/01/2020	1	
23/01/2020	1	
25/01/2020	4	
27/01/2020	8	
28/01/2020	4	

G	
01/01/2020	New Y
02/04/2020	Good I
05/04/2020	Easter
03/05/2020	May D
31/05/2020	Spring
30/08/2020	Summ
25/12/2020	Christr
26/12/2020	Boxing
27/12/2020	Christr
28/12/2020	Boxing

NETWORKDAYS Function

12. You can calculate the working days.

Start Date	End Date	No of Working Days
01/01/2020	03/04/2020	=NETWORKDAYS(B20,C20,$G$5:$G$14
07/01/2020	14/01/2020	NETWORKDAYS(start_date, end_date, [holidays])
09/01/2020	13/01/2020	

Use WORKDAY.INT and NETWORKDAYS.NTL

13. You can use those function to specify which days is your weekend

=WORKDAY.INTL(B6,C6,7,$G$5:$G$14

WORKDAY.INTL(start_date, days, [weekend], [holidays])

14. From the list select the number corresponding to your country weekend.

For example, 7 for Fri and Sat.

Exercise 05 B: More Date Formula

1. In the table you must fill the columns using date functions to get the result from the **date** column.

Date and Time Formulas

Date	Invoice Total	DayNum	DayName	MonthNum	MonthName	Year	Weekday	Is Weekend?
10/12/2020	£ 5,095							
15/01/2020	£ 5,484							

2. Use function **DAY**

Date	Invoice Total	DayNum
10/12/2020	£ 5,095	=DAY(A4
15/01/2020	£ 5,484	DAY(serial_number)

Date	Invoice Total	DayNum
10/12/2020	£ 5,095	10
15/01/2020	£ 5,484	15
18/12/2020	£ 9,134	18
20/07/2020	£ 6,162	20

3. Use **TEXT** Function to get the day

Date	Invoice Total	DayNum	DayName
10/12/2020	£ 5,095	10	=TEXT(A4,"ddd"
15/01/2020	£ 5,484	15	TEXT(value, format_text)

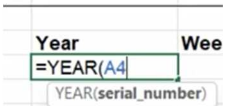
Date	Invoice Total	DayNum	DayName
10/12/2020	£ 5,095	10	Thu
15/01/2020	£ 5,484	15	

4. ddd → Mon …. dddd →Monday
5. Get Month Number using **MONTH** Function

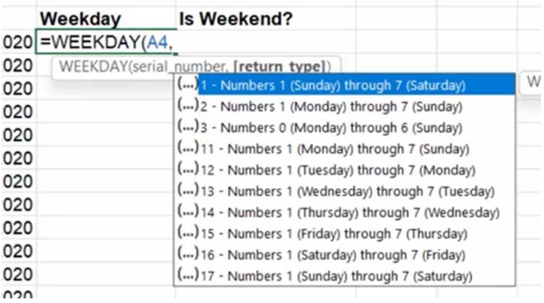
Date	Invoice Total	DayNum	DayName	MonthNum
10/12/2020	£ 5,095	10	Thu	=MONTH(A4
15/01/2020	£ 5,484	15	Wed	MONTH(serial_number)

6. Use **TEXT** Function to get the Month Name.

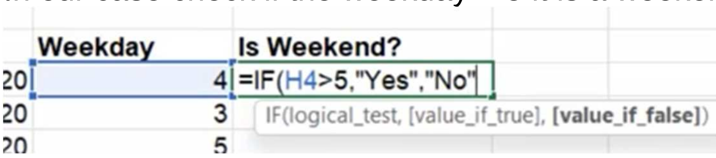
7. MMM➔ Jan MMMM➔January.
8. Use **YEAR** function to get the year.



9. Use a **WEEKDAY** Function to get the day of the week.



10. Choose from the list which one is Day Number 1.
11. For example, I would choose 2 for (Mon to Sun).
12. As the first day of the week is Monday.
13. Use the resulting number to decide if it was a **weekend?**
14. In our case check if the weekday > 5 it is a weekend.



Exercise for you:

15. Try again with the first day of the week, Saturday.

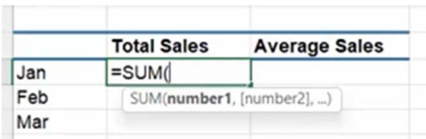
Chapter 6 Hyperlinks and 3D Reference

Exercise 6 A: 3D Reference

1. If you have sheets that have **EXACTLY** the same structure in your workbook, you can use the 3D reference to facilitate Calculations.
2. In this workbook we have as you can see:

	Sales	
Jan	£	4,512.00
Feb	£	9,300.00
Mar	£	7,908.00
Apr	£	6,275.00
May	£	4,067.00
Jun	£	6,350.00
Jul	£	4,336.00
Aug	£	3,699.00
Sep	£	7,732.00
Oct	£	8,541.00
Nov	£	9,507.00
Dec	£	7,691.00
Total	**£**	**79,918.00**

3. 4 worksheets for **UK, USA, Japan, China** with the same table structure.
4. We want to sum all data in the **Total Sales** sheet.
5. Open the **Total Sales** sheet and calculate the total sales.
6. I want total sales for **January** across all countries.
7. Create SUM Calculation.



	Total Sales	Average Sales
Jan	=SUM(	
Feb		
Mar		

8. Click on **UK** worksheet.
9. Notice that the formula has added the UK to your calculation.

	A	B	C	D
3		Sales		
4	Jan	£ 4,512.00		
5	Feb	[illegible]		
6	Mar	£ 7,908.00		
7	Apr	£ 6,275.00		
8	May	£ 4,067.00		
9	Jun	£ 6,350.00		
10	Jul	£ 4,336.00		
11	Aug	£ 3,699.00		
12	Sep	£ 7,732.00		
13	Oct	£ 8,541.00		
14	Nov	£ 9,507.00		
15	Dec	£ 7,691.00		
16	Total	£ 79,918.00		

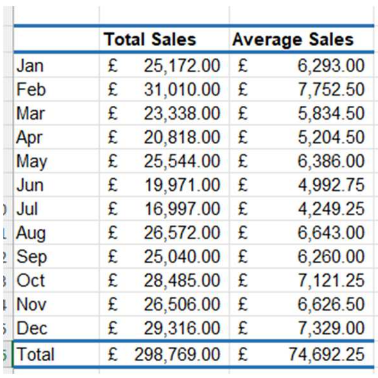
10. Now select the value for Jan (B4).

11. Hold **SHIFT** Key and Select **China** Worksheet.
12. That would select all the worksheets of countries.

13. Notice that the function has changed to say select B4 for all worksheets from UK to China.

14. Press Enter to complete the calculation.
15. You can fill the rest of the columns cell even for **Total** cell.

16. Try again with AVERAGE Function.



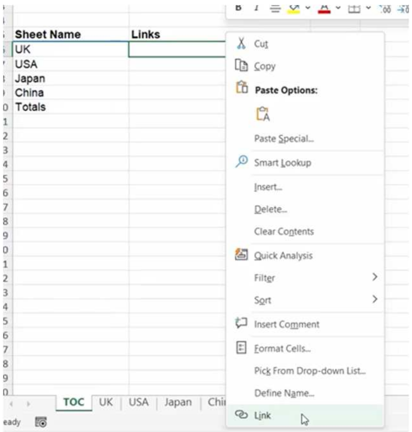
	Total Sales	Average Sales
Jan	£ 25,172.00	£ 6,293.00
Feb	£ 31,010.00	£ 7,752.50
Mar	£ 23,338.00	£ 5,834.50
Apr	£ 20,818.00	£ 5,204.50
May	£ 25,544.00	£ 6,386.00
Jun	£ 19,971.00	£ 4,992.75
Jul	£ 16,997.00	£ 4,249.25
Aug	£ 26,572.00	£ 6,643.00
Sep	£ 25,040.00	£ 6,260.00
Oct	£ 28,485.00	£ 7,121.25
Nov	£ 26,506.00	£ 6,626.50
Dec	£ 29,316.00	£ 7,329.00
Total	£ 298,769.00	£ 74,692.25

Exercise 06 B: Inserting Hyperlinks to Worksheets

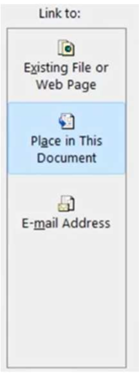
1. Create a new worksheet TOC before the countries worksheets like the one in figure.

2. We want to make it simple for anyone to navigate through.
3. We will create hyperlinks for every worksheet.
4. Right click the cell **B6** and select Link.

5. Or select the link from Inse➔Links➔Link

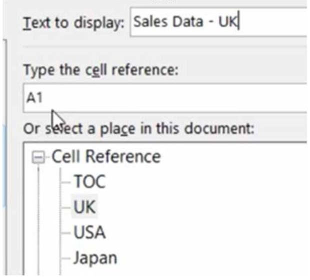


6. You can also sue the key shortcut (**CTRL+K**).
7. Notice that you can insert link for this workbook or other workbooks or files or mail.

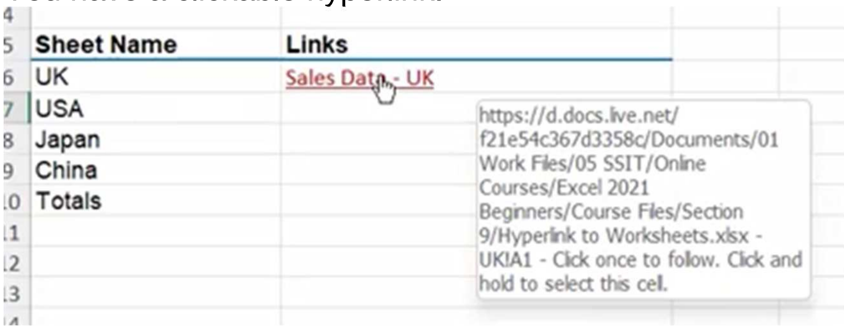


8. Select Place in this document.

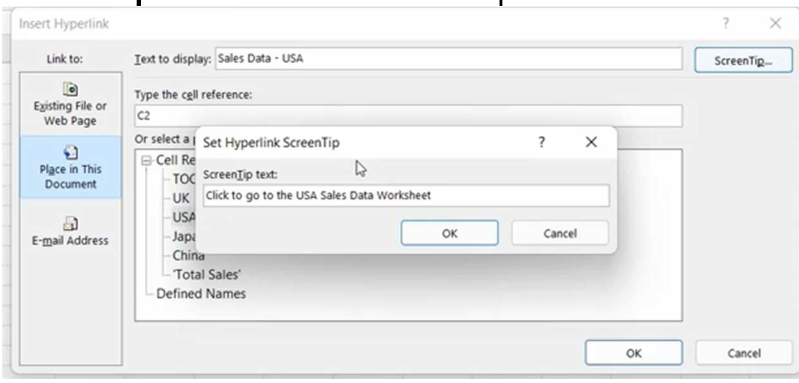
9. Create a hyperlink for UK Sales.



10. You have a clickable hyperlink.



11. Click to test.
12. Create a Hyperlink to USA Sales make the cell reference C2 and click **screen tip** button to create a tool tip to the link when user hover.



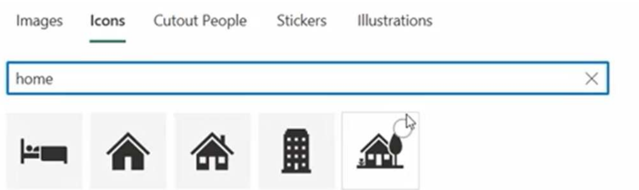
13. Notice the tooltip when you hover over the link.

14. Click on the link and notice that your curser is on **C2** of USA worksheet.
15. Complete all other links to other pages and test them.

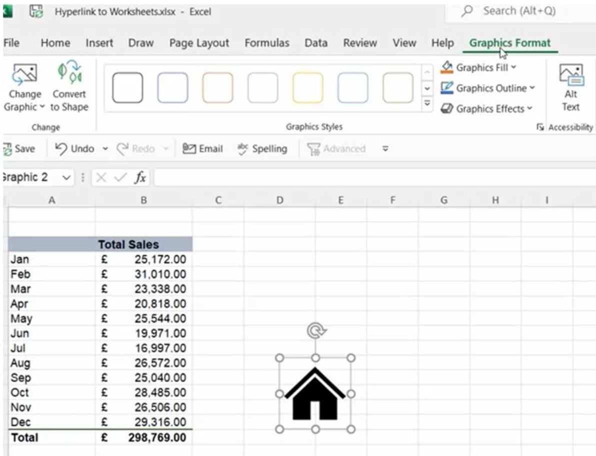
Using Image as Hyperlink

16. You can use pictures, Shape or icon as a hyperlink.
17. I want to put an icon that get the user back to **TOC** worksheet.
18. Insert➔illustrations➔Icons.

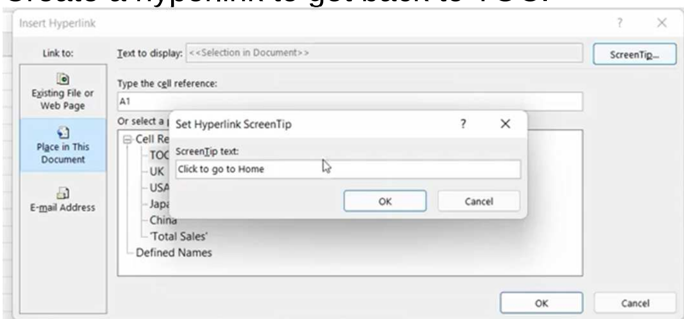
19. Search with **Home** word.



20. Insert the icon and notice you have a new tab in the ribbon **Graphic Format**.



21. Change the icon color to light blue.
22. Select the icon then press CTRL+K.
23. Create a hyperlink to get back to TOC.



24. Copy the icon to all other worksheets.

Exam 5: 3D References

3D REFERENCING

1. Use 3D Referencing to complete the summary table on the **'Total_Sales'** worksheet.

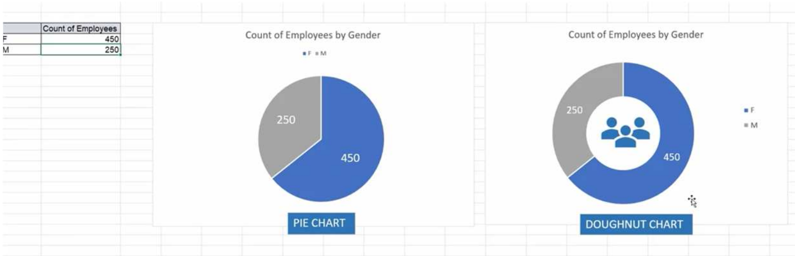
TOTAL SALES (JAN - APR 2019)				
	North	South	East	West
Jan	$48,279	$50,759	$47,063	$45,893
Feb	$40,173	$36,003	$44,219	$49,253
Mar	$46,637	$45,988	$48,998	$35,176
Apr	$51,834	$45,274	$54,787	$52,495

Chapter 7: Charts

Demo 7: Choosing the right chart

1. Charts are a great way to represent Data.
2. You must decide which type of chart suits the data you want to analyze.

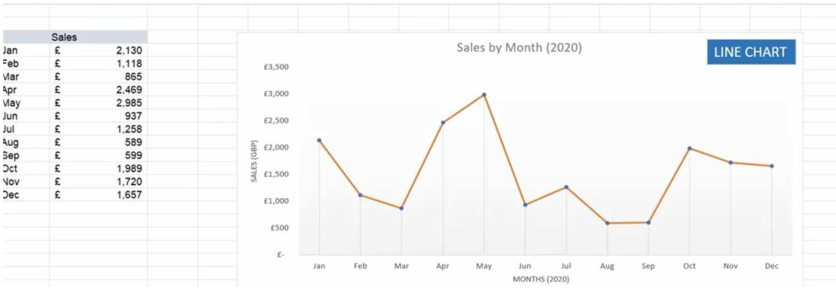
Pie and Donut Charts



3. Good if you have small data segments.
4. Not suitable with more segments.

Line Chart

5. Suitable to represent trends and changes in values over time.
6. It works best for time series data.

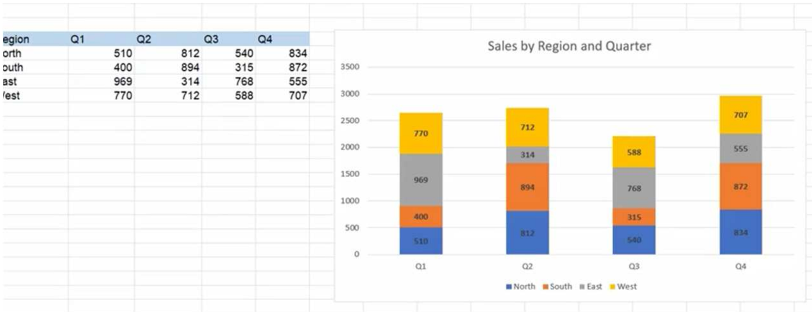


Column and Bar Charts

7. Suitable and clear for most data types.

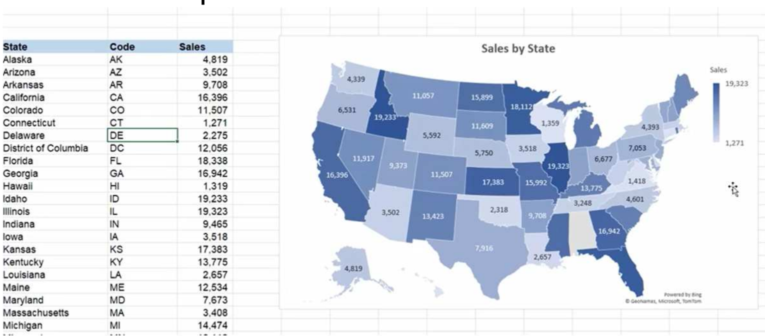
Stacked column Chart

8. Part of column chart family.
9. Good for comparing values.
10. Good for data with 2 dimensions.

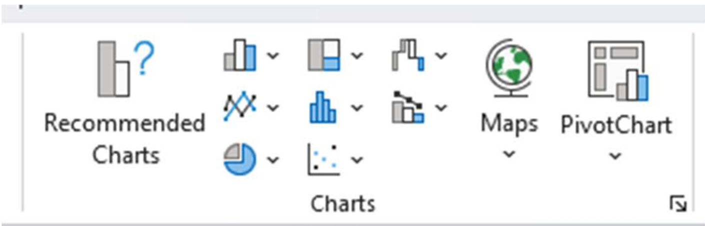


Field Map Chart

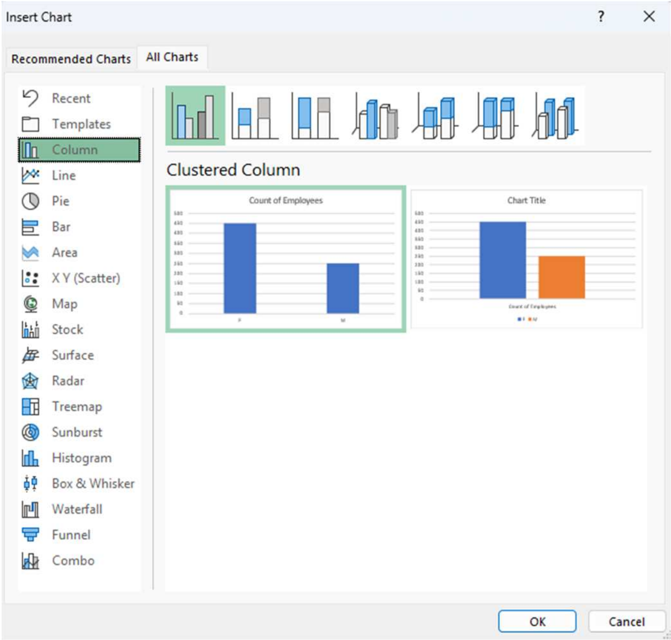
11. Effective way to display geographical data.
12. The shade depends on the values.



13. The darker the shading the higher the sales.
14. You can find many chart types available in Excel.
15. Insert➔Charts



16. Click on the small arrow in bottom right to get all the charts by categories.



Exercise 07 A Creating Charts

1. In this exercise we will create charts.

Insert a Chart into Your Worksheet

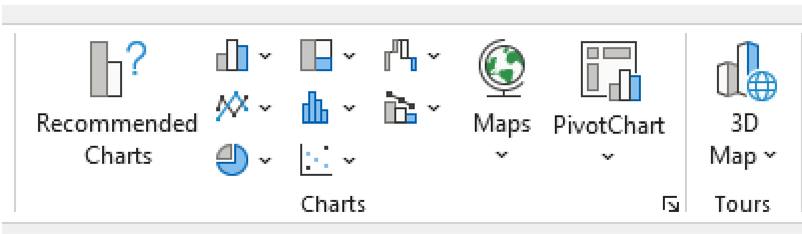
2. In worksheet “**Chart Basics**” You have a “monthly sales per item” data set and we want to visualize.

Chart Basics

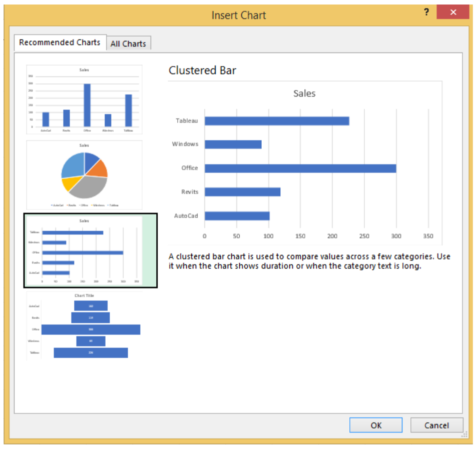
Monthly Sales per Items

Item	**Sales**
AutoCad	102
Revits	119
Office	300
Windows	89
Tableau	226

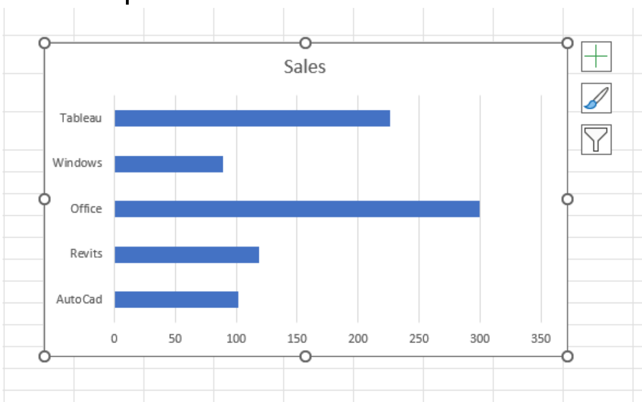
3. You can Go to Insert ➔Charts➔Chose any chart you want.
4. Highlight the data set or even select one cell inside.
5. If you are not which chart to pick, Click on **Recommended Charts.**
6. Insert➔Charts➔Recommended Chart.



7. Excel Gives you some Proposals.



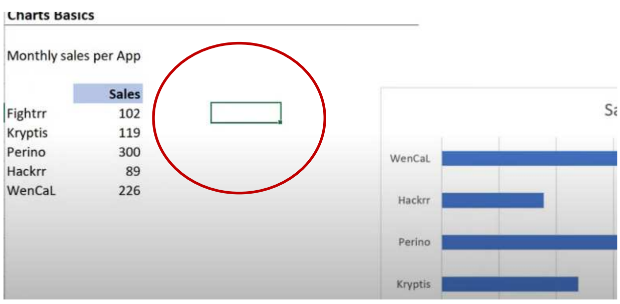
8. Select Bar Chart and press OK.



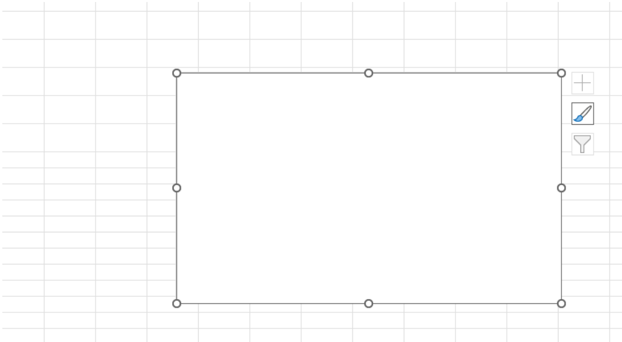
9. Bar chart appears on your worksheet.
10. That is because I have highlighted the data before.

What if you did not highlight the data set?

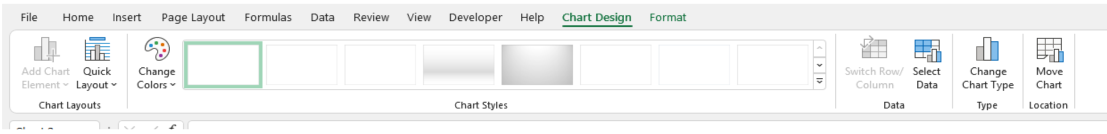
11. Select any empty cell outside the range of your data.



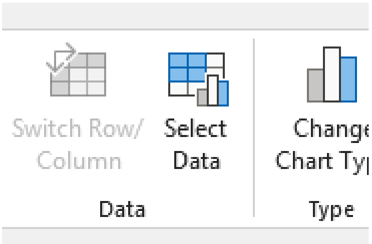
12. Insert➔Charts➔Clustered Column Chart.



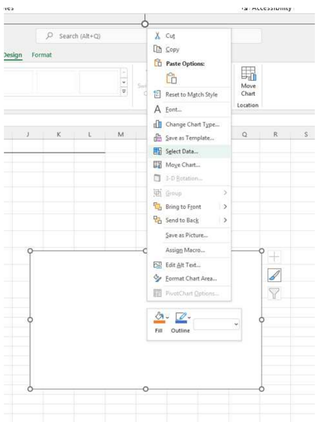
13. You get an **empty Canvas**.
14. Notice you have no data in this chart.
15. When you insert chart, notice that you have 2 new taps **Chart Design** and **Format** in the ribbon.



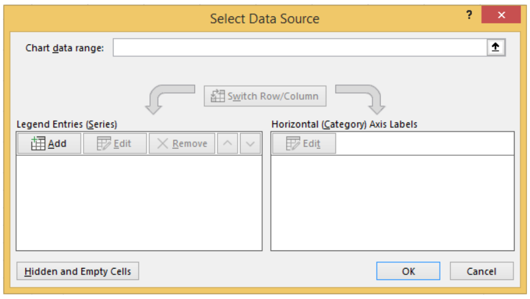
16. You can add Data to your chart.
17. Chart Design➔Select Data.



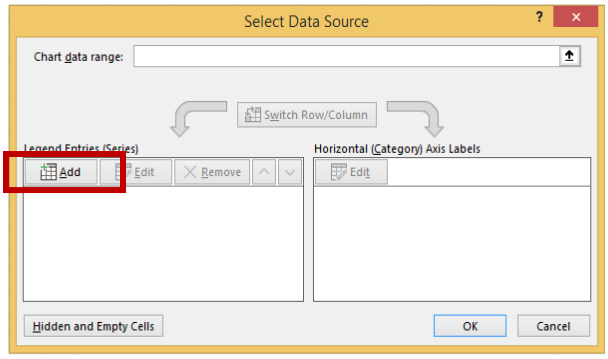
18. Or right click the empty chart and choose "**select Data**".



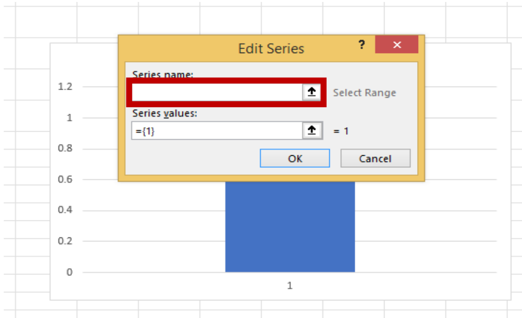
19. As you can see, In Excel you have many ways to get to the same place.
20. You might think there are a lot of options, but in fact they are just repeated.
21. **Select Data source** dialogue box appears.
22. Choose **Add** Button



23. You can add different elements manually.

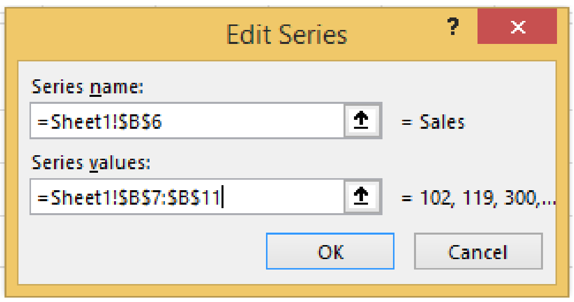


24. Add the item names (Series Names)

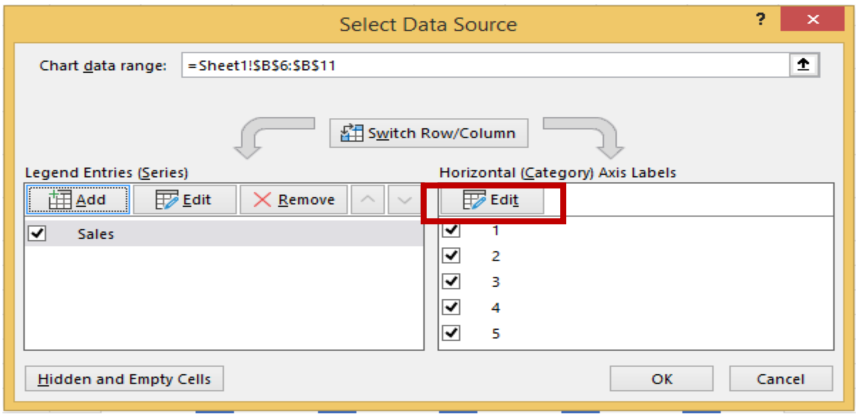


25. Select Cell (=Sheet1!$B$6) Sales.
26. Add Series Values.
27. In Values delete what in text box first then select the value Range (=Sheet1!$B$7:$B$11).
28.

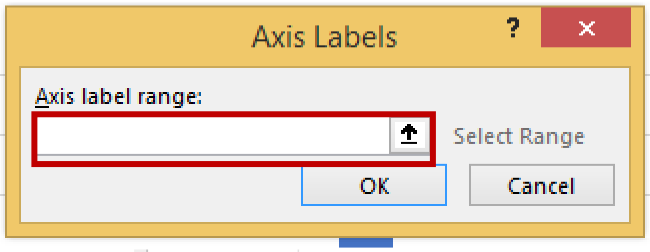
Sales
102
119
300
89
226

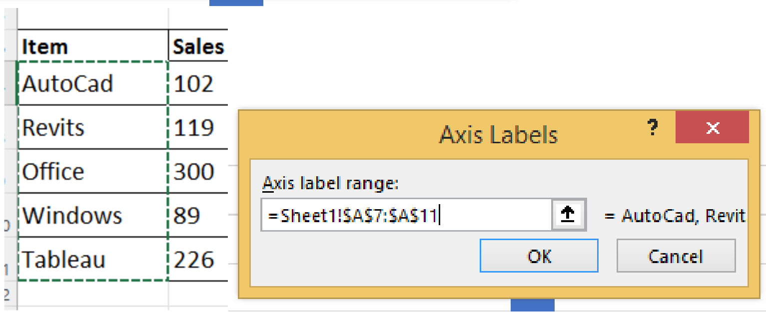


29. As you can see, item labels are missing.

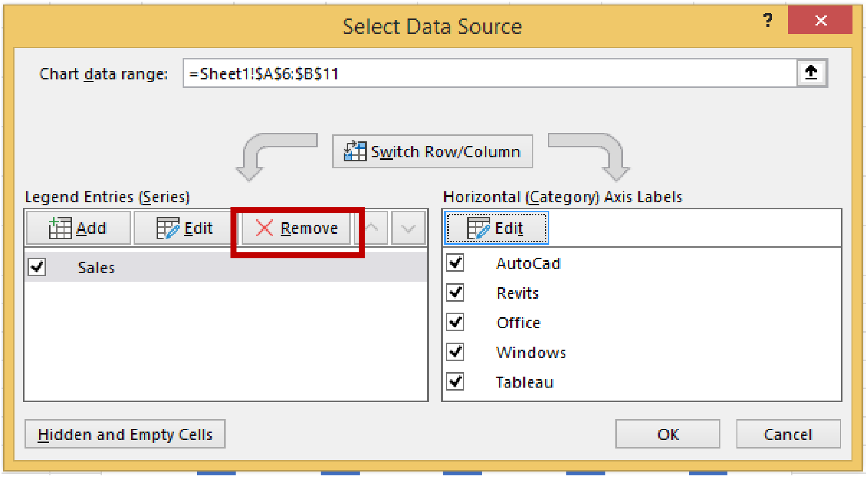


30. In Horizontal Category pane click Edit and chose labels range (=Sheet1!$A$7:$A$11).
31.





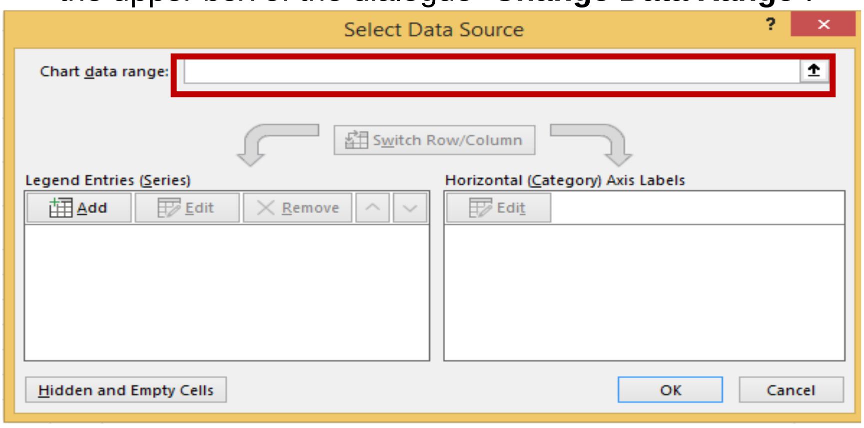
32. As you can see, this is the way and place where you can add , edit or remove your series.



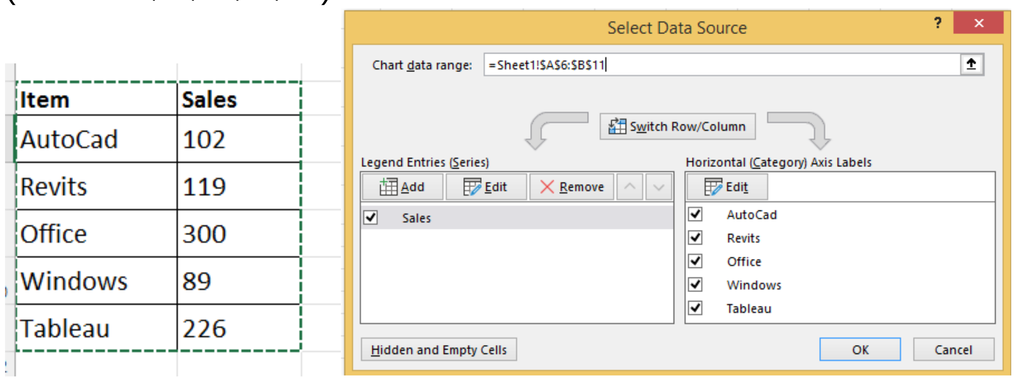
33. Click **Remove** to remove your data series.

Another way easier to add Data Series

34. The fastest way to add series and names values and names is to use the upper box of the dialogue “**Change Data Range**”.



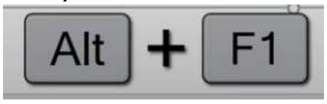
35. click on Chart Data Range Box and choose the data area (=Sheet1!$A$6:$B$11).



36. **Delete** your Chart (to try something else).

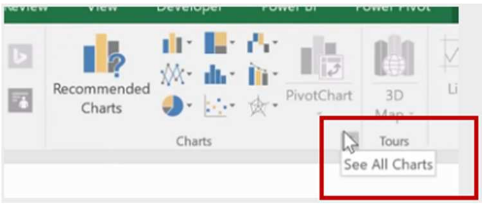
Shortcut to Add the default chart

37. The easiest way to add the chart is to select the data and then press (Alt+F1).
38. Excel insert the default chart for you.
39. Highlight your data➔(Alt+F1)➔Your default data appear.

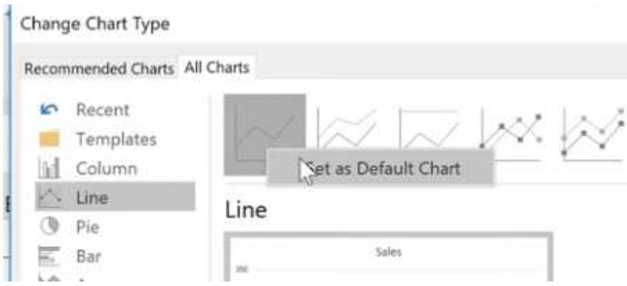


Change Default Chart

40. You can change your default Chart.
41. Go to Insert ➔ Charts➔ click in small arrow in the bottom left corner (see All Charts).



42. Go to line charts.
43. Right click the first one and choose: Set as default Chart.

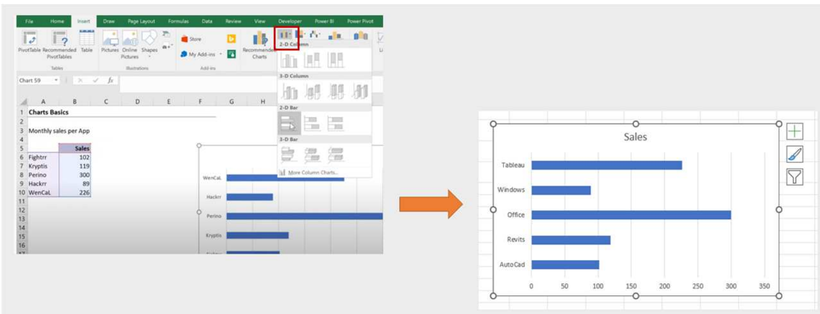


44. Delete your current Chart and try your new default Chart (Alt+F1).

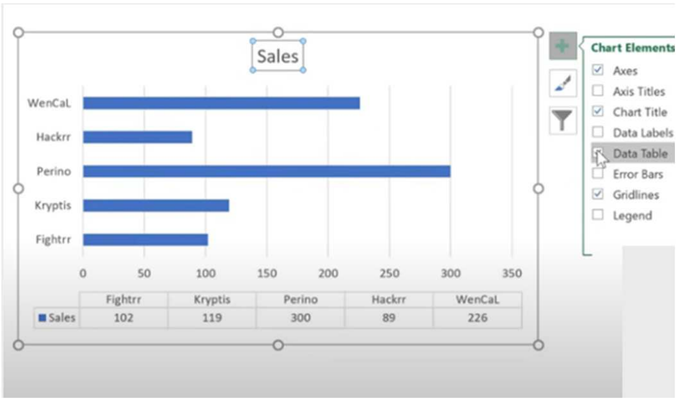
45. Get your default Chart Back to Column Chart.
46. Delete All Charts in the Sheet.

Exercise 07 B Adjusting Charts

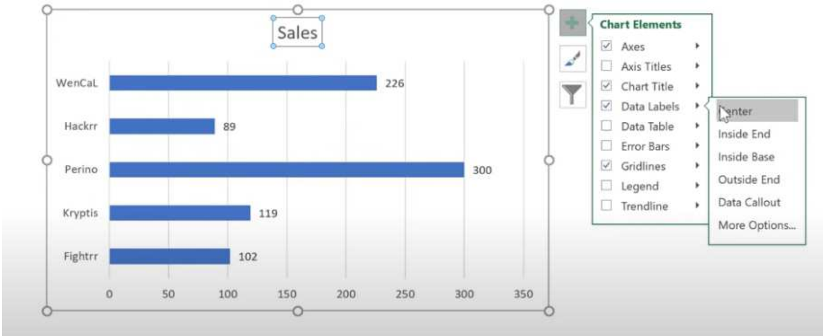
1. In the same sheet **Chart Basics**.
2. If I decided to use Bar chart.
3. Go to Insert and Chose Bar chart.



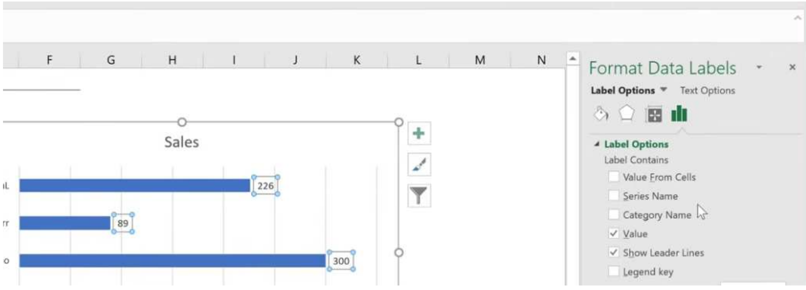
4. Notice that your chart is comprised of different chart elements.
5. Click on **bars** (data series) this is an element.
6. Click on **vertical Grid** (this is an element).
7. Click on **Title** (an element).
8. To add or remove elements to your Chart Use the (**+**) on right of your chart.
9. Click + ➔ Data table.



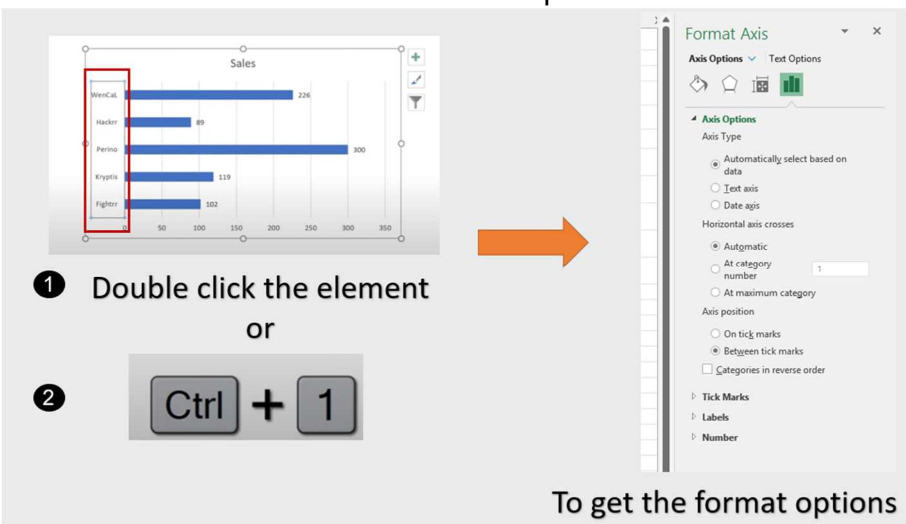
10. Remove Data Table.
11. Add **Data Labels** and click the arrow next to select the **position** (inside, outside, etc....).



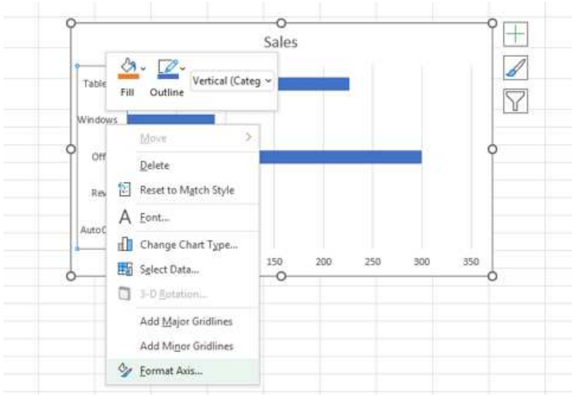
12. Click on **more options** to open Format Data labels on the right.



13. Close the Dialogue box.
14. Depending on what element you select you have different options.
15. You can also get this dialogue box by **double clicking** on the element.
16. Select **Y** Access then double click to open **Format Axis**.



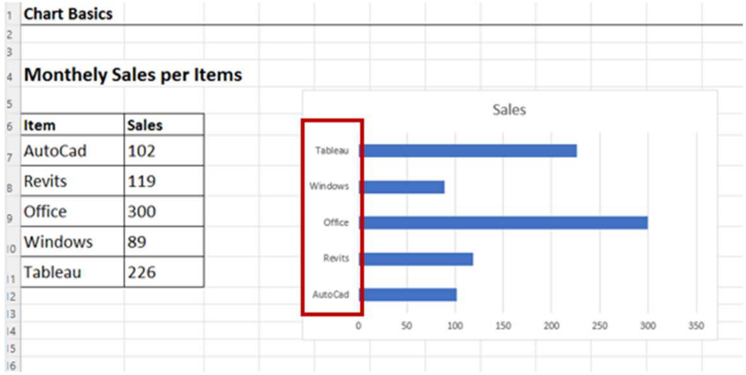
17. Close the Dialogue box.
18. You can also use the shortcut key (**Ctrl+1**).
19. Select **Data series** and press **Ctrl+1**.
20. You can also right click the **bars** and choose **Format Data Series**.



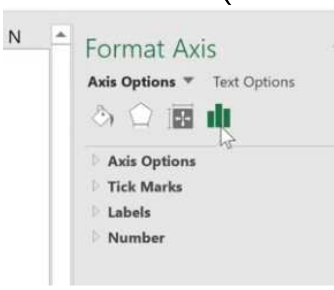
21. **Move** to different **elements** and notice how dialogue box changes.
22. Keep the **format option** dialogue box activated and move from one element to another to change its options.

Reverse Sorting of Y Access

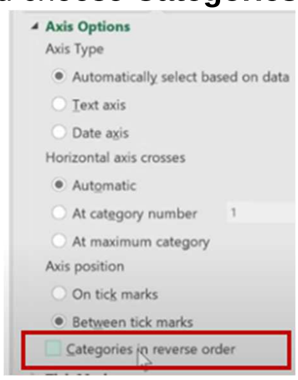
23. If My boss wanted me to reverse the sorting of the Y access.
24. Select the **Y** access.



25. In **format Access** choose the last icon (**Axis Options**).

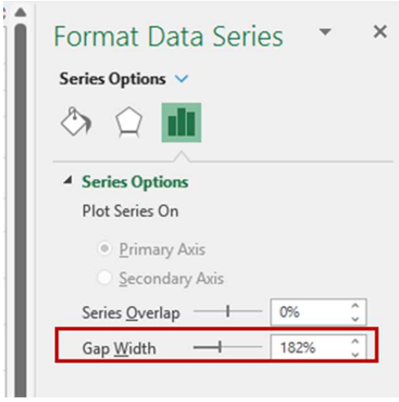


26. Expand Axis Options and choose **Categories in reverse Order**.



Change Gab Width

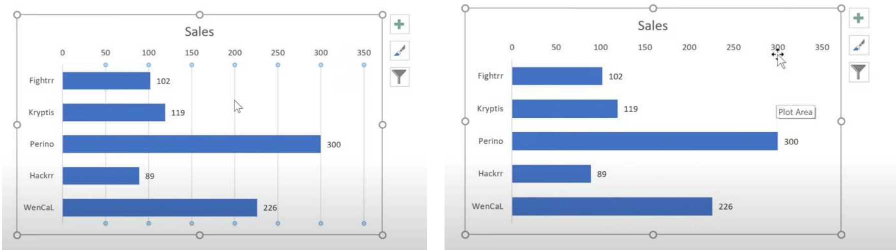
27. Click on **bars**.
28. In format data series choose **Series Options**.



29. Decrease Gap Width to 80%.

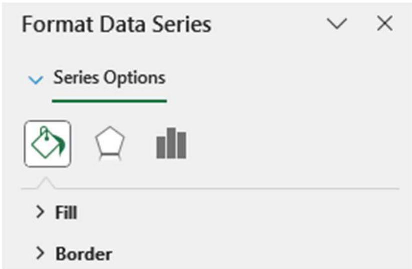
Delete an Element of Chart

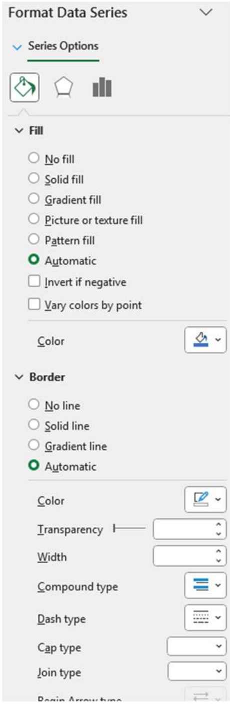
30. If you want to delete an element, just click and press **delete** in your Keyboard.
31. Choose **vertical Grid** and press Delete.



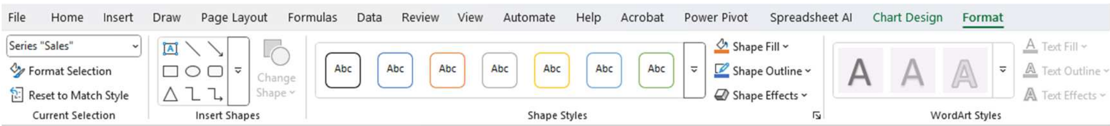
Exercise 7 C: Improving the Chart

1. In the same sheet **Chart Basics.**
2. To format any element, you can choose first and then in Format Dialogue box choose **File & line** option (first button).

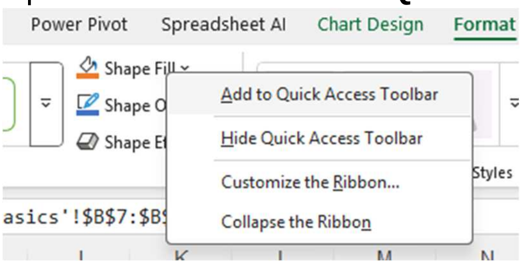




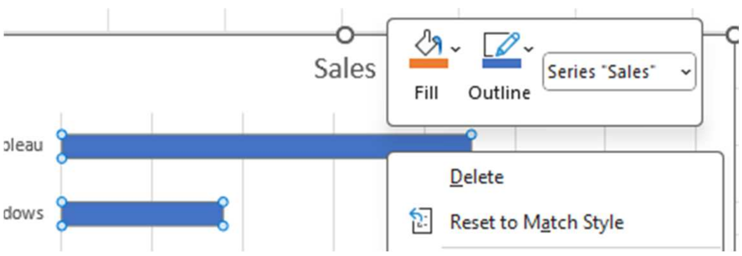
3. Chose **Bars**.
4. In format data series you can format **fill** and **border** options.
5. You can also use the **Format** tab in Ribbon.



6. If you use an option a lot, you can add to your **quick access** tool bar.
7. Right click on the option and choose **Add to Quick Access** tool bar.

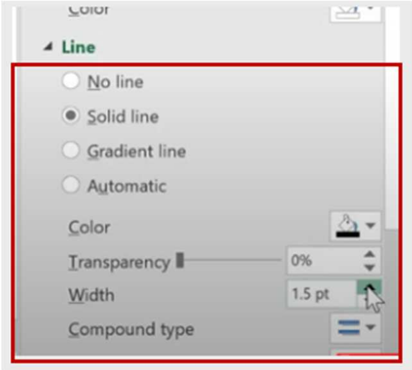


8. Right click your bars, here you can directly format Fill and outline from the **flying menu** appears.

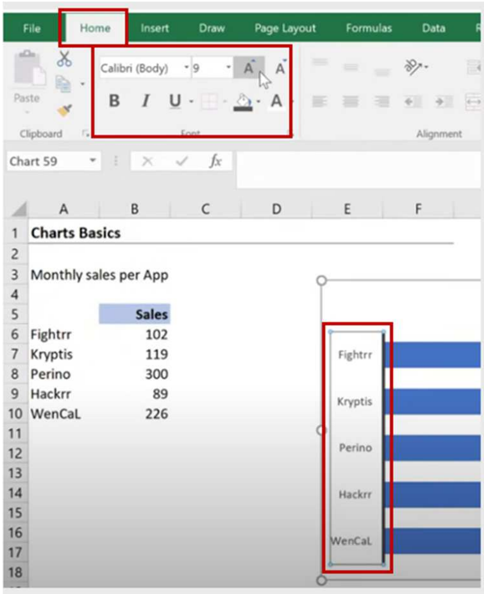


9. I want to make my **Y** access **Thicker**.
10. Choose **Y** access.
11. Go to **Fill & line** option.
12. On line option choose **solid line**.
13. Chose **Black** color from **Color**.

14. Make **width 2pt**.

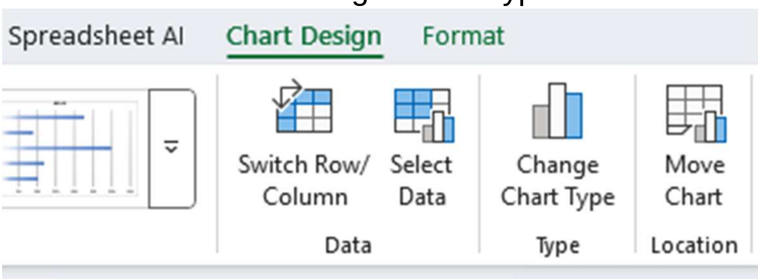


15. To format the **text** of the **Axes**, select and then from **Home** tab format it. Increase **font size** to 10 and make it **bold**.



Exercise 7 D: Add more Series

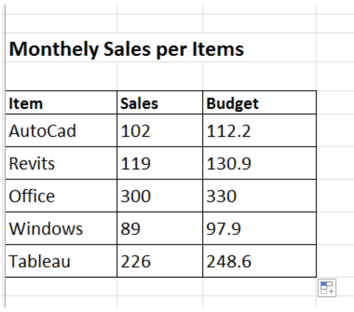
1. Convert your Chart to **column type**.
2. Select and choose Change Chart Type.



3. I want to add **budget** data series.
4. Let us add a new column **budget** besides sales.
5. Budget = Sales *1.1.

Monthly sales per App

	Sales	Budget
Fightrr	102	=B6*1.1
Kryptis	119	
Perino	300	
Hackrr	89	
WenCaL	226	

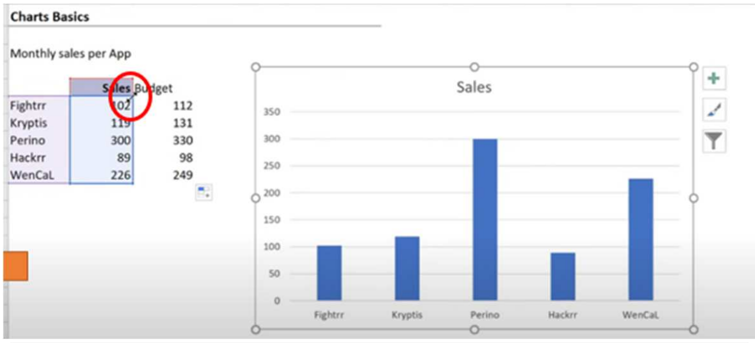


Monthely Sales per Items

Item	Sales	Budget
AutoCad	102	112.2
Revits	119	130.9
Office	300	330
Windows	89	97.9
Tableau	226	248.6

Expand the Data Graphically

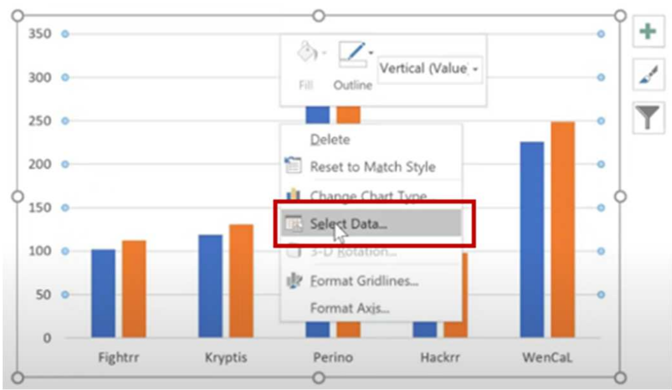
6. The easiest way is to select the chart and **expand** the data in the table to include the new column.



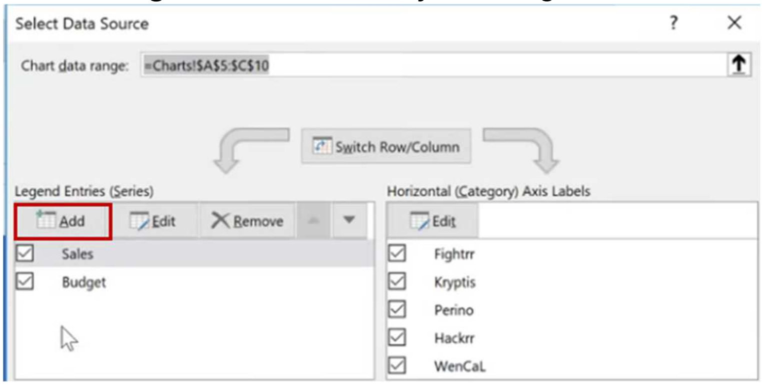
7. You must **click on the whole chart** before you expand.
8. Click on Data series and try to expand ...you fail.
9. **You must click on the chart Area First**.

Use Select Data Option

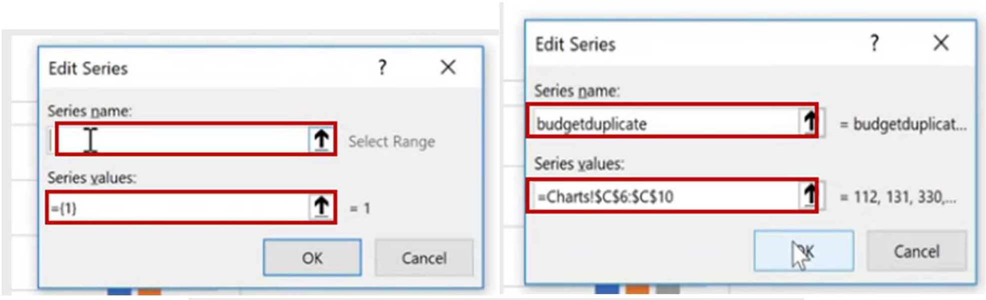
10. Another way is to right click on the chart and choose to select data.
11. Right click on the chart and choose **select data**.

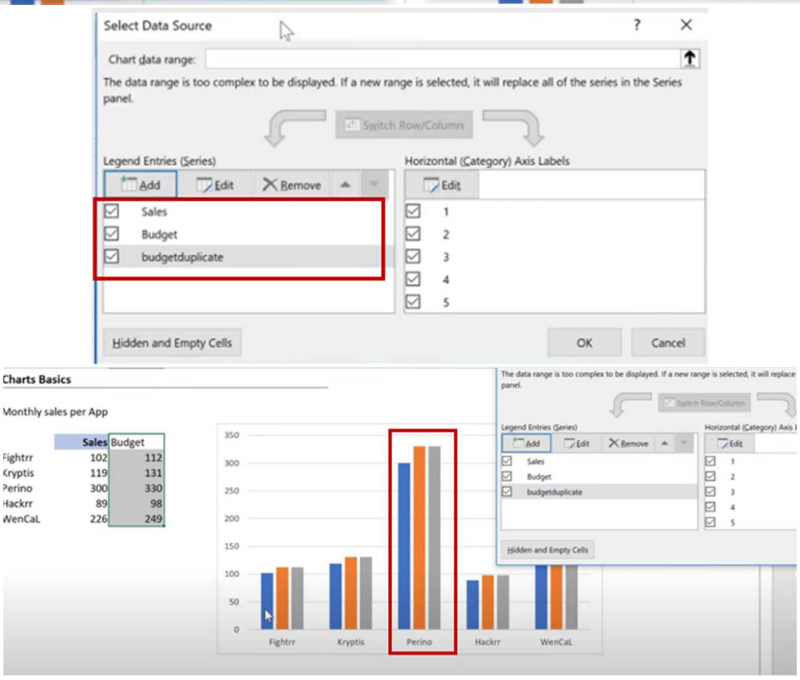


12. Here you can manually add your series.
13. Let us add the **budget series** manually once again with another name.

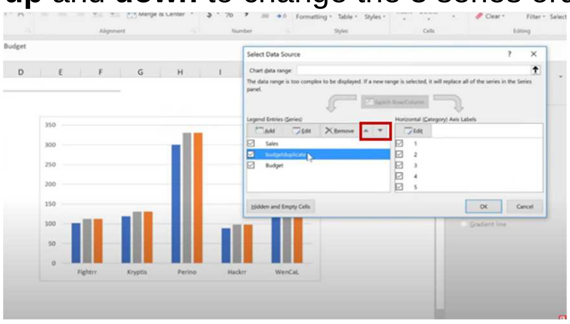


14. Click **Add** and in series name write: **Budget Duplicated**.

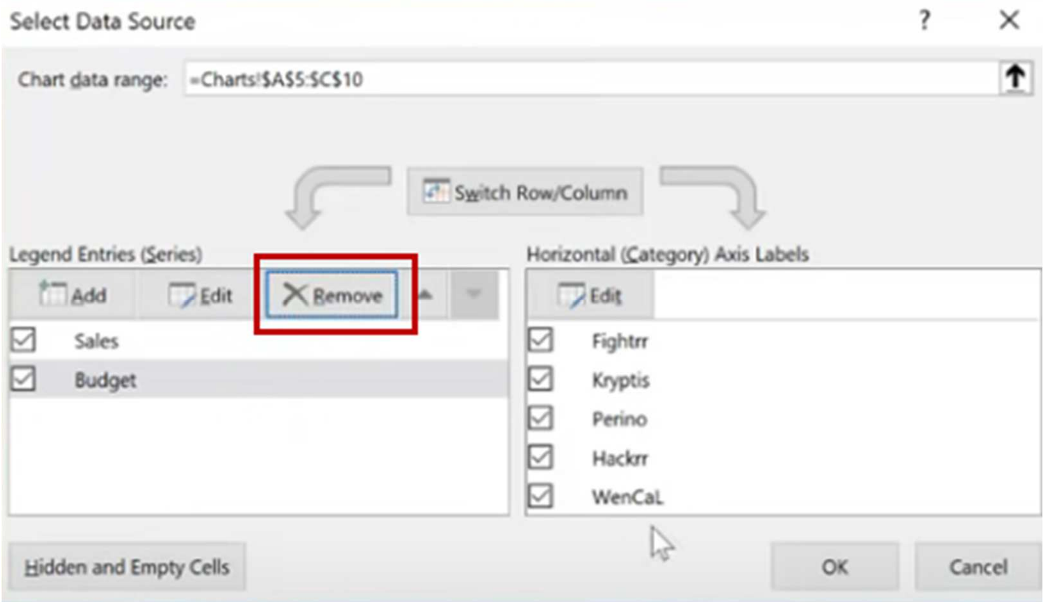




15. In **Values** select the **series values**
16. Use the arrows **up** and **down** to change the 3 series order.



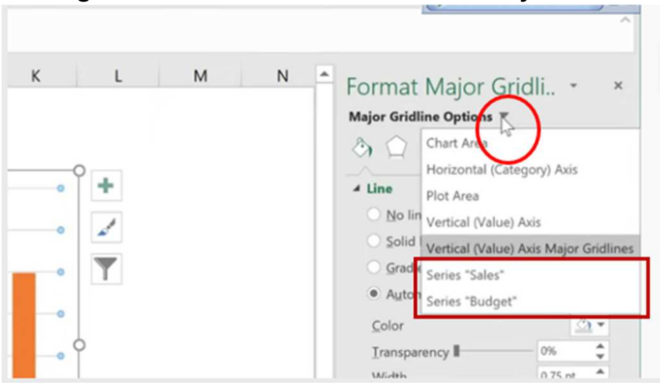
17. Select the **duplicated series** and delete it.



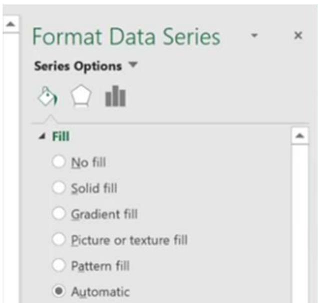
18. Click **OK**

Change the Series Color and select if not Appears

19. Sometimes you cannot find your series if it has no data or have white color.
20. Change the new series color of fill and border to **white**.
21. The series disappears.
22. To find the series just click in the arrow of the objects in the **Format** Dialogue box and choose the series you want to select.



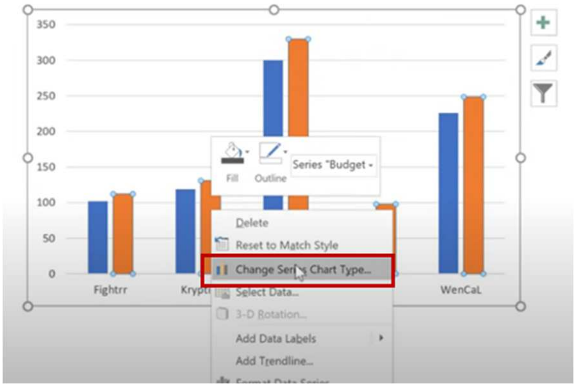
23. Notice that this list contains all the elements you have in your chart.
24. Here you can choose your series and format it.



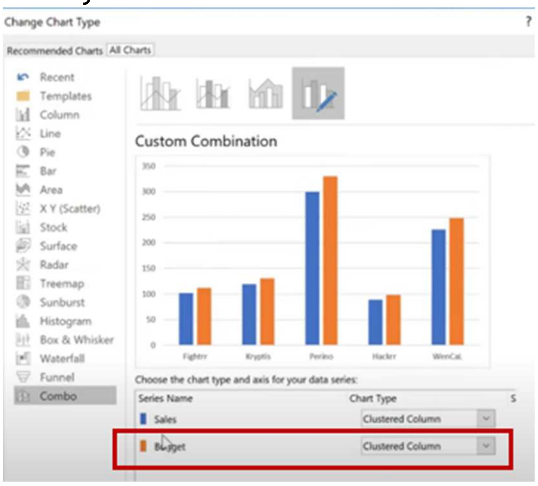
25. Always use this list to reach your elements directly.

Exercise 7 E: Creating Combination Chart

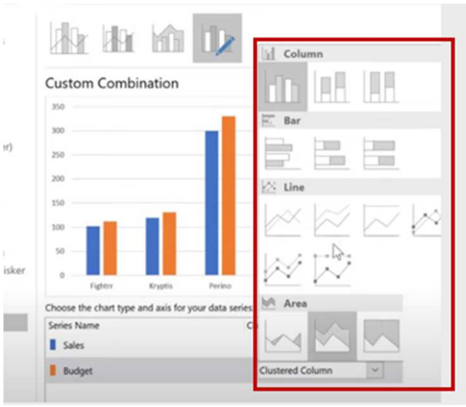
1. My boss asked me to show **Budget** as **line** series.
2. Then he changed his mind and asked me to show him only as **markers**.
3. Select any **series**, right click and choose **change series chart type**.

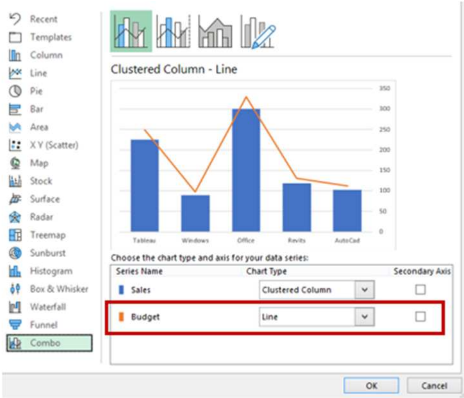


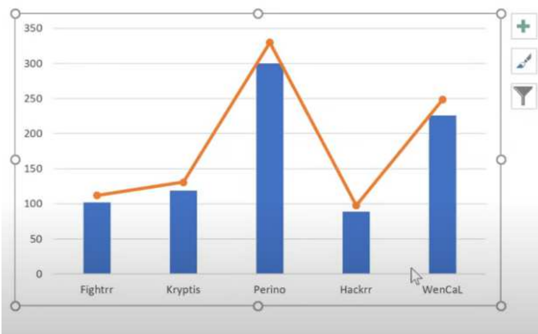
4. Excel takes you directly to **Combo** view.



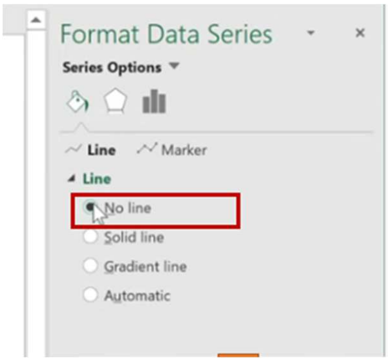
5. Here you can see all your series and you can change the chart type of each from the drop-down list.
6. Change **Budget series** to **Line with markers** and click OK.

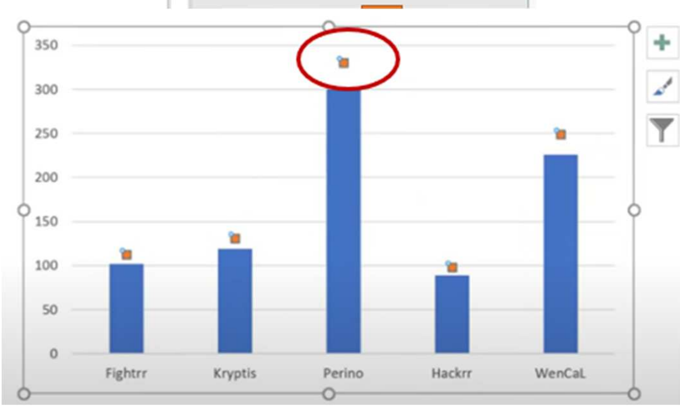




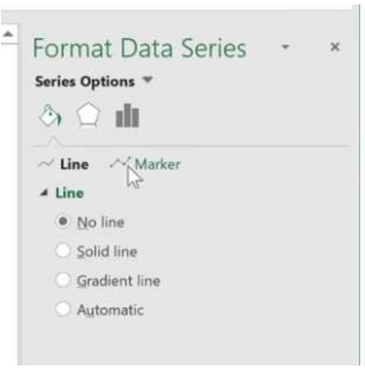


7. Select the **line** and in Format Dialogue box in **Fill & line**.
8. Under line chose **no line**.

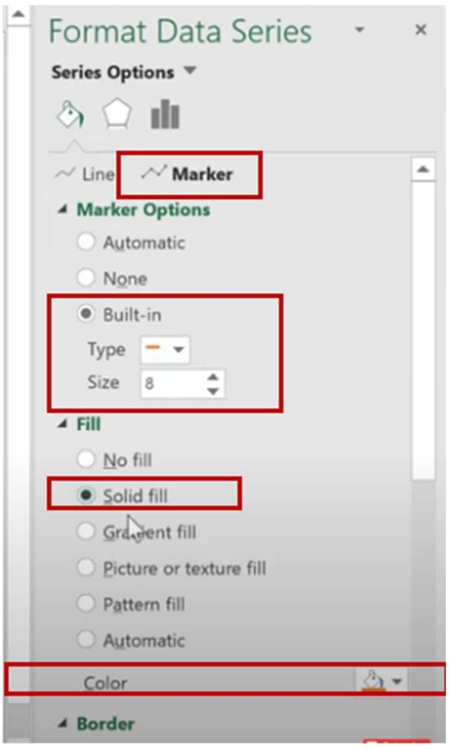


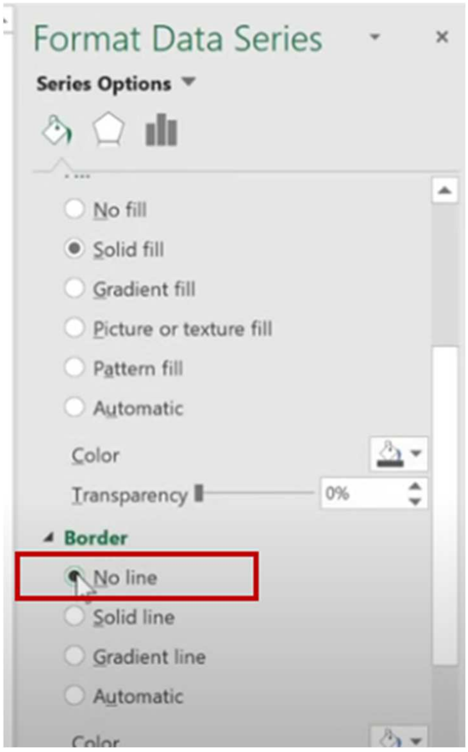


9. Click the **marker** on the top.

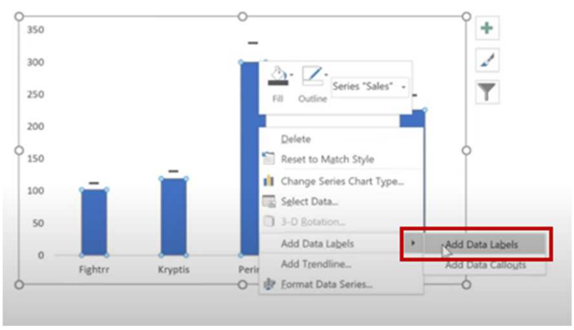


10. Under the marker option choose Built in.
11. Chose line (**Dash**).
12. Increase size to **8**.
13. Change the fill to **Orange**.
14. no **Border** chose **no line**.

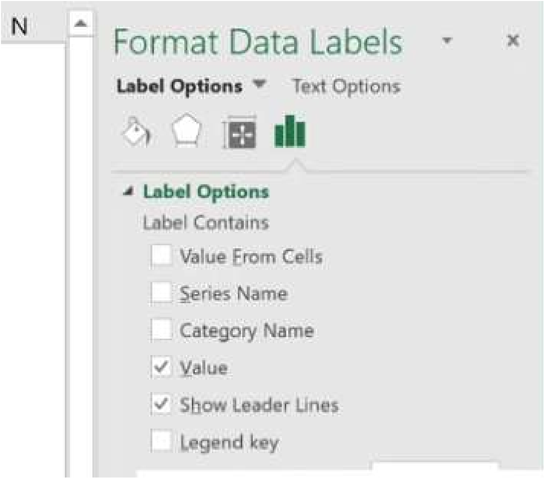




15. Click on **Sales** series and add **data labels**.

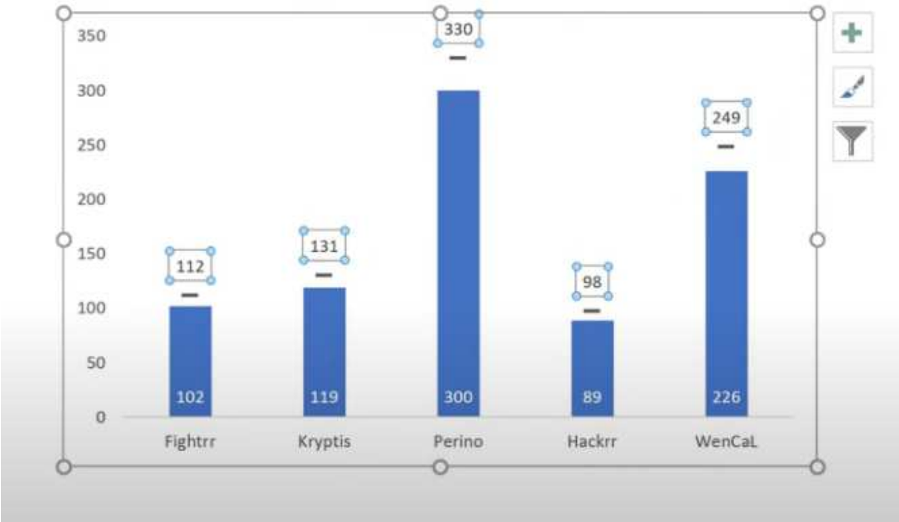


16. On format dialogue box ➔ options (last button).
17. Under label options ➔ label position = inside base.



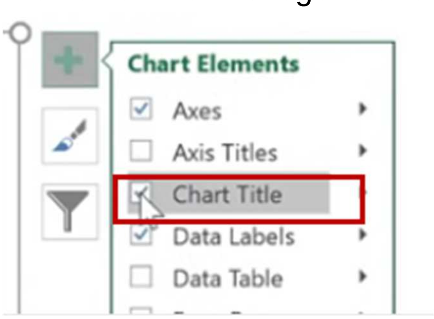
18. From home tab in ribbon change color to white.

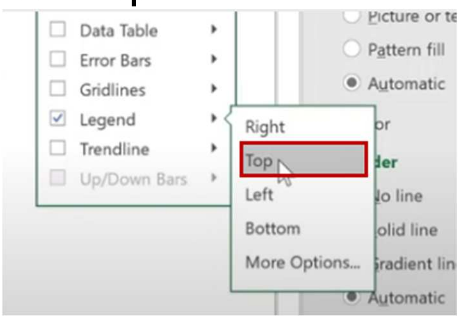
19. Select the other series (**Budget**).
20. Right click and add data label.
21. Make the labels on Top.



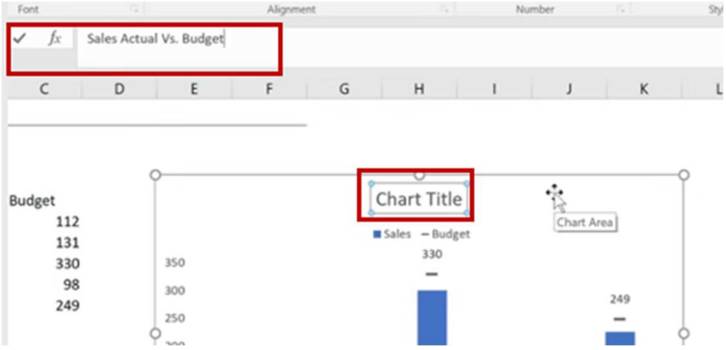
Add a chart Title and Legend

22. Click (**+**) on the right of the chart and check **Title** and **Legend**.
23. Click on arrow next to legend and choose **Top**.

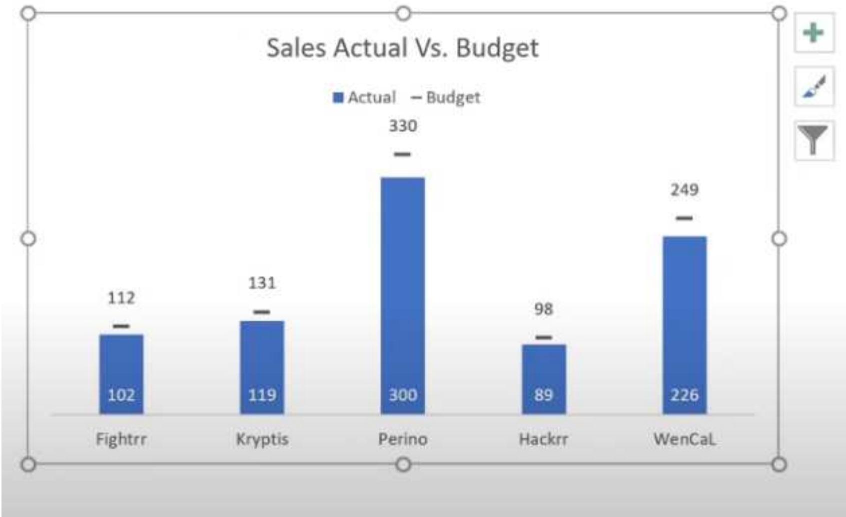




24. Select the **Title**.
25. In formula box write the title: **Sales Actual Vs. Budget**



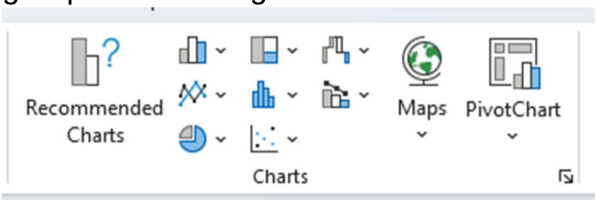
26. Select the **legend**.
27. I want to change **Sales** to **Actual**.
28. Go to the table of Data and change the **column Header** to Actual.
29. Click on **Y access** and **delete**.



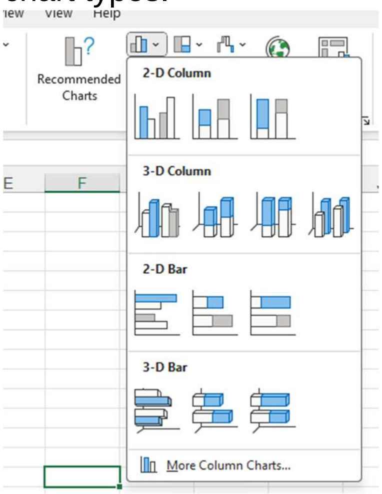
Exercise 07 F: Create and Format Types of Charts

Create Pie Chart

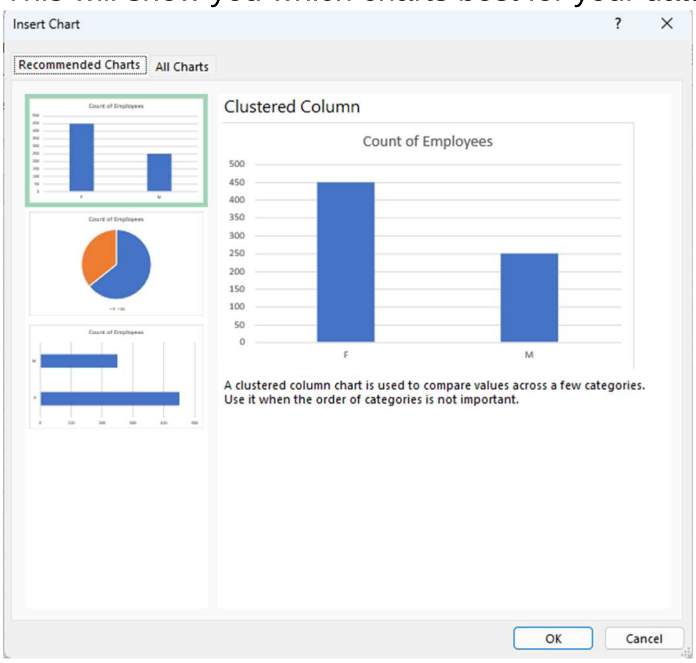
47. We have date divided into the number male and female employees.
48. It is better to use pie chart to present.
49. Select your data set.
50. In the insert ribbon tab in Charts group, you have many chart types grouped into categories.



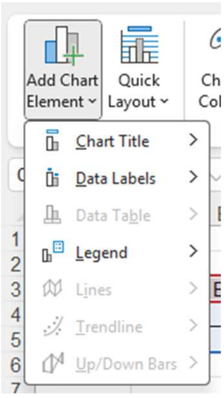
51. If you click any small arrow before any of them it will show you more chart types.



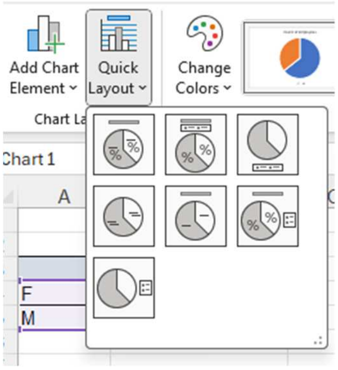
52. If you are not sure which chart type suitable for your data you can click on the **Recommended Charts** button.
53. This will show you which charts best for your data.



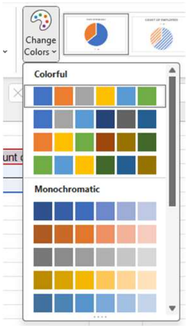
54. Select Pie Chart and click OK.
55. A pie chart inserted into your worksheet and 2 contextual ribbon tabs appears (**Chart Design** and **Format**).
56. In Chart Design you can:
    a. **Add chart elements**.



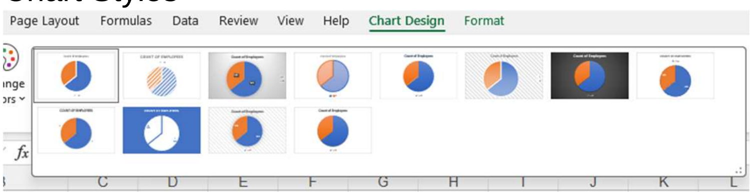
    b. **Quick layout** to create chart with some elements



    c. Change color scheme

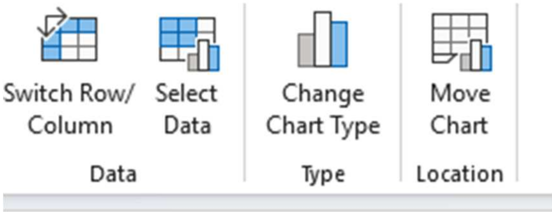


    d. Chart Styles



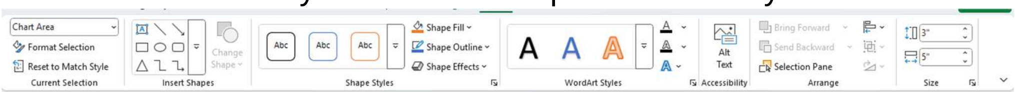
it is predefined format. Try them to see by yourself

e. You also have buttons to

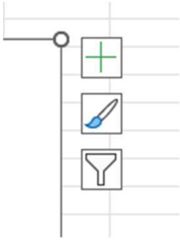


switch row/column , select data , change chart type and move the chart to another worksheet.

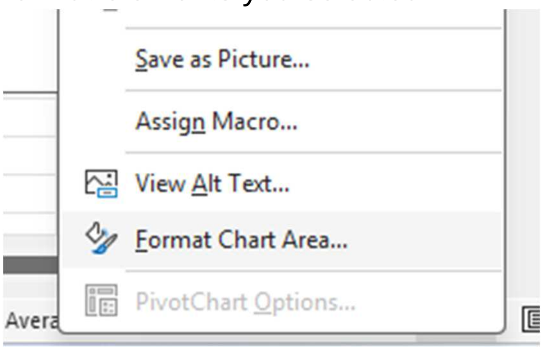
57. In the Format ribbon you will have all option to format your chart.



58. Select Female section in the Pie chart and change it to **Green**.
59. Use Shape Style➔Shape Fill.



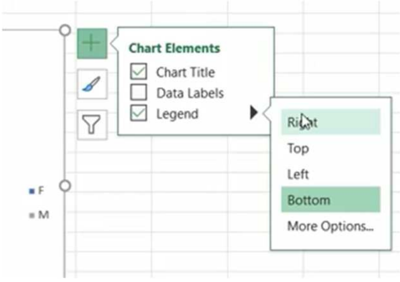
60. With every chart you will find 3 icons on the top right of the chart.
    a. Chart Elements.
    b. Chart Styles.
    c. Chart Filter.
61. If you right click your chart, you can reach format window on the right to format elements you selected.



Format Pie Chart

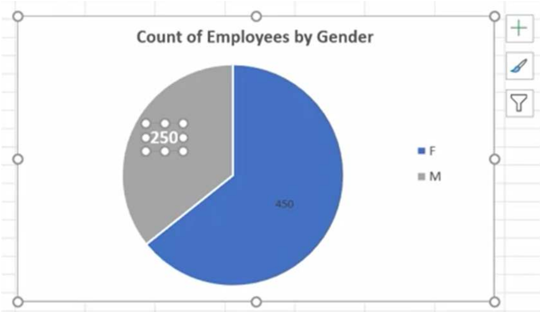
1. Chart Design ➔Change Colors➔Colorful Pallet 2

2. Change title to: **Count of employees by gender**.
3. Select title, **CTRL+B** to make it bold.
4. Move legend to the right.



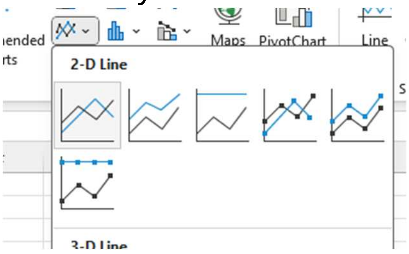
5. You can click on legend and move it to the place you want.
6. Increase the font size of the legend.
7. Add Data labels.

8. Make the labels at the end and then select each label and move.
9. Make label Font size bigger.
10. Change the font color to white and make it bold.



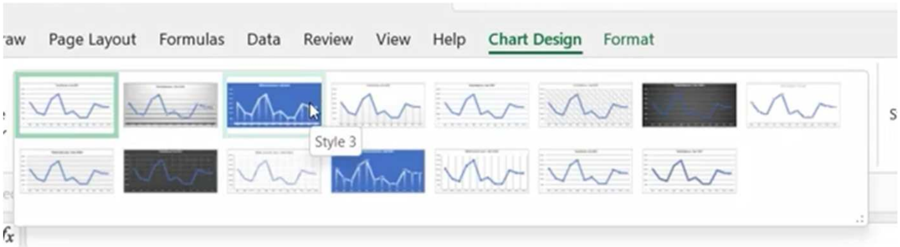
Create Line Chart

62. Select any cell in data and select 2D Chart.

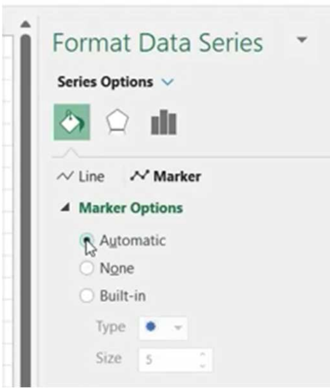


Format Line Chart

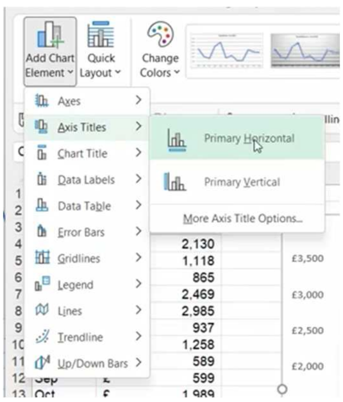
1. Change Title: **Total Sales Jan-Dec 2020** and make them bold.
2. In Chart Design try every style hover to see the result before you choose.

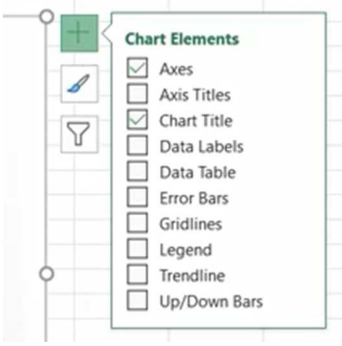


3. Choose Style 4.
4. I want add markers.
5. Click on the line, right click➔**Format Data Series** to open format panel.



6. Data Fill➔Marker➔Automatic.
7. Fill➔Solid Fill (Orange).
8. I want add an access title.



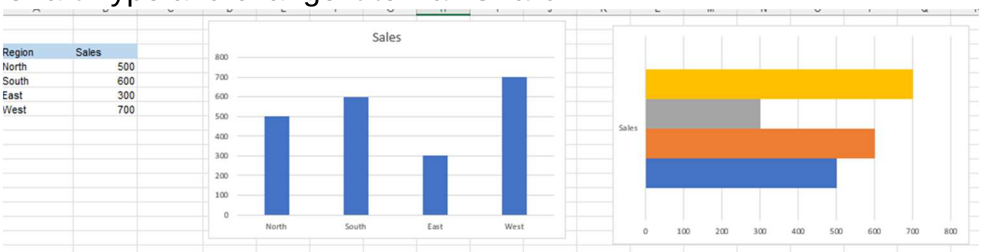


9. You can use + Sign or Chart Design➔Add Chart elements.
10. Add **Primary Horizontal** and **Primary Vertical**.
11. Or simply click **+** ➔ Access Titles.
12. Change Access Titles to **Months** and **Sales**.
13. I want to add an outline for the chart.
14. Format➔Shape Outline.

15. Add black outline.

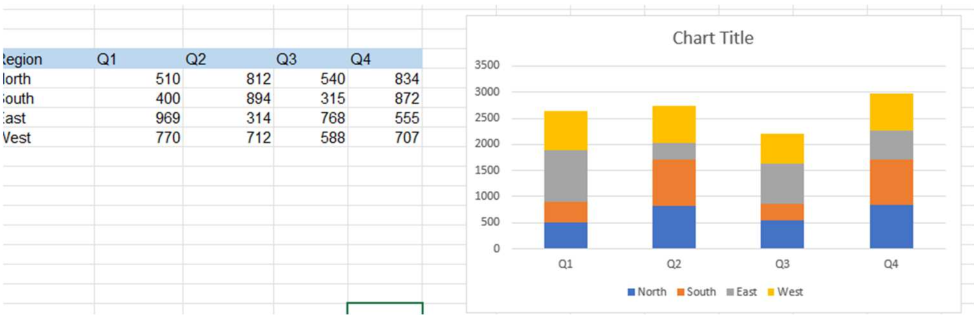
Create Column Chart and Bar Chart

63. Select your data.
64. Insert 2D clustered chart.
65. Select the chart and use shortcut Key **CTRL+D** to duplicate the chart.
66. Select the second chart and from the Chart design➔Type➔Change Chart Type and change it to **Bar Chart.**



Stacked Column Chart

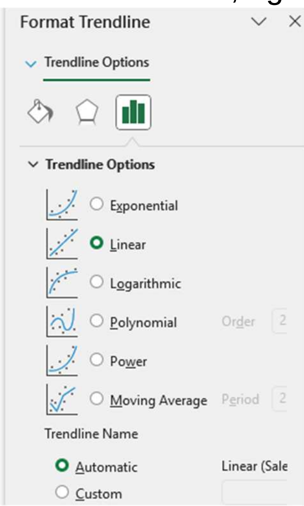
67. Select Data.
68. Create Stacked Column Chart.



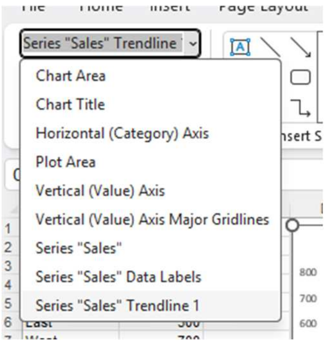
Format Column Chart

1. Change Column Chart Title to: **Sales by Regions**.
2. Select Bars.
3. Change bars to green Format➔Shape Fill.
4. I want to decrease the gaps between bars.
5. Right click bars and select **Format Data Series**.
6. In series options decrease the **Gap Width**.

7. Add Data labels ✚ ➔Data Labels.
8. Make labels inside end, increase font size and make it **white** and **Bold**.
9. Add trend line ✚ ➔ Trend line.
10. Select trend line, right click, format trend line.



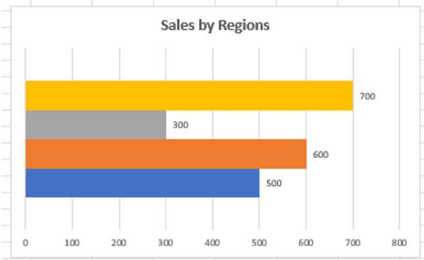
11. You can get the parts of your chart to select if you cannot click through Format ➔Custom Selection.



12. Select grid lines and change their color.

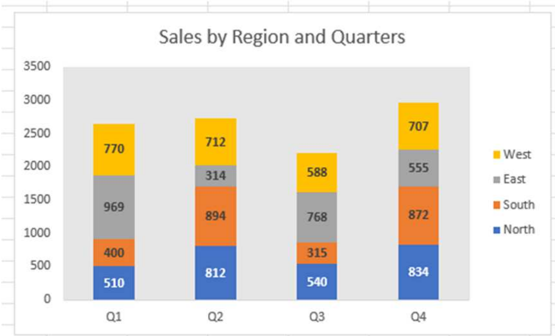
Format Bar Chart

13. Add title: **Sales by Region**.
14. Add Legend and move it to the top.
15. Add data labels outside end.
16. Select the Y access title and remove.



Format Stacked Column Chart

17. Add title: **Sales by Region and quarters**.
18. Add Data labels.
19. Make all data labels bold.
20. Move legend to the right.
21. Select plot area
22. Format ➔Shape Fill and choose a **light grey** color.
23. Turn off **Grid lines.**

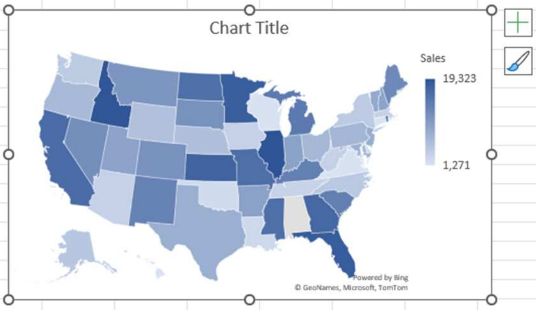


Create Filled Map Chart

69. You must be sure that name of cities and states or countries are written correctly.

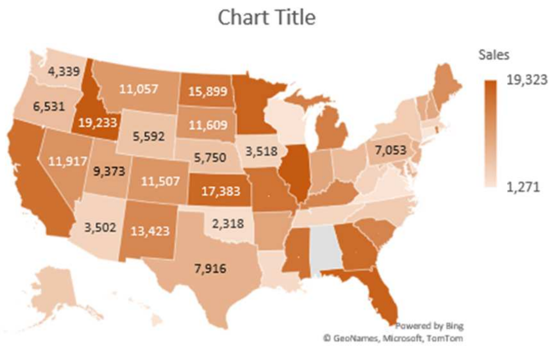
70. You must be connected to the internet.

tate	Code	Sales
laska	AK	4,819
rizona	AZ	3,502
rkansas	AR	9,708
alifornia	CA	16,396
olorado	CO	11,507
onnecticut	CT	1,271
elaware	DE	2,275
istrict of Columbia	DC	12,056
lorida	FL	18,338
eorgia	GA	16,942
awaii	HI	1,319
daho	ID	19,233
linois	IL	19,323
ndiana	IN	9,465
owa	IA	3,518



Format Map Chart

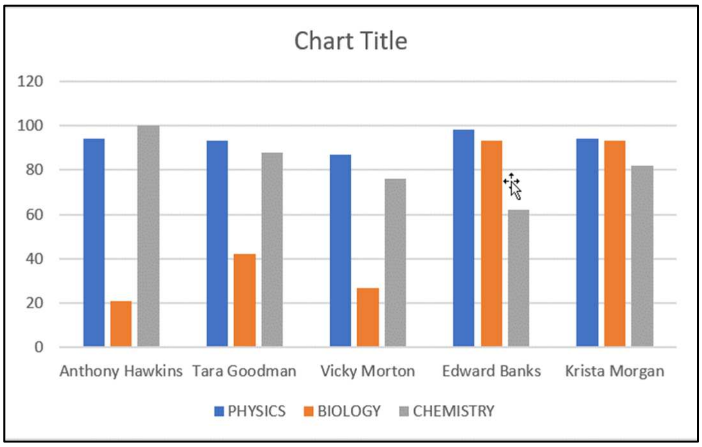
24. Change title: **Sales by State**.
25. Try the **styles** you have in **Chart Design** tab.
26. Add data labels.
27. Change color pallet.
28. Chart Design➔Change Color➔ Color pallet 3.



Exam 06: Charts

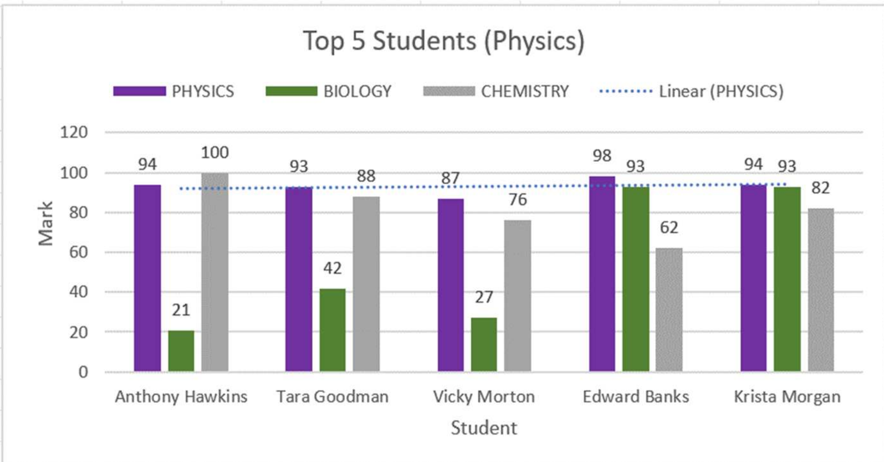
CREATING A CHART

1. On the '**Charts**' worksheet, format the data as a table.
2. Name the table '**Top_5_Physics**'.
3. Apply a filter to the '**Physics**' column to show the Top 5 students.
4. Create a '**2D Clustered Column**' chart.



EDITING A CHART

1. Add the Chart Title '**Top 5 Students (Physics)**'
2. Move the '**Legend**' to the top.
3. Add '**Axis Labels**' to the horizontal and vertical axis.
4. Add '**Data Labels**' to the outside edge.
5. Change the color of the data bars.
6. Add a trendline to show the trend of the '**Physics**' mark.



Chapter 8 Advanced Logical Functions

Exercise 08 A: Nested IF

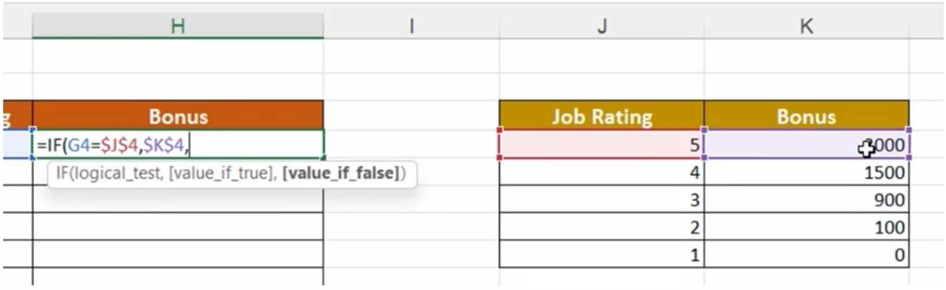
1. It is IF statement inside other IF Statement.
2. We want to add bonuses to employees according to their job rating.
3. You have a table on the rite for bonus amount for each job rating.

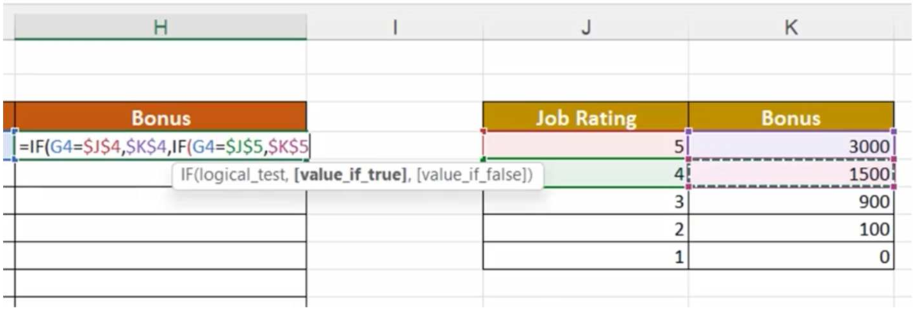

Nested IFs

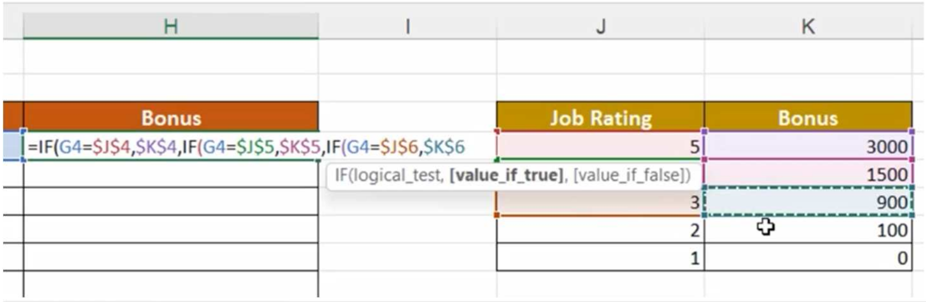
Employee Name	Building	Department	Hire Date	Years	Salary	Job Rating	Bonus
Page, Lisa	West	ADC	12/24/2001	20	$ 60,981.00	1	
Taylor, Hector	West	ADC	2/3/2014	7	$ 60,915.00	4	
Dawson, Jonathan	West	ADC	2/16/2010	11	$ 97,071.00	5	
Duran, Brian	Taft	ADC	8/13/2015	6	$ 115,421.00	3	

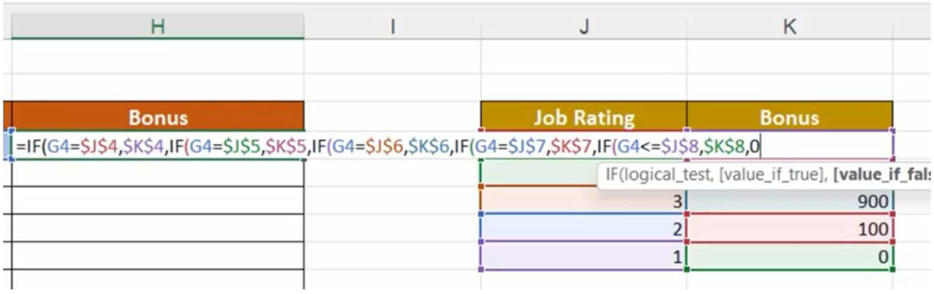
Job Rating	Bonus
5	3000
4	1500
3	900
2	100
1	0

4. So, I need to perform multiple logical tests.
5. Now go step by step creating the logic









6. Add 4 closing brackets.

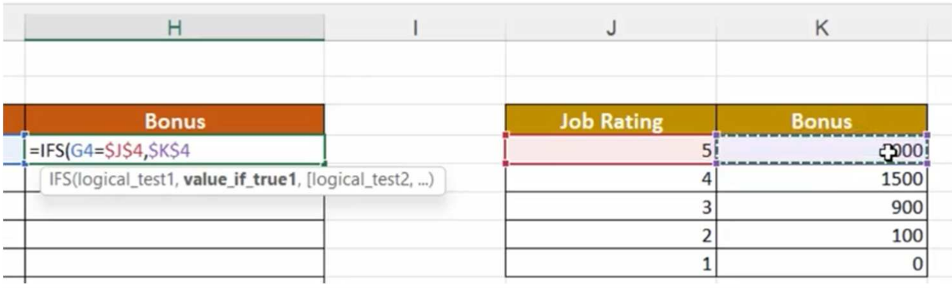
7. It is so complex

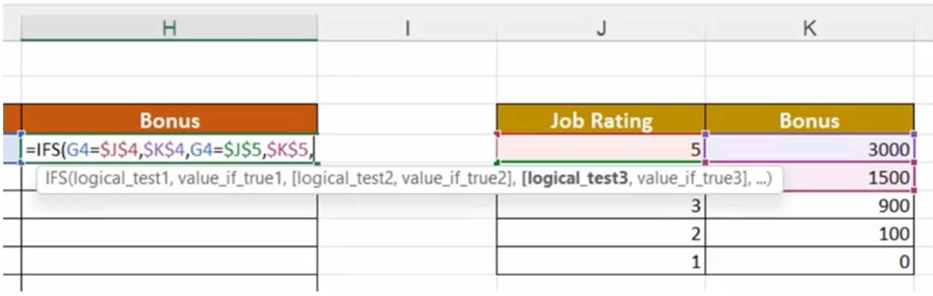
8. If you see errors, select all the columns (**CTRL+SHFT+↓**).
9. Select ignore error.

Exercise 08 B: IFS Statement

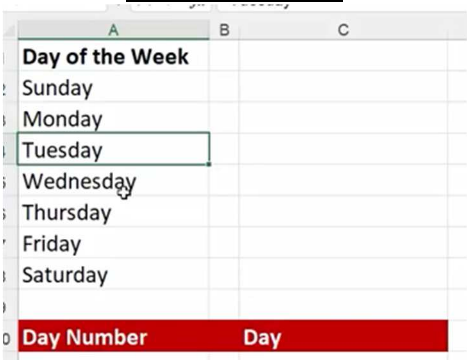
1. In sheet **IFS Example 1** you have the same problem.

2. Let us try to solve it using **IFS** function.

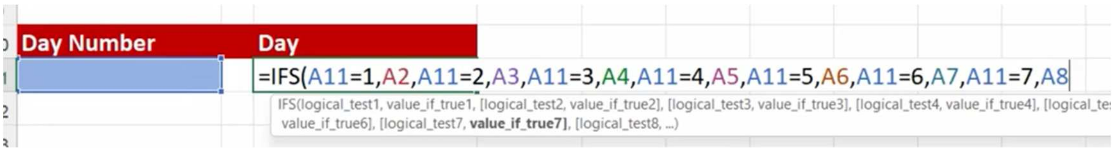




3. In sheet **IFS Example 2**.



4. I have list of Days Sunday to Saturday (1 to 7).
5. I want if I put the day number I get the name of the day.
6. For example, 1 gives me Sunday and 7 gives me Saturday.
7. Let us create a formula for that.



8. Notice that you have an error **NA** when you enter number > 7 or leave the cell empty.
9. You can handle this win **IFNR** or **IFERROR** function or add a **TRUE** value for the last check in the **IFS** statement.

Exercise 08 C: Conditional IF

COUNTIF, SUMIF, AVERAGEIF

1. We have a table of Sales team member names with their companies' job titles and total sales.

Sales Team	Company	Job Title	Total Sales
Angelina Webb	Computech	Sales Rep	$ 7,270.00
Alison Bryant	MicroWorld	Sales Manager	$ 9,250.00
Jenna Norman	MicroWorld	Sales Rep	$ 6,201.00
Abel Hoffman	Computech	Sales Manager	$ 7,480.00
Ervin Ray	MicroWorld	Sales Rep	$ 7,139.00

2. We want the total sales according to some conditions.

CRITERIA			SUMIF(S)	
MicroWorld				*Total Sales for MicroWorld*
MicroWorld	Sales Rep			*Total Sales for all Sales Reps at MicroWorld*
MicroWorld	Sales Rep	>7000		*Total Sales for all Sales Reps at MicroWorld with more than 7000 in*

3. For example, Total Sales for **MicroWorld** company.
4. So we want to **SUM** total sales but only **IF** the company in the company column = **MicroWord**.
5. So, we will use **SUMIF** function

SUMIF Function

6. It needs **2** required arguments and **1** optional.

```
=SUMIF(
SUMIF(range, criteria, [sum_range])
```

7. MUST= **Range** to check in and **Criteria** to check.
8. OPTIONAL = SUM Range , that we want to sum.
9. In our case the
   a. Range = Company Range.
   b. Criteria = MicroWord
   c. Sum_range = Total Sales.

Company	Job Title	Total Sales
Computech	Sales Rep	$ 7,270.00
MicroWorld	Sales Manager	$ 9,250.00
MicroWorld	Sales Rep	$ 6,201.00
Computech	Sales Manager	$ 7,480.00
MicroWorld	Sales Rep	$ 7,139.00

CRITERIA			SUMIF(S)
MicroWorld			=SUMIF(B6:B17,F6,D6:D17) SUMIF(range, criteria, [sum_range])
MicroWorld	Sales Rep		
MicroWorld	Sales Rep	>7000	

```
=SUMIF(B6:B17,F6,D6:D17)
```

COUNTIF Function

10. In table 2 on the right, I want to count the total Number of employees in CompuTech company.

11. Use COUNTIF function

Company	Job T
Computech	Sales
MicroWorld	Sales
MicroWorld	Sales
Computech	Sales
MicroWorld	Sales
MicroWorld	Sales
Computech	Sales
Computech	Sales
MicroWorld	Sales
Computech	Sales
Computech	Sales
MicroWorld	Sales

CRITERIA			COUNTIF(S)
Computech			=COUNTIF(B6:B17,F14)
Computech	Sales Manager		COUNTIF(range, criteria)
Computech	Sales Manager	<9000	

=COUNTIF(B6:B17,F14)

AVERAGEIF

12. I want to know average sales of computech.

CRITERIA			AVERAGEIF(S)
Computech			=AVERAGEIF(B6:B17,F20,D6:D17)
Computech	Sales Rep		AVERAGEIF(range, criteria, [average_range])

Exercise 08 D: Multiple Conditional IFS

1. What if we want to use more than one Criteria?

SUMIFS

2. I want total sales of Reps at MicroWorld company.

=SUMIFS(

SUMIFS(sum_range, criteria_range1, criteria1, ...)

   a. First we give the range we want to sum
   b. Then we give the Criteria Range1 , and Criteria 1
   c. Criteria Range 2 and Criteria 2
   d. And so on …..

MicroWorld	Sales Rep		=SUMIFS(D6:D17,B6:B17,F7,C6:C17,G7)

Company	Job Title
Computech	Sales Rep
MicroWorld	Sales Manager
MicroWorld	Sales Rep
Computech	Sales Manager
MicroWorld	Sales Rep
MicroWorld	Sales Manager
Computech	Sales Manager

3. For 3 conditions:

MicroWorld	Sales Rep	>7000	=SUMIFS(D6:D17,B6:B17,F8,C6:C17,G8,D6:D17,H8)

COUNTIFS

Computech	Sales Manager		=COUNTIFS(B6:B17,F15,C6:C17,G15)

Computech	Sales Manager	<9000	=COUNTIFS(B6:B17,F16,C6:C17,G16,D6:D17,H16)

COUNTIFS(criteria_range1, criteria1, [criteria_range2, criteria2], |

AVERAGEIFS

Computech	Sales Rep		=AVERAGEIFS(D6:D17,B6:B17,F21,C6:C17,G21)

AVERAGEIFS(average_range, criteria_range1, criteria1, [criteria_rang

Try Yourself:

MINIFS, **MAXIFS**

Exam 07 Advanced Logical Functions

Ex 01

EXERCISE 01 - Logical Functions

Athlete Name	Country	Gender	Medal	Prize Money
Page, Lisa	USA	F	Bronze	
Taylor, Hector	USA	M	Gold	
Dawson, Jonathan	Canada	M	Gold	
Duran, Brian	USA	M	Silver	
Weber, Larry	UK	M	Silver	
Pratt, Erik	UK	M	Bronze	
O'Connor, Kent	South Africa	M	Bronze	
Spencer, Lucy	Australia	F	Gold	
Wiggins, Frank	Australia	M	Silver	
Tanner, Claire	Australia	F	Gold	
Strickland, Rajean	USA	M	Bronze	
Chase, Susan	UK	F	Bronze	

Medal	Prize Money
Bronze	100
Silver	250
Gold	500

Using **Nested IFs** or **IFS**, complete the 'Prize Money' field in the table.

Using **Nested IFs** or **IFS**, complete the 'Prize Money' field in the table.

Ex 02

Athlete Name	Country	Gender	Medal	Prize Money
Page, Lisa	USA	F	Bronze	$ 100
Taylor, Hector	USA	M	Gold	$ 500
Dawson, Jonathan	Canada	M	Gold	$ 500
Duran, Brian	USA	M	Silver	$ 250
Weber, Larry	UK	M	Silver	$ 250
Pratt, Erik	UK	M	Bronze	$ 100
O'Connor, Kent	South Africa	M	Bronze	$ 100
Spencer, Lucy	Australia	F	Gold	$ 500
Wiggins, Frank	Australia	M	Silver	$ 250
Tanner, Claire	Australia	F	Gold	$ 500

Criteria	Result
Number of Athletes from the USA	
Number of Athletes from Australia and Female	
Number of Atheletes from the UK, Male, Silver	

Criteria	Result
Total Prize Money for USA	
Total Prize Money for all Female, Canadian Athletes	
Total Prize Money for North America	

Complete the results column using the criteria.

Chapter 9 Advanced Lookup Information Techniques

Exercise 09 A: Lookups with INDEX and MATCH

1. VLOOKUP must look up in the first row of the table you give.
2. What if the column you want to look up in is not the 1st one in table?
3. There is more than work around this one of them is using INDEX and MATCH functions.
4. In our example we want to look in **App** Column (Column 2) and get back values from **Category** (column 1), **Profit** (column 4) and **Revenue** (Column 3).

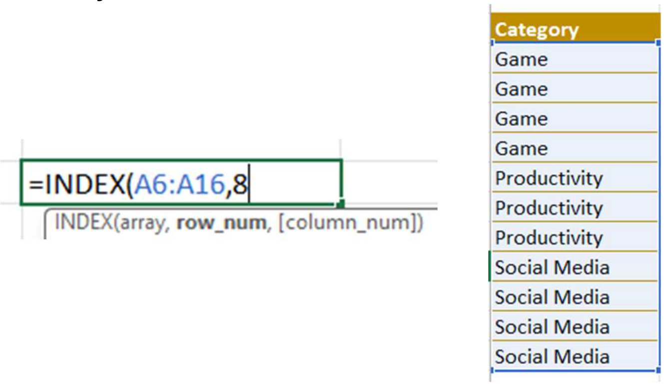
Category	App	Revenue	Profit
Game	Temple Run	$ 11,649.00	$ 802.00
Game	Candy Crush	$ 7,718.00	$ 876.00
Game	Doom	$ 15,033.00	$ 469.00
Game	Bejewelled	$ 18,700.50	$ 984.90
Productivity	Office Lens	$ 14,432.00	$ 240.00
Productivity	Google Docs	$ 17,990.00	$ 1,166.00
Productivity	OneDrive	$ 11,022.00	$ 550.00
Social Media	Twitter	$ 17,760.00	$ 800.00
Social Media	Instagram	$ 30,399.60	$ 786.80
Social Media	Facebook	$ 20,400.00	$ 614.40
Social Media	TikTok	$ 60,000.00	$ 10,000.00

Select App:	Google Docs
Category	
Profit	
Revenue	

5. Before we use them, both let us know what each one do alone.

INDEX Function

6. It Takes: **Array (Range)** and the **row index**
7. It Returns: The **value** in this Row
8. For example, if you give INDEX the **Category** column array and **8** it will return **Social Media**.
9. Let us try it:

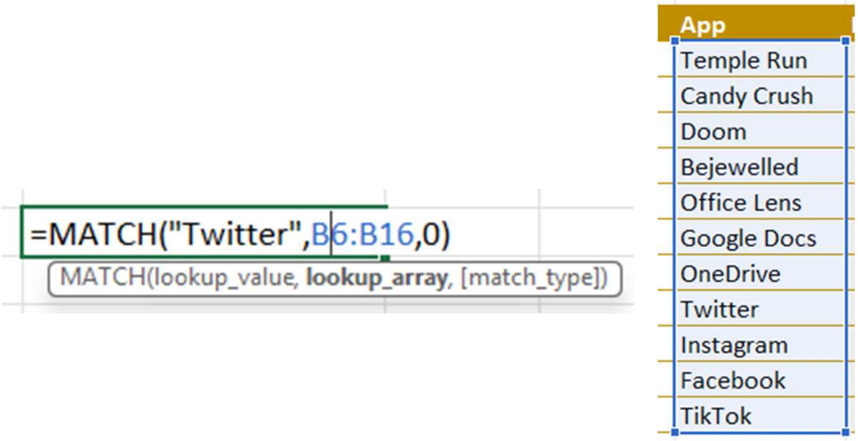


10. But I do not know the Number of Row to give?
11. That is the rule of MATCH Function

MATCH Function

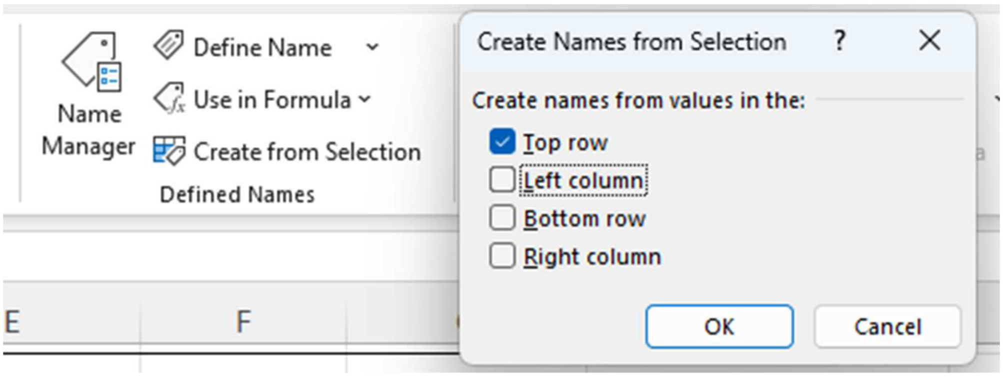
12. It takes: A **value** to lookup and **Array** (**Range**) to look in.
13. It Returns: The **Index** of this Value in the Array.

14. Let us try it

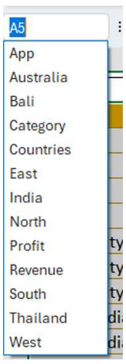


15. If I gave it **Twitter** and give it the **App** Range it will return **8**.
16. Now I can use both function:
    a. Give Value to **MATCH** to get back the **Index**.
    b. Give the Index I got to **INDEX** to get the value corresponds.
17. Let us now solve our Problem this way:

18. First Convert all Column Ranges to **Named Ranges** to make it easier to work with.
19. Select the whole table.
20. Formulas➔Defined Names➔Create from Selection.



21. Use Top Row.
22. Check now the Named ranges in the Name drop box.



23. Now let us use it in our calculation.

24. Find Category correspond to App Selected:

Select App:	Google Docs
Category	=INDEX(Category,MATCH(H5,App,0))

Category	App
Game	Temple Run
Game	Candy Crush
Game	Doom
Game	Bejewelled
Productivity	Office Lens
Productivity	Google Docs
Productivity	OneDrive
Social Media	Twitter
Social Media	Instagram
Social Media	Facebook
Social Media	TikTok

Select App:	Google Docs
Category	Productivity
Profit	
Revenue	

25. Find Revenue correspond to App selected:

Select App:	Google Docs
Category	Productivity
Profit	=INDEX(Profit,MATCH(H5,App,0)

26. Find Revenue correspond to App Selected.

Revenue	=INDEX(Revenue,MATCH(H5,App,0)

27. Check your result

Select App:	Google Docs
Category	Productivity
Profit	$ 1,166.00
Revenue	$ 17,990.00

Productivity	Google Docs	$ 17,990.00	$ 1,166.00

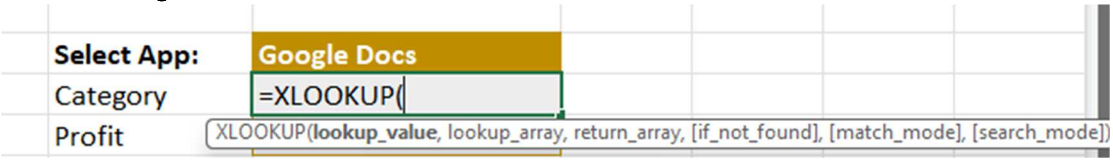
Exercise 09 B: XLOOKUP

1. It is a brand-new function in Excel 2021.
2. Let us solve the same problem again.

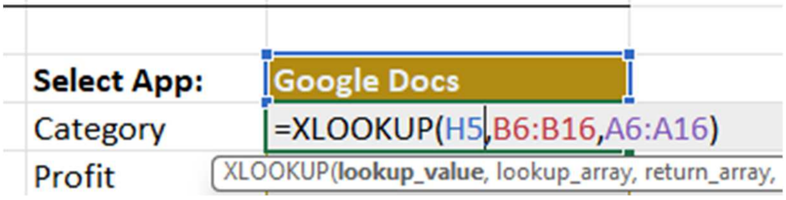
XLOOKUP

3. Get Category corresponding to selected App:

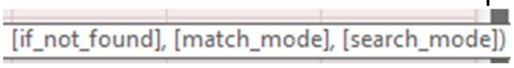
4. Start writing the XLOOKUP function.



5. The 1st Argument is **Lookup_value** the one I want to look up by which **H5** cell that contain the **App** Name.
6. 2nd argument is the **lookup_array** ,the column that it will search in which is the range (B6:B15) the App Column.
7. 3rd argument is the **return_array** the column it will return the value from, here it is range (A6:A16) the **Category** column.
8. If you press enter after closing the baratheas it will work.



9. But the new function has new 3 options let us explore.



   a. **If_not_found** : what value should retune if the search return nothing.
   b. **Match_mode**
      i. Exact Match,
      ii. Exact Match or nearest smaller item
      iii. Exact Match or nearest larger item
      iv. Wild character match.
   c. **Search_Mode**:
      i. Search last-to-first.
      ii. Search first-to-last.
      iii. Binary search ascending.
      iv. Binary search Descending.

10. Get the Profit value.
11. Let us make it easier and convert your data set into table.
12. Name table **App_List**.
13. Now create the XLookup function:
14. Start with first argument H5
15. In second argument write the name of your table the [
16. The would open the list of table column to select from
17. Select App column. (**App_List[App]**)

18. For the 3rd argument use App_List[Profit]

Select App:	Google Docs
Category	Productivity
Profit	=XLOOKUP(H5,App_List[App],App_List[Profit],,0,1

XLOOKUP(lookup_value, lookup_array, return_array, [if_not_found], [match_mode], **[search_mode]**)

19. Use (**,,**) to bypass the **not_found** argument.
20. Use **0** form **match_mode** for exact match.
21. Use **1** for **search_mode** to search from top to bottom.

Select App:	Google Docs	
Category	Productivity	
Profit	$	1,166.00

22. Get Revenue
23. Try to write the formula the same way.
24. But this time select the column from table when you write 2nd and 3rd argument.
25. You will get the same formula automatic instead of writing the name of the table first then [.
26. Your formula should look like this.

Revenue	=XLOOKUP(H5,App_List[App],App_List[Revenue],,0,1

XLOOKUP(lookup_value, lookup_array, return_array, [if_not_found], [match_mode], **[search_mode]**)

Select App:	Google Docs	
Category	Productivity	
Profit	$	1,166.00
Revenue	$	17,990.00

27. What if **Google Docs** repeated in the App column in the table?
28. Add the following column to the end of your table.

Utility	Google Docs	$	50,000.00	$	40,000.00

29. Now your table has two Google Doc.
30. Notice that all values in our XLOOKUP functions did not change, as all use the **search_mode is** <u>from top to bottom</u> (Default mode).
31. Now go to revenue formula and change the **search_mode** from 1 to -1.

Revenue	=XLOOKUP(H5,App_List[App],App_List[Revenue],,0,-1)

XLOOKUP(lookup_value, lookup_array, return_array, [if_not_found], [match_mode], **[search_mode]**)

Revenue	$	50,000.00

32. Notice now that the result reflect the last value of Google Doc.

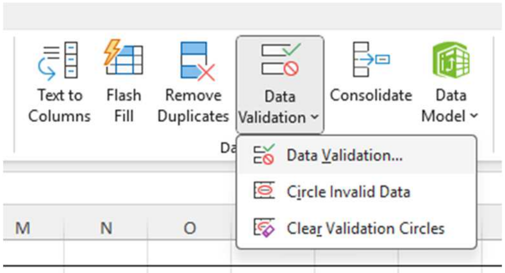
Exercise 09 C: Create Drop Down List

1. I have some data about employees, and I want to create a drop-down list on the small table on the right so I can choose to get the information of each employee.

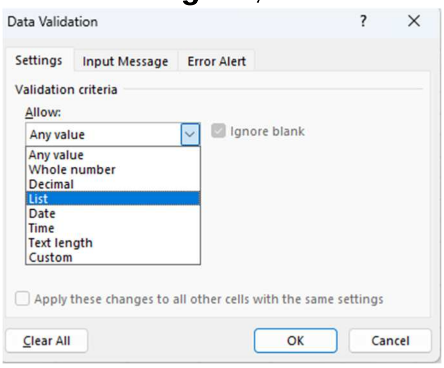
Name	Department	Salary
Sarah Jones	IT	$ 55,328
Lucy Edwards	Sales	$ 26,567
Mike Woods	Sales	$ 49,938
Carl Taylor	Marketing	$ 35,228
Michael Parks	Marketing	$ 30,716
Vin Patel	Finance	$ 33,361
Ming Sia	IT	$ 31,815
Phllip Lim	IT	$ 34,222
Olivia Porter	Finance	$ 33,219
Sam Spencer	Sales	$ 55,619
Marcus Green	Marketing	$ 50,441

Name	
Department	
Salary	

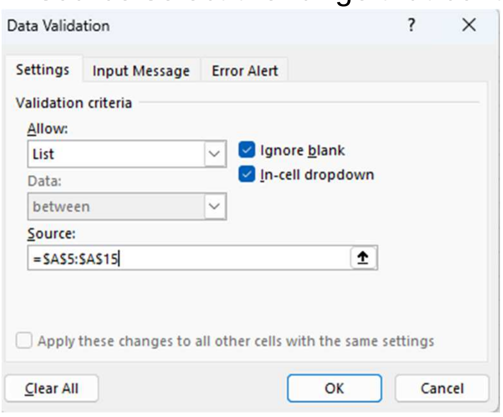
2. Data➔Data Tools➔Data Validation➔Data Validation…



3. In the **Setting** tab, in the **Allow**, select **List.**



4. In source select the range that contains the names.

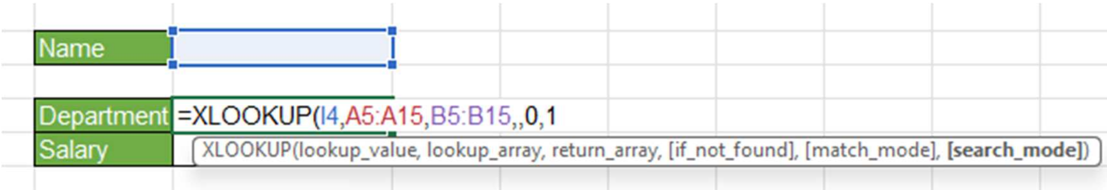


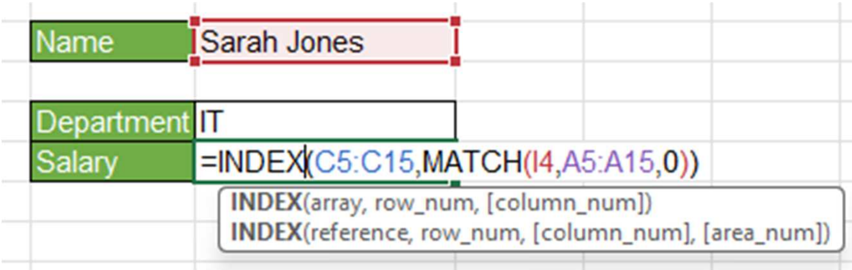
5. Click OK.

6. Now you have the list to select from.

Do it Yourself:

7. Create Lookup functions to retrieve **Department** and **Salary** for each employee.
8. Use **VLOOKUP** or **XLOOKUP** or **INDEX** + **MATCH.**





9. Now try your drop-down list.
10. Notice you have **NA** error when you do not select any employee.
11. Handel the error using **IFNA** or **IFERROR** functions.

Handle Duplicate in the list

12. What if there is a duplicate in the employee's name?
13. Add the same name at the end of the employees' list.
14. Notice that the new employee does not appear on the list.
15. Go and update the range in the Data Validation windows.

16. Notice that the new employee appears on the list only once.

17. In the new versions of Excel, they have solved the problem for you, and you do not have to take any action.
18. If you have an old version of Excel, use UNIQUE Function at any cell in the worksheet or another hidden worksheet.

19. Use the new list as the range of your data validation.
20. You can **hide** this list or move it to another worksheet.
21. Delete the new list and you will notice that the data was deleted from the drop-down list.

Handle the update of new values

22. What if your list is always updated?
23. You can convert your data into a table
24. Click **CTRL+T** and convert your data into a table.
25. Try to add new data to the table, it will expand.
26. Now go and select the range of the new column in Data Validation list.
27. Now as soon as you add a new record it will appear on the list.

Exam 08 Advanced Lookup information

Lookup Functions

Athlete Name	Bib Number	Event	Route	Position
Page, Lisa	1012	Full Marathon	East Loop	8
Duran, Brian	1491	Full Marathon	East Loop	6
O'Connor, Kent	1858	Full Marathon	East Loop	10
Spencer, Lucy	1920	Full Marathon	East Loop	1
Tanner, Claire	1789	Full Marathon	East Loop	2
Wilkins, Jesse	1997	Full Marathon	East Loop	4
Holland, Donald	1090	Full Marathon	East Loop	7
Burton, Cam	1958	Full Marathon	East Loop	3
Todd, Steven	1471	Full Marathon	East Loop	5

Athelete	
Bib Number	
Route	
Position	

In the worksheet:

1. Create a **data validation** drop-down list in cell **H3** that lists the athletes.
2. Using a lookup function of your choice, return the results in column H.
3. Add error checking to the formula to account for any #N/A errors that may occur.

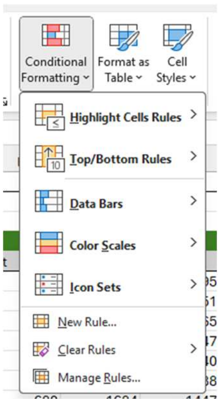
Chapter 10: Conditional Formatting

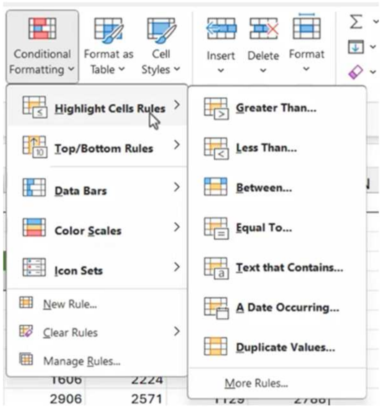
Conditional Formatting in Excel allows you to change the appearance of cells based on their values. This helps you quickly highlight important patterns and trends in your data.

Exercise 10 A: Highlight Cells Rules

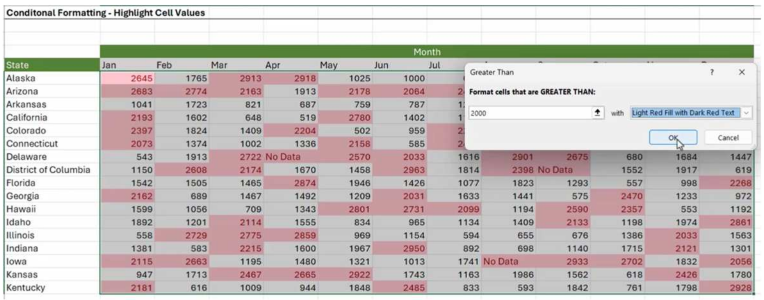
Highlight Cell Rules

1. We want to highlight cells that meet specific criteria in the sheet.
2. You can find all conditional formatting options on Home➔Style➔Conditional Formatting.

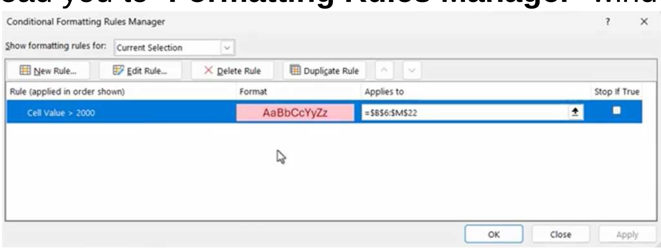




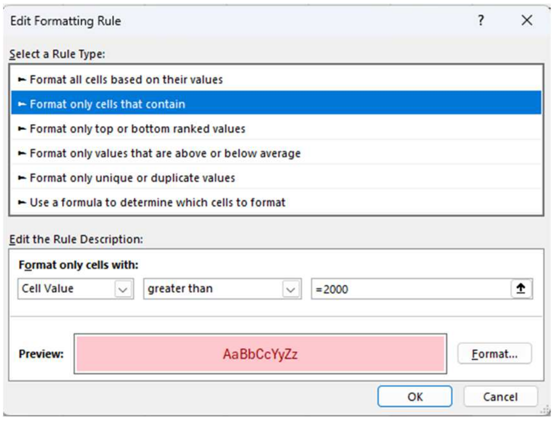
3. "**Highlight Cell Rules**" option has many options in a secondary menu.
4. I want highlight values that is > 2000.
5. Select all the range.
6. Highlight Cell Rules ➔ Greter Than.
7. Write 2000.



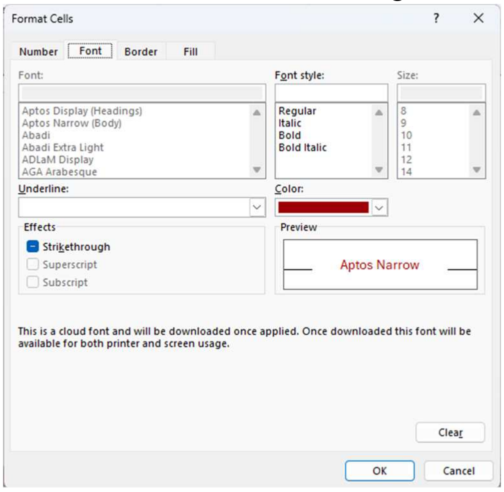
8. Notice that "**No Data**" Cells is highlighted too.
9. To solve that let us go back to our rule.
10. Conditional Formatting➔Mange Rules.
11. That would lead you to "**Formatting Rules Manager**" window.



12. Click **Edit Rule** button.

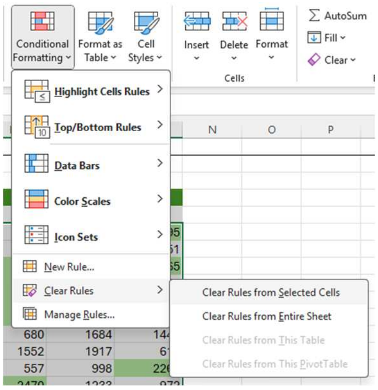


13. The Edit Formatting Rule window appears.
14. It is where you can change the “**Rule Type**” or edit the “**Rule Description**”.
15. And you can Click the **Format** Button to change the Format Style.



Clear Rule

16. To Clear the Rule Use clear rule option.

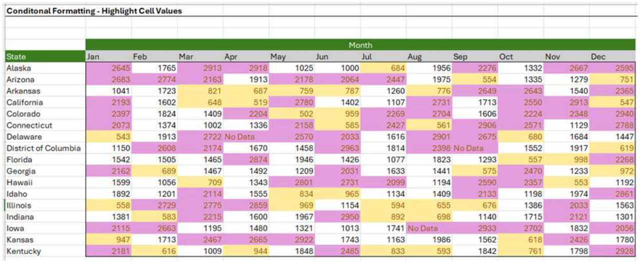


17. Select **Clear Rule from Entire Sheet**.
18. Repeat steps to highlight values **< 1000** with **yellow** color.

Applying Multiple Rules

19. Highlight the cells and go to Rule manager.
20. Duplicate the rule of <1000 with yellow color

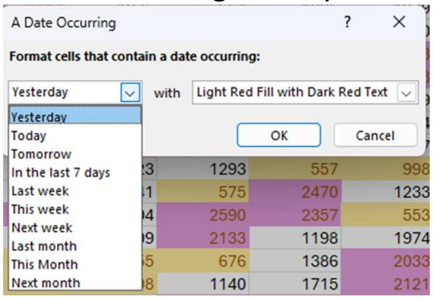
21. Edit new Rule to > 2000 with red color.



Conditonal Formatting - Highlight Cell Values

	Month											
State	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Alaska	2645	1765	2913	2918	1025	1000	684	1956	2276	1332	2667	2595
Arizona	2683	2774	2163	1913	2178	2064	2447	1975	554	1335	1279	751
Arkansas	1041	1723	821	687	759	787	1260	776	2649	2643	1540	2365
California	2193	1602	648	519	2780	1402	1107	2731	1713	2550	2913	547
Colorado	2397	1824	1409	2204	502	959	2269	2704	1606	2224	2348	2940
Connecticut	2073	1374	1002	1336	2158	585	2427	561	2906	2571	1129	2788
Delaware	543	1913	2722	No Data	2570	2033	1616	2901	2675	680	1684	1447
District of Columbia	1150	2608	2174	1670	1458	2963	1814	2398	No Data	1552	1917	619
Florida	1542	1505	1465	2874	1946	1426	1077	1823	1293	557	998	2268
Georgia	2162	689	1467	1492	1209	2031	1633	1441	575	2470	1233	972
Hawaii	1599	1056	709	1343	2801	2731	2099	1194	2590	2357	553	1192
Idaho	1892	1201	2114	1555	834	965	1134	1409	2133	1198	1974	2861
Illinois	558	2729	2775	2859	969	1154	594	655	676	1386	2033	1563
Indiana	1381	583	2215	1600	1967	2950	892	698	1140	1715	2121	1301
Iowa	2115	2663	1195	1480	1321	1013	1741	No Data	2933	2702	1832	2056
Kansas	947	1713	2467	2665	2922	1743	1163	1986	1562	618	2426	1780
Kentucky	2181	616	1009	944	1848	2485	833	593	1842	761	1798	2928

22. You can use other options in the Highlight cells:
    a. **Between:** Make you highlight cells between two values
    b. **Equal too**: For specific number.
    c. **Text Contains**: For specific Text.
    d. **Date Occurring**: Date periods.



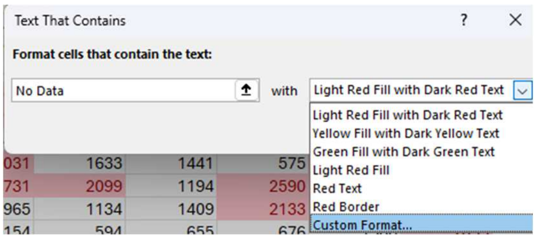
    e. **Duplicate Values**: This will highlight and duplicate values in your selected range.
23. Use option "**Text Contains**" to Add New Rule to Highlight cells with "**No Data**" with **Green** Color.

Highlight Duplicates

24. Clear All Rules from the sheet.
25. Highlight the Duplicate Values with red background color.
26. The cells that are repeated are highlighted.
27. Clear all Rules.

Solve the No Data Highlighted with > 2000 Problem

28. To solve the problem, we will add 2 rules.
29. Add Rule that highlights cells > 2000.
30. Add another Rule with Options "**Text Contains**": **No Data**
    a. Select **Custom Format** this time.

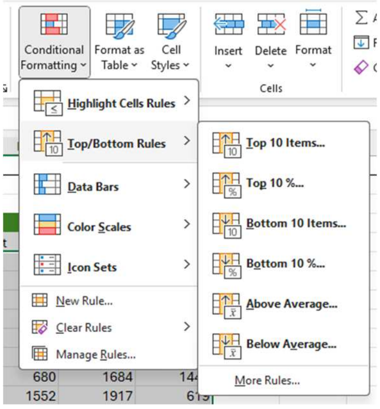


    b. Choose **Fill**: **No Color**.
    c. Choose **Font**: **Automatic**.
31. The cells with No Data will not be highlighted.

Exercise 10 B: Data Bars

Top Bottom Rules

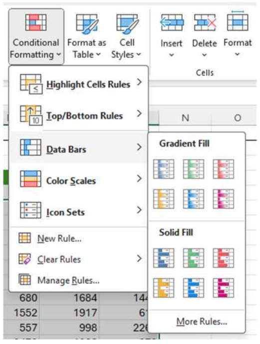
1. Try the Options in this menu



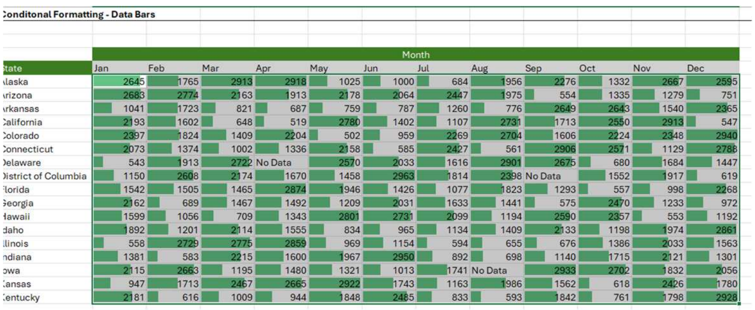
2. For example, try **Top 10 %** you can change color and the %.

Bar Chart

3. Bar Chart is a Visual Representation of "**in cell bar chart** ".
4. You have two Groups:
    a. Gradient Fill.
    b. Solid Fill.
5. You can also Click "**More Rules**" to customize how it looks.



6. Highlight the data.
7. Select Solid Fill➔Green.



Conditonal Formatting - Data Bars

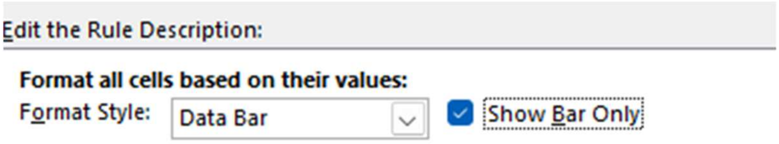
	Month											
tate	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
laska	2645	1765	2913	2918	1025	1000	684	1956	2276	1332	2667	2595
rizona	2683	2774	2163	1913	2178	2064	2447	1975	554	1335	1279	751
rkansas	1041	1723	821	687	759	787	1260	776	2649	2643	1540	2365
alifornia	2193	1602	648	519	2780	1402	1107	2731	1713	2550	2913	547
olorado	2397	1824	1409	2204	502	959	2269	2704	1606	2224	2348	2940
onnecticut	2073	1374	1002	1336	2158	585	2427	561	2906	2571	1129	2788
elaware	543	1913	2722	No Data	2570	2033	1616	2901	2675	680	1684	1447
istrict of Columbia	1150	2608	2174	1670	1458	2963	1814	2398	No Data	1552	1917	619
lorida	1542	1505	1465	2874	1946	1426	1077	1823	1293	557	998	2268
eorgia	2162	689	1467	1492	1209	2031	1633	1441	575	2470	1233	972
awaii	1599	1056	709	1343	2801	2731	2099	1194	2590	2357	553	1192
daho	1892	1201	2114	1555	834	965	1134	1409	2133	1198	1974	2861
llinois	558	2729	2775	2859	969	1154	594	655	676	1386	2033	1563
ndiana	1381	583	2215	1600	1967	2950	892	698	1140	1715	2121	1301
owa	2115	2663	1195	1480	1321	1013	1741	No Data	2933	2702	1832	2056
ansas	947	1713	2467	2665	2922	1743	1163	1986	1562	618	2426	1780
entucky	2181	616	1009	944	1848	2485	833	593	1842	761	1798	2928

Manage Rules

8. Open "**Conditional Formatting Rules Manager**".
9. Click Edit Rule.

Show Bars Only

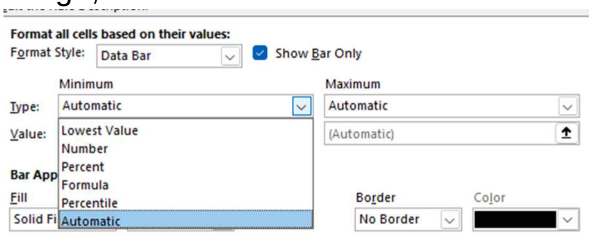
10. Select option: **Show Bars Only.**



11. This will show only bars without values in cells.
12. This is good if you are presenting a “**Confidential Data**”.

Min and Max Value

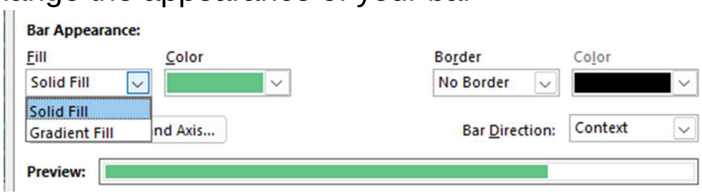
13. Go back to “**Conditional Formatting Rule Manager**”.
14. Clear option Show Bar Only.
15. Notice that Min and Max Value are set to Automatic.
16. That means that **the length of the bar** will depend on your data Max and Min values.
17. If you want to define the values you can change depending on **Type**: Number, Percentage, ….. etc.



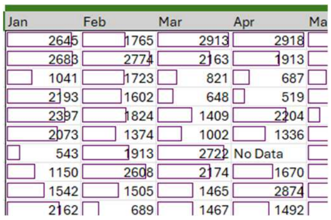
18. Try this:
    a. Change **Type** of Min and Max to **Number**.
    b. Set Min=**0** and Max = **3000.**
    c. Click OK.
    d. Try again with Values **500** and **6000**
    e. Click OK
    f. This Time you will have values near 500 will barely have bar.
19. Be aware the Max and Min Values will affect the length of your bars.
20. Change Max and Min Values again to **Automatic**.

Bar Appearance

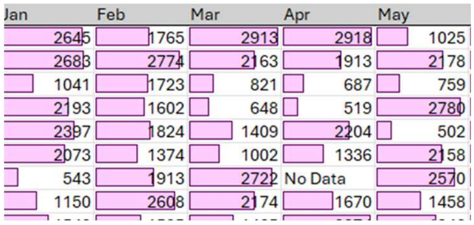
29. Go back to “**Conditional Formatting Rule Manager**”.
30. You can Change the appearance of your bar



31. Try This
    a. Choose **White** for **Fill**
    b. Choose **Solid Border** with **Dark Purple**.
    c. Click OK.
    d. It gives a different effect.
    e. Try again with fill **Light pink** instead of White and see.



Jan	Feb	Mar	Apr	Ma
2645	1765	2913	2918	
2683	2774	2163	1913	
1041	1723	821	687	
2193	1602	648	519	
2397	1824	1409	2204	
2073	1374	1002	1336	
543	1913	2722	No Data	
1150	2608	2174	1670	
1542	1505	1465	2874	
2162	689	1467	1492	



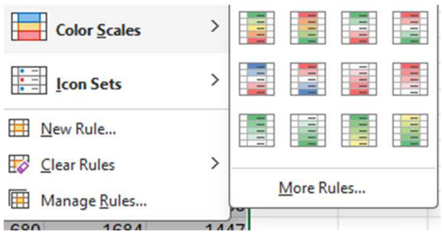
Jan	Feb	Mar	Apr	May
2645	1765	2913	2918	1025
2683	2774	2163	1913	2178
1041	1723	821	687	759
2193	1602	648	519	2780
2397	1824	1409	2204	502
2073	1374	1002	1336	2158
543	1913	2722	No Data	2570
1150	2608	2174	1670	1458

Exercise 10 C: Color Scales

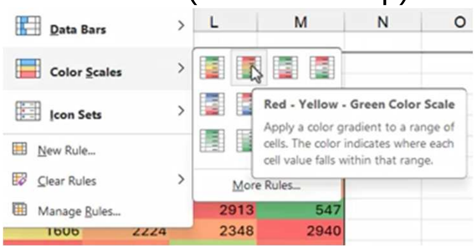
1. We can create Heatmap that shows the trends.
2. Select the whole data.
3. If the data is huge you can use:

Ctrl + Shift + → then Ctrl + Shift + ↓

4. Choose Color Scale
5. You have 12 galleries of color scales.



6. There are:
    a. 2 color Scale and
    b. 3 color scales.
7. Choose Red Yellow Green one (2nd one on top).



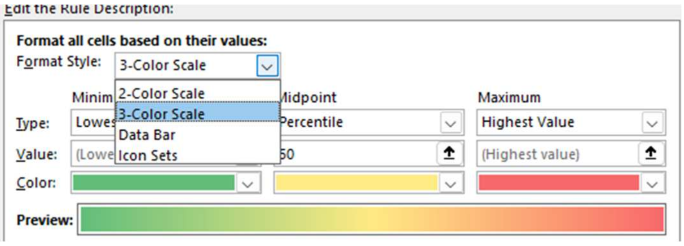
8. Read the tooltip.
9. The representation will be:
    a. **Highest** Value is represented with **Red**.
    b. **Green** is the **Lowest**.
10. You can clearly see that the highest value is **2963** in June.
11. The colors will be graded from **Red** to **Yellow** to **Green**.

	Month											
tate	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
laska	2645	1765	2913	2918	1025	1000	684	1956	2276	1332	2667	2595
rizona	2683	2774	2163	1913	2178	2064	2447	1975	554	1335	1279	751
rkansas	1041	1723	821	687	759	787	1260	776	2649	2643	1540	2365
alifornia	2193	1602	648	519	2780	1402	1107	2731	1713	2550	2913	547
olorado	2397	1824	1409	2204	502	959	2269	2704	1606	2224	2348	2940
onnecticut	2073	1374	1002	1336	2158	585	2427	561	2906	2571	1129	2788
elaware	543	1913	2722	No Data	2570	2033	1616	2901	2675	680	1684	1447
istrict of Columbia	1150	2608	2174	1670	1458	2963	1814	2398	No Data	1552	1917	619
lorida	1542	1505	1465	2874	1946	1426	1077	1823	1293	557	998	2268
eorgia	2162	689	1467	1492	1209	2031	1633	1441	575	2470	1233	972
awaii	1599	1056	709	1343	2801	2731	2099	1194	2590	2357	553	1192
daho	1892	1201	2114	1555	834	965	1134	1409	2133	1198	1974	2861
linois	558	2729	2775	2859	969	1154	594	655	676	1386	2033	1563
ndiana	1381	583	2215	1600	1967	2950	892	698	1140	1715	2121	1301
owa	2115	2663	1195	1480	1321	1013	1741	No Data	2933	2702	1832	2056
ansas	947	1713	2467	2665	2922	1743	1163	1986	1562	618	2426	1780
entucky	2181	616	1009	944	1848	2485	833	593	1842	761	1798	2928

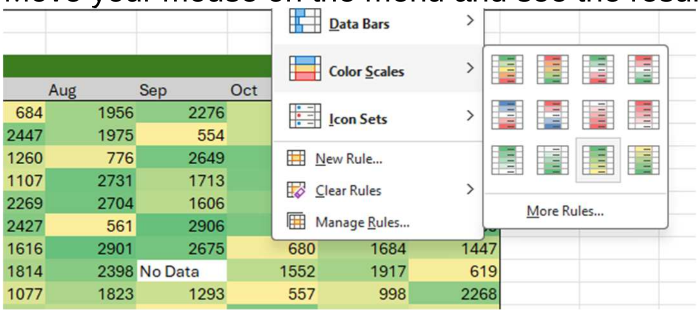
Change the color

12. You can go to "**Conditional Formatting Rule Manager**" and:

a. Change the Number of colors.
b. Choose colors.



13. Try this:
    a. Use **2** colors and see the result.
    b. Change the two colors and see result.
    c. Get back to **3** colors: light **blue-yellow-purple** and see result.
    d. Crazy colors 😊
    e. Go now and Choose from the default options.
    f. Move your mouse on the menu and see the results.



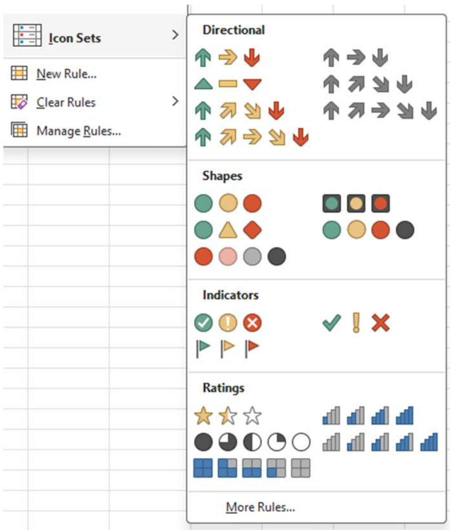
    g. Select **Yellow-Green** one.
14. Notice that **No Data** has no color.

Exercise 10 D: Icon Set

1. You have list of companies with Budget, Actual, Difference columns

Company	Budget	Actual	Difference
Company A	143	409	266
Company B	471	230	-241
Company C	305	272	-33
Company D	152	254	102
Company E	132	216	84
Company F	188	272	84
Company G	215	320	105

2. The difference shows if the company **Over Budget** like Company A or **Under Budget** like Company B.
3. Select values of **Difference** column.
4. Select an Icon Set.

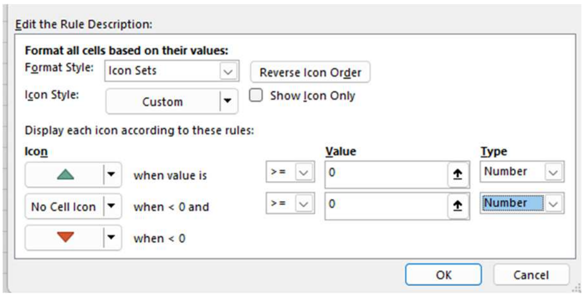


5. You have
    a. Groups: Directional, Shapes, Indicators, and Ratings
    b. And you have 3, 4, and 5 Icons
6. Select **3 Direction Arrows**.

7. Excel tries to divide your values into 3.
8. Excel Applies Green up arrow for Top one and red down arrow for Bottom values.
9. The values in the middle are yellow arrows.
10. Try:
    a. Use only Arrows without data and see the result.
    b. Get Values with arrows back.

Edit Icon Values

11. It is better to make the Green Arrow for positive values and , red one for negative ones.
12. Try This:
    a. Choose Different 3 color Style.
    b. Change Value Types to Numbers.
    c. I need only 2 colors, so choose the middle one and select no cell icon.
    d. You will have only > 0 and < 0 arrows.



Company	Budget	Actual	Difference
Company A	143	409	▲ 266
Company B	471	230	▼ -241
Company C	305	272	▼ -33
Company D	152	254	▲ 102
Company E	132	216	▲ 84
Company F	188	272	▲ 84
Company G	215	320	▲ 105
Company H	361	388	▲ 27
Company I	199	168	▼ -31
Company J	162	127	▼ -35
Company K	397	204	▼ -193
Company L	494	256	▼ -238
Company M	458	277	▼ -181
Company N	365	470	▲ 105
Company O	258	394	▲ 136
Company P	402	461	▲ 59
Company Q	391	269	▼ -122

13. Try this:
    a. Choose to show icon only.
    b. Adjust the icons in the middle of the cell.
    c. That makes it so clear to see if difference is negative or positive.

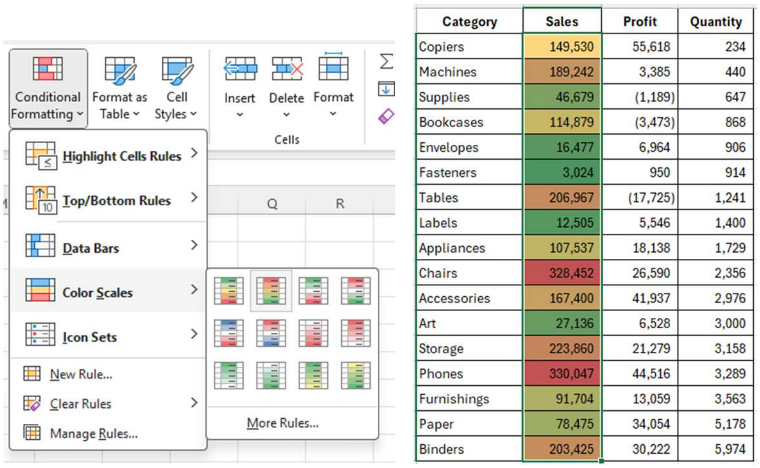
Company	Budget	Actual	Difference
Company A	143	409	▲
Company B	471	230	▼
Company C	305	272	▼
Company D	152	254	▲
Company E	132	216	▲
Company F	188	272	▲
Company G	215	320	▲
Company H	361	388	▲
Company I	199	168	▼
Company J	162	127	▼
Company K	397	204	▼
Company L	494	256	▼
Company M	458	277	▼
Company N	365	470	▲
Company O	258	394	▲
Company P	402	461	▲
Company Q	391	269	▼

Exercise 10 E: Format Sales Report

In this exercise we will reconsolidate what we have learn to a Sales Report

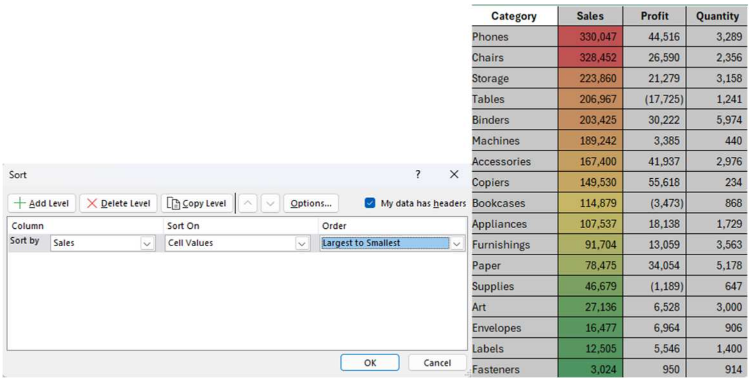
Step 1 – Apply Background Color to the Sales Column

1. Select all the cells in the **Sales** column (excluding the header).
2. Go to the **Home** tab on the Ribbon.
3. In the **Styles** group, click **Conditional Formatting**.
4. From the menu, choose **Color Scales**.
5. Change Background color to **Light Blue**.
6.



Category	Sales	Profit	Quantity
Copiers	149,530	55,618	234
Machines	189,242	3,385	440
Supplies	46,679	(1,189)	647
Bookcases	114,879	(3,473)	868
Envelopes	16,477	6,964	906
Fasteners	3,024	950	914
Tables	206,967	(17,725)	1,241
Labels	12,505	5,546	1,400
Appliances	107,537	18,138	1,729
Chairs	328,452	26,590	2,356
Accessories	167,400	41,937	2,976
Art	27,136	6,528	3,000
Storage	223,860	21,279	3,158
Phones	330,047	44,516	3,289
Furnishings	91,704	13,059	3,563
Paper	78,475	34,054	5,178
Binders	203,425	30,222	5,974

7. Choose a blue gradient color scale (for example: **Blue – White – Blue**).
    - In Excel, darker colors will represent higher values, lighter colors will represent lower values.
8. Sort Sales Z to A to see the gradient.



Category	Sales	Profit	Quantity
Phones	330,047	44,516	3,289
Chairs	328,452	26,590	2,356
Storage	223,860	21,279	3,158
Tables	206,967	(17,725)	1,241
Binders	203,425	30,222	5,974
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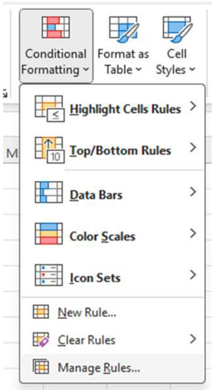
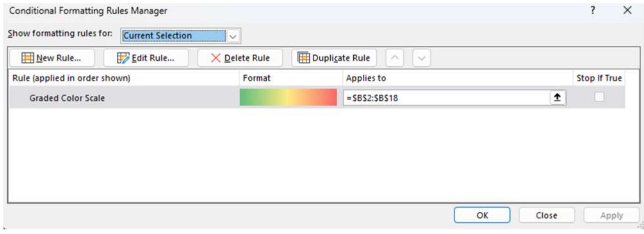
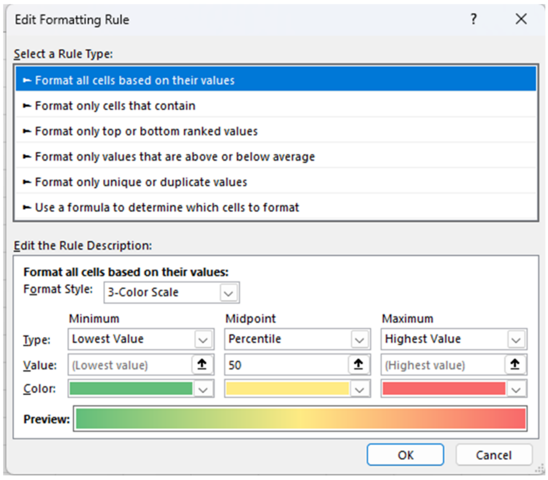
Art	27,136	6,528	3,000
Envelopes	16,477	6,964	906
Labels	12,505	5,546	1,400
Fasteners	3,024	950	914

9. To customize the colors:
    - Go back to **Conditional Formatting → Manage Rules…**.
    - Select your color scale rule and click **Edit Rule…**.
    - Click the **Color** drop-down menus to choose your preferred colors.
    - Click **OK** twice.

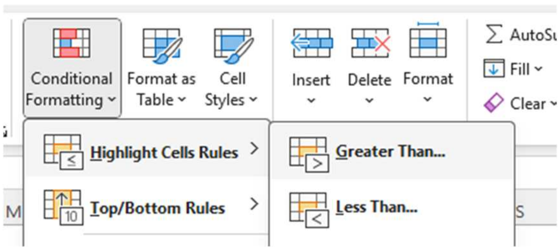




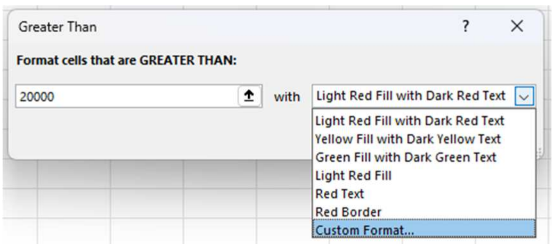


Step 2 – Apply Font Color Conditional Formatting to the Profit Column

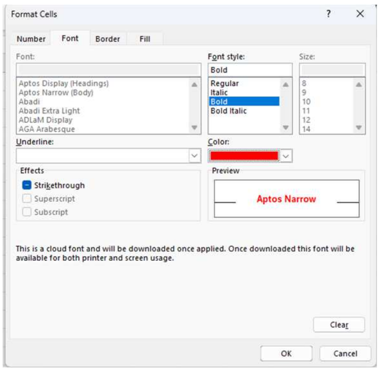
1. Select all the cells in the **Profit** column (excluding the header).
2. Go to **Home** → **Conditional Formatting** → **Highlight Cells Rules** → **Greater Than…** (or **Less Than…** depending on what you want to highlight).



3. In the dialog box, enter the value or select a cell to compare against.
4. Enter 20000.
5. In the formatting drop-down, choose **Custom Format…**.



6. Go to the **Font** tab, choose a **Dark Red** color, and click **OK**.

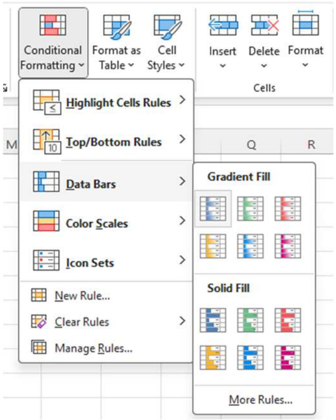


Category	Sales	Profit	Quantity
Phones	330,047	44,516	3,289
Chairs	328,452	26,590	2,356
Storage	223,860	21,279	3,158
Tables	206,967	(17,725)	1,241
Binders	203,425	30,222	5,974
Machines	189,242	3,385	440
Accessories	167,400	41,937	2,976
Copiers	149,530	55,618	234
Bookcases	114,879	(3,473)	868
Appliances	107,537	18,138	1,729
Furnishings	91,704	13,059	3,563
Paper	78,475	34,054	5,178
Supplies	46,679	(1,189)	647
Art	27,136	6,528	3,000
Envelopes	16,477	6,964	906
Labels	12,505	5,546	1,400
Fasteners	3,024	950	914

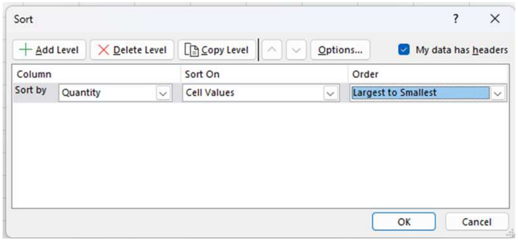
7. Click **OK** again to apply.

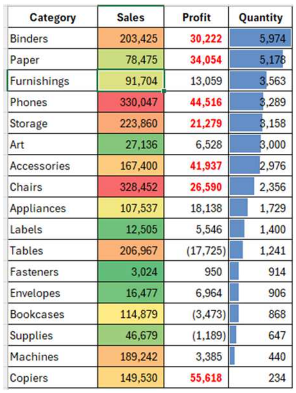
Step 3 – Apply Data Bars to the Quantity Column

1. Select all the cells in the **Quantity** column (excluding the header).
2. Go to **Home** → **Conditional Formatting** → **Data Bars**.
3. Choose a solid fill color (e.g., green or blue).



4. Sort Quantity in descending order to see the largest bars at the top:
   - Go to **Data** → **Sort Z** → **A**.

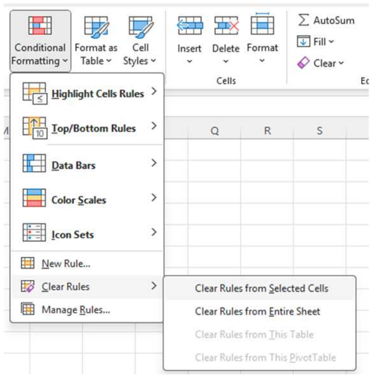




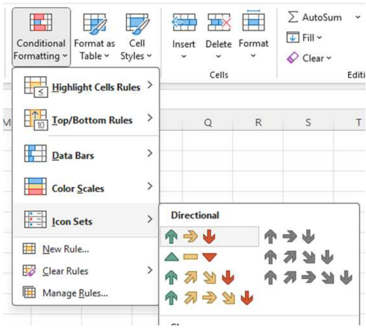
Category	Sales	Profit	Quantity
Binders	203,425	30,222	5,974
Paper	78,475	34,054	5,178
Furnishings	91,704	13,059	3,563
Phones	330,047	44,516	3,289
Storage	223,860	21,279	3,158
Art	27,136	6,528	3,000
Accessories	167,400	41,937	2,976
Chairs	328,452	26,590	2,356
Appliances	107,537	18,138	1,729
Labels	12,505	5,546	1,400
Tables	206,967	(17,725)	1,241
Fasteners	3,024	950	914
Envelopes	16,477	6,964	906
Bookcases	114,879	(3,473)	868
Supplies	46,679	(1,189)	647
Machines	189,242	3,385	440
Copiers	149,530	55,618	234

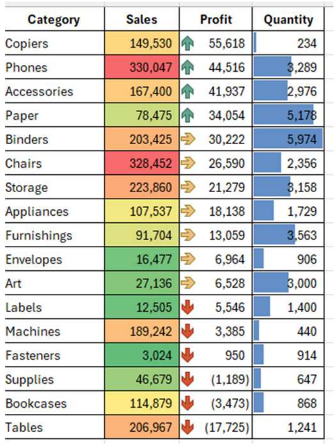
Step 4 – Apply Icon Sets to the Profit Column

1. First, remove the font color rule from the Profit column:
   - Select the Profit column.
   - Go to **Home → Conditional Formatting → Clear Rules → Clear Rules from Selected Cells**.



2. Now apply an icon set:
   - Select the Profit column again.
   - Go to **Conditional Formatting → Icon Sets**.
   - Choose a 3-icon set (e.g., Green Circle, Yellow Triangle, Red Diamond).





Category	Sales	Profit	Quantity
Copiers	149,530	55,618	234
Phones	330,047	44,516	3,289
Accessories	167,400	41,937	2,976
Paper	78,475	34,054	5,178
Binders	203,425	30,222	5,974
Chairs	328,452	26,590	2,356
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Machines	189,242	3,385	440
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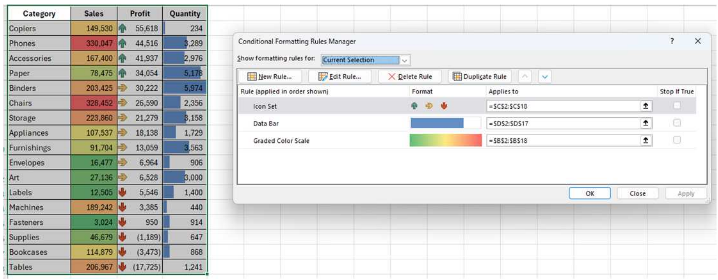
Supplies	46,679	(1,189)	647
Bookcases	114,879	(3,473)	868
Tables	206,967	(17,725)	1,241

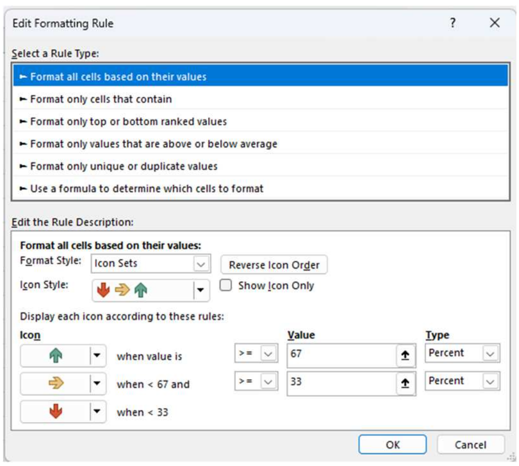
3. Sort the Profit column in descending order to see high profits with green icons on top.

Customize the Icon Set

1. Go to **Conditional Formatting → Manage Rules…**.



2. Select your icon set rule and click **Edit Rule…**.
3. You can:
   - Change icons to different shapes or symbols.
   - Switch from **percentages** to **specific numeric values** for thresholds.



4. Click **OK** to save changes.

Review Your Final Table
By now, you should see:

- **Sales Column** → Blue gradient background (dark blue for high sales).
- **Profit Column** → Green, yellow, or red icons showing profitability.
- **Quantity Column** → Data bars showing volume visually.

Your table should clearly highlight trends and key values, making it easier to analyze your data at a glance.

Category	Sales	Profit	Quantity
Copiers	149,530	55,618	234
Phones	330,047	44,516	3,289
Accessories	167,400	41,937	2,976
Paper	78,475	34,054	5,178
Binders	203,425	30,222	5,974
Chairs	328,452	26,590	2,356
Storage	223,860	21,279	3,158
Appliances	107,537	18,138	1,729
Furnishings	91,704	13,059	3,563
Envelopes	16,477	6,964	906
Art	27,136	6,528	3,000
Labels	12,505	5,546	1,400
Machines	189,242	3,385	440
Fasteners	3,024	950	914
Supplies	46,679	(1,189)	647
Bookcases	114,879	(3,473)	868
Tables	206,967	(17,725)	1,241

Exam 9: Conditional Formatting

You have a table for the Rainfall for 10 cities.

Rainfall for 10 cities: 2021 - 2023

Color Scales - Add a 3-color scale to the dataset

City	2021	2022	2023
New York	43.3	42.5	43.6
Los Angeles	24.5	24.9	25.1
Chicago	37.7	37.6	37.8
Houston	45.1	44.2	45.3
Phoenix	12.6	12.9	13.1
Philadelphia	39.4	39.0	39.3
San Antonio	31.4	31.0	31.2
San Diego	22.0	22.1	22.3
Dallas	36.7	36.8	36.9
San Jose	20.7	20.8	20.9

Highlight all cells where rainfall is between 35.0 and 45.0

City	2021	2022	2023
New York	43.3	42.5	43.6
Los Angeles	24.5	24.9	25.1
Chicago	37.7	37.6	37.8
Houston	45.1	44.2	45.3
Phoenix	12.6	12.9	13.1
Philadelphia	39.4	39.0	39.3
San Antonio	31.4	31.0	31.2
San Diego	22.0	22.1	22.3
Dallas	36.7	36.8	36.9
San Jose	20.7	20.8	20.9

Icon Sets: Show a green traffic light for positive values. Red traffic light for negative values.

City	2022	2023	Difference
New York	42.5	43.6	1.1
Los Angeles	24.9	25.1	0.2
Chicago	37.6	35.6	-2.0
Houston	44.2	45.3	1.1
Phoenix	12.9	13.1	0.2
Philadelphia	39.0	39.3	0.3
San Antonio	31.0	28.2	-2.8
San Diego	22.1	22.3	0.2
Dallas	36.8	36.9	0.1
San Jose	20.8	18.6	-2.2

1. **Table 1**: Apply a 3-color scale.
2. **Table 2**: Highlight cells with values between 35.0 inches and 45.0 inches using a yellow fill.
3. **Table 3**:
   - Use icon sets to display a **red traffic light** for negative values and a **green traffic light** for positive values in the "Difference" column.
   - Show only the icon (no text) and center-align it in the cell.

Your final answer should look like this:

Rainfall for 10 cities: 2021 - 2023

Color Scales - Add a 3-color scale to the dataset

City	2021	2022	2023
New York	43.3	42.5	43.6
Los Angeles	24.5	24.9	25.1
Chicago	37.7	37.6	37.8
Houston	45.1	44.2	45.3
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Highlight all cells where rainfall is between 35.0 and 45.0

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Houston	45.1	44.2	45.3
Phoenix	12.6	12.9	13.1
Philadelphia	39.4	39.0	39.3
San Antonio	31.4	31.0	31.2
San Diego	22.0	22.1	22.3
Dallas	36.7	36.8	36.9
San Jose	20.7	20.8	20.9

Icon Sets: Show a green traffic light for positive values. Red traffic light for negative values.

City	2022	2023	Difference
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Los Angeles	24.9	25.1	
Chicago	37.6	35.6	
Houston	44.2	45.3	
Phoenix	12.9	13.1	
Philadelphia	39.0	39.3	
San Antonio	31.0	28.2	
San Diego	22.1	22.3	
Dallas	36.8	36.9	
San Jose	20.8	18.6	